

THE EYE

VOLUME 4

Engineering Specification

The controlled build, contract, runtime, assurance, and release specification for The Eye

OBSERVE / UNDERSTAND / PREDICT / SIMULATE / DECIDE / LEARN

CONTROLLED ENGINEERING BASELINE / V1.0

Inherited from Volume 0 — Product Constitution and Volume 3 — Technical Architecture

1 August 2026

Document Control

Volume 4 is the controlled engineering realization of The Eye. It converts frozen constitutional and architectural authority into testable construction, runtime, security, reliability, delivery, and acceptance obligations.

CONTROL	VALUE
Product	The Eye
Tagline	The Operating System for Strategic Intelligence
Controlled document	Volume 4 — Engineering Specification
Edition	1.0
Issue date	1 August 2026
Status	Controlled Engineering Baseline
Binding sources	Volume 0 — Product Constitution v1.0; Volume 3 — Technical Architecture v1.0
Primary audience	CTO, principal and staff engineers, architecture, AI and data engineering, security, privacy, SRE, release engineering, quality, governance, product and programme leadership
Normative scope	Deployable behavior, state ownership, APIs and events, object semantics, workflows, agents, models, trust, resilience, operations, delivery, tests, conformance, and acceptance
Downstream inheritance	Volume 5 — AI Architecture; Volume 6 — Infrastructure Architecture; Volume 7 — Data Platform; implementation repositories and runbooks

Normative language

SHALL, MUST, SHALL NOT, and MUST NOT are binding. MAY permits an implementation choice inside a frozen semantic boundary. Requirements carry stable ES-CC-NNN identities. Conformance requires verifiable evidence for the exact artifact, configuration, environment, deployment profile, and scope under review.

CONTROLLING RULE

Volume 0 controls constitutional meaning. Volume 3 controls architecture. Volume 4 controls engineering realization. Lower-level specifications add precision but cannot weaken, rename, bypass, or silently reinterpret higher authority.

Engineering Charter

The engineering baseline is complete only when every material behavior has one owner, one authoritative state boundary, a versioned contract, explicit failure semantics, measurable quality, deployment-parity evidence, and a governed release path.

ENGINEERING COMMITMENT	REQUIRED CONSTRUCTION CONSEQUENCE
One strategic operating system	Services, agents, models, interfaces, and surfaces implement one connected product contract; they do not become independent intelligence products.
Ten canonical layers	Every deployable unit maps to stable layer and component identities while preserving distinct authority and failure boundaries.
Knowledge Graph centrality	Shared semantic identity and relationships are resolved through governed graph contracts, never ad hoc identifiers or private relationship stores.
Temporal and provenance completeness	Canonical products preserve time, truth state, source evidence, transformation, model, agent, human, policy, correction, and supersession lineage.
Human final authority	Consequential institutional decisions and irreversible effects stop at an authenticated human gate.
Bounded intelligence execution	Agents, models, tools, and workflows execute under explicit purpose, capability, quality, budget, supervision, and stop conditions.
Safe degradation	Failure may reduce capability but never fabricates success, currency, confidence, completeness, authorization, provenance, or approval.
Immutable delivery	One signed candidate advances across environments and deployments with reproducible evidence, staged rollout, monitoring, and rollback.
Deployment semantic parity	SaaS, Private Cloud, and On-Premise pass the same canonical conformance fixtures and preserve customer portability.

Definition of engineering complete

- Controlling requirements, interfaces, schemas, policies, models, agents, workflows, configuration, and migrations are versioned and owned.
- Normal, partial, abstained, stale, denied, failed, recovering, and terminal behavior is explicit and testable.
- Security, privacy, sovereignty, audit, quality, resilience, recovery, and observability evidence closes for the intended consequence and deployment scope.
- Operations can diagnose, constrain, recover, explain, challenge, export, and retire the capability without uncontrolled access or hidden state.
- A signed operational acceptance record identifies exact scope, independent authorities, open conditions, review, and withdrawal criteria.

CONSTITUTIONAL INHERITANCE / C-004 · C-005 · C-008 · C-035 · C-042 · C-049 · C-052

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Native Word heading styles provide navigation. Chapter and requirement identifiers are stable references for engineering, assurance, operations, and downstream volumes.

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Executive Engineering Statement

Volume 4 defines the engineering system that makes The Eye persistent, explainable, recoverable, governable, and safe to operate at enterprise and sovereign scale.

The central engineering problem is not model invocation. It is preserving institutional meaning and authority while a continuous system observes heterogeneous sources, constructs connected knowledge, maintains digital representations, forecasts and simulates uncertain futures, prepares decision packages, records human decisions, governs commitments, measures outcomes, and learns without rewriting history.

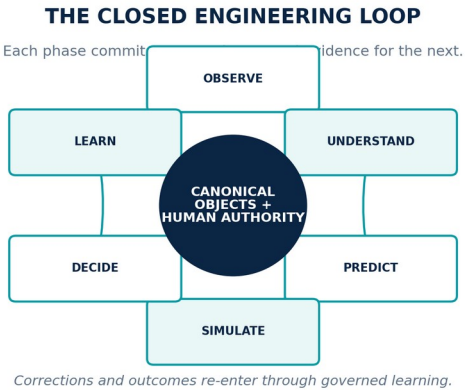


Figure 1. The Closed Engineering Loop. Each phase commits governed state and evidence for the next.

SYSTEM PLANE	ENGINEERING OBLIGATION
Canonical product plane	Typed intelligence objects preserve identity, version, temporal truth, truth state, provenance, policy, quality, correction, and lifecycle.
Execution plane	Services, events, workflows, human gates, agents, models, and tools coordinate through durable and versioned contracts.
State plane	One accountable component owns each canonical lifecycle; projections remain rebuildable and cross-owner writes are prohibited.
Trust plane	Identity, authorization, classification, rights, privacy, sovereignty, cryptography, audit, supply chain, and isolation are enforced at material boundaries.
Reliability plane	Quality, SLOs, capacity, failure domains, degradation, recovery, observability, incidents, and lifecycle operations describe customer-visible product state.
Delivery plane	Repositories, configuration, tests, immutable promotion, packaging, acceptance, and governance preserve conformance from source to operation.

ENGINEERING THESIS

The Eye may evolve in technology, deployment, and implementation. It remains one product only when canonical meaning, human authority, evidence, control, temporal reconstruction, and portability are invariant and continuously proven.

CONSTITUTIONAL INHERITANCE / C-001 · C-004 · C-005 · C-008 · C-035 · C-042 · C-052

PART I

Engineering Authority

Normative scope, conformance, decomposition, and inheritance.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

1 Engineering Authority and Scope

Volume 4 converts the controlled architecture into a testable engineering system. It specifies behavior, ownership, contracts, state transitions, failure handling, and evidence required before any implementation may represent itself as The Eye.

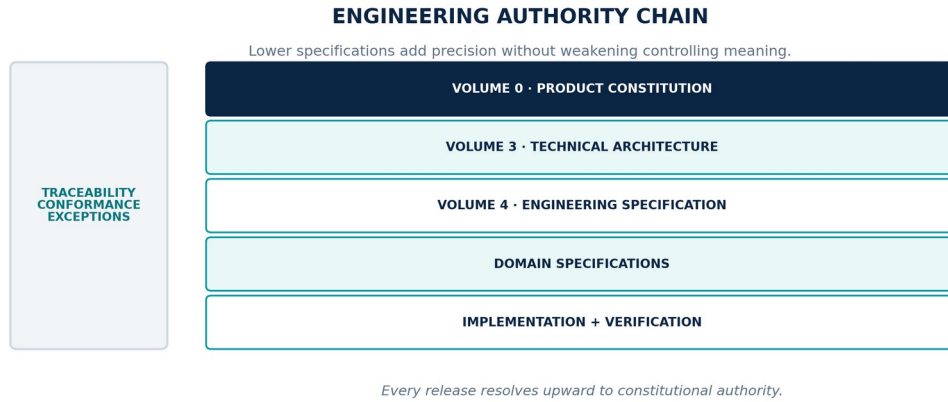


Figure 2. Engineering Authority Chain. Lower specifications add precision without weakening controlling meaning.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Authority	Volume 0 controls constitutional invariants; Volume 3 controls architecture; Volume 4 controls engineering realization.
Applies to	All first-party services, agents, models, data products, user surfaces, extensions, marketplace packages, deployment modes, and operational tooling.
Does not authorize	A new product category, renamed canonical layer, alternate truth system, autonomous final decision right, or deployment-specific semantic fork.
Conformance result	Conformant, conditionally conformant under a recorded exception, or non-conformant. Silence is never approval.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-01-001	Every deployable unit SHALL identify its owning canonical layer, component ID, state boundary, contracts, quality obligations, and accountable engineering owner.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.
ES-01-002	Every material behavior SHALL trace to an approved requirement in this volume and to the controlling architecture chapter or invariant.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-01-003	Implementation convenience SHALL NOT collapse canonical layers, bypass human authority, erase provenance, or create a second source of strategic truth.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.
ES-01-004	A product surface MAY aggregate capabilities, but the underlying ownership, authorization, evidence, and failure states SHALL remain separable and testable.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-01-005	A release SHALL be rejected when conformance evidence is absent, stale, unverifiable, or scoped to a different artifact version.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Unmapped component or service	Block production registration until ownership and inheritance are recorded.
Architecture conflict	Stop implementation promotion and open a governed exception or amendment workflow.

FAILURE CONDITION	REQUIRED BEHAVIOR
Undeclared customer-visible behavior	Treat as non-conformant product behavior and remove or place behind a disabled controlled flag.

Acceptance evidence

- Component inventory proves complete mapping to the ten canonical layers.
- Release manifest identifies requirement, interface, object, policy, and test versions.
- Architecture conformance review records no unresolved category or authority conflict.
- Production registry rejects unsigned or unmapped workloads and packages.

CHAPTER ACCEPTANCE

Engineering Authority and Scope is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-001 · C-003 · C-008 · C-051 · C-052

2 Specification Hierarchy and Conformance

Engineering decisions inherit downward through an explicit control chain. Lower-level specifications may add precision, but they may not silently weaken, reinterpret, or supersede higher authority.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Control order	Product Constitution -> Technical Architecture -> Engineering Specification -> domain specifications -> implementation -> verification evidence.
Normative artifact	A versioned document, schema, policy, test suite, manifest, or decision record that is approved by its accountable authority.
Conflict rule	The highest controlling authority prevails until an explicit amendment is ratified and propagated.
Exception rule	Exceptions are narrow, owned, time-bounded, monitored, reversible, and prohibited from changing constitutional meaning.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-02-001	Every normative artifact SHALL declare authority, owner, status, version, effective time, supersession, and dependent artifacts.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.
ES-02-002	Every implementation repository SHALL expose a machine-readable conformance manifest at build and release time.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-02-003	A conditional exception SHALL carry scope, rationale, risk acceptance, compensating controls, expiry, and closure criteria.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-02-004	No component owner MAY approve an exception that expands that component's authority or removes an independent control.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.
ES-02-005	Conformance SHALL be re-evaluated when controlling specifications, schemas, policies, models, deployment constraints, or dependencies change.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Conflicting specifications	Resolve against the hierarchy; quarantine the lower artifact until corrected.
Expired exception	Fail the affected release or workload closed unless an authorized renewal is recorded before expiry.
Missing evidence	Return an indeterminate conformance result and deny production promotion.

Acceptance evidence

- Automated checks validate manifest references and version closure.
- Exception registry proves owner, expiry, monitoring, and closure status.
- Dependency impact tests identify artifacts affected by a controlling change.
- Independent assurance verifies separation of proposal, approval, and verification duties.

CHAPTER ACCEPTANCE

Specification Hierarchy and Conformance is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-004 · C-035 · C-049 · C-051 · C-052

3 Normative Requirement Model

Requirements are stable engineering objects, not prose annotations. Each requirement has a durable identity, an owning authority, explicit verification, and a lifecycle that survives document and implementation revisions.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Identifier	ES-CC-NNN, where CC is the two-digit chapter and NNN is the ordered requirement within that chapter.
Modality	SHALL and MUST are mandatory; SHALL NOT and MUST NOT are prohibitions; MAY is an allowed choice inside the stated boundary.
Verification	Inspection, analysis, automated test, fault injection, security test, operational exercise, or accountable review.
Lifecycle	Proposed -> reviewed -> accepted -> implemented -> verified -> released -> monitored -> superseded or retired.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-03-001	A mandatory requirement SHALL have at least one executable or independently reviewable verification method before implementation is accepted.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-03-002	Requirement text SHALL state one controlling obligation and avoid hidden compound conditions that cannot be tested independently.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-03-003	Derived requirements SHALL reference their parent requirement and SHALL NOT weaken its acceptance threshold.	Negative conformance test	Adversarial fixture proving the prohibited path is denied and audited.
ES-03-004	A superseded requirement SHALL remain resolvable for Decision Replay, incident analysis, audit, and historical conformance reconstruction.	Operational exercise	Time-stamped exercise record with telemetry, operators, impact, recovery, and closure evidence.
ES-03-005	Metrics used as acceptance evidence SHALL declare population, window, unit, aggregation, exclusions, threshold, and data lineage.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Ambiguous requirement	Return it to the owning authority before implementation or test design begins.
Untestable requirement	Classify as incomplete and block acceptance until a verification method exists.
Conflicting acceptance evidence	Preserve both results, open a challenge, and withhold conformance.

Acceptance evidence

- Requirement registry contains unique IDs and no unresolved duplicate authority.
- Every mandatory requirement maps to one or more verification records.
- Historical versions remain queryable with valid and system time.
- Release evidence shows the exact requirement set evaluated for the artifact.

CHAPTER ACCEPTANCE

Normative Requirement Model is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-005 · C-011 · C-035 · C-049 · C-052

4 Product Boundary and Engineering Decomposition

The Eye is one Strategic Intelligence Operating System. Engineering decomposition creates independently owned and deployable units without creating feature islands, parallel truth, or deployment-specific products.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Canonical layer	The stable responsibility and state boundary defined by Volume 3.
Logical component	A stable architecture capability identified by Lx-Cyy.
Deployable unit	One or more processes released, scaled, secured, and observed as a unit while preserving logical ownership.
Projection	A rebuildable access representation; never an undeclared replacement for canonical state.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-04-001	Each logical component SHALL have exactly one accountable owning team even when several deployable units realize it.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-04-002	A deployable unit MAY implement multiple components only when state ownership, contracts, policy decisions, telemetry, and failure isolation remain distinguishable.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-04-003	Direct database coupling across component ownership boundaries SHALL be prohibited; integration SHALL use a declared contract or governed projection.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.
ES-04-004	User interfaces, chat, dashboards, reports, and APIs SHALL remain access modes over the operating system and SHALL NOT become independent product truth stores.	Negative conformance test	Adversarial fixture proving the prohibited path is denied and audited.
ES-04-005	Shared libraries SHALL contain non-authoritative reusable mechanics; they SHALL NOT conceal cross-layer business state or decision authority.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Circular state ownership	Refactor ownership before production; no shared-write compromise is permitted.
Undeclared shared database	Stop writes, identify the authoritative owner, and migrate consumers to contracts or projections.
Access surface creates state	Route the state transition through its owning component and preserve the surface as caller context.

Acceptance evidence

- Dependency graph contains no cross-boundary database write path.
- Service catalog maps every deployable unit to logical components and owners.
- Projection rebuild tests prove canonical recovery without the projection store.
- Access-mode tests show identical authority and truth behavior across UI, API, and agent calls.

CHAPTER ACCEPTANCE

Product Boundary and Engineering Decomposition is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-001 · C-003 · C-007 · C-008 · C-038

5 Engineering Principles

The engineering system preserves the closed strategic loop by making truth, time, provenance, authority, uncertainty, failure, and learning explicit in every relevant component and transition.

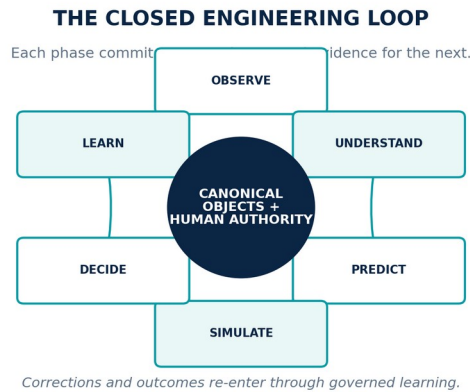


Figure 3. The Closed Engineering Loop. Each phase commits governed state and evidence for the next.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Truthful state	Observed, claimed, inferred, assessed, decided, synthetic, withdrawn, and superseded information remain distinguishable.
Human authority	AI prepares, reasons, forecasts, simulates, recommends, and explains; a named human role makes the final consequential decision.
Permanent memory	Versioned institutional state remains reconstructable under lawful retention and deletion controls.
Safe change	Learning proposes change; evaluation, approval, release, monitoring, rollback, and tenant isolation govern adoption.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-05-001	Every material output SHALL expose its evidence boundary, truth state, time scope, uncertainty, policy state, producing versions, and accountable owner.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-05-002	No automated path SHALL translate a recommendation into a final consequential decision without an authenticated human approval contract.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.
ES-05-003	Correction SHALL create new history and impact propagation; it SHALL NOT overwrite the record used by an earlier decision.	Negative conformance test	Adversarial fixture proving the prohibited path is denied and audited.
ES-05-004	Degraded capability SHALL be visible at the point of use and SHALL NOT present stale, partial, or substituted output as healthy.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-05-005	Learning SHALL produce governed proposals and SHALL NOT mutate production models, prompts, agents, policies, ontologies, or decision logic silently.	Negative conformance test	Adversarial fixture proving the prohibited path is denied and audited.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Unknown truth state	Quarantine the object from decision-active use until classified or explicitly abstained.
Human approval unavailable	Pause, expire, or escalate the workflow; never infer approval from timeout.

FAILURE CONDITION	REQUIRED BEHAVIOR
Lineage discontinuity	Mark dependent products unfit and trigger impact analysis and repair.

Acceptance evidence

- End-to-end fixtures reconstruct source-to-outcome lineage.
- Decision tests prove an authenticated human gate for every final consequential transition.
- Correction replay reproduces prior state without hindsight contamination.
- Learning release tests prove evaluation, approval, monitoring, and rollback controls.

CHAPTER ACCEPTANCE

Engineering Principles is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-002 · C-004 · C-005 · C-010 · C-025 · C-031 · C-043 · C-044

6 Cross-Volume Traceability

Traceability is an operational dependency graph linking constitutional authority, architecture, engineering requirements, code, schemas, policies, tests, releases, runtime evidence, decisions, and outcomes.

TRACEABILITY MESH						
	ARCH	REQ	CONTRACT	CODE	TEST	EVIDENCE
IDENTITY	ENFORCE	VERIFY	ENFORCE	VERIFY	ENFORCE	VERIFY
TIME	VERIFY	ENFORCE	VERIFY	ENFORCE	VERIFY	ENFORCE
TRUTH	ENFORCE	VERIFY	ENFORCE	VERIFY	ENFORCE	VERIFY
POLICY	VERIFY	ENFORCE	VERIFY	ENFORCE	VERIFY	ENFORCE

A broken link is an indeterminate conformance result.

Figure 4. Traceability Mesh. Requirements remain connected from authority through production evidence.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Upstream	Controlling invariants, architecture chapters, component and interface IDs, object definitions, and ADRs.
Engineering	Requirement IDs, design records, code modules, schemas, policies, configurations, test suites, and owners.
Release	Immutable artifact digests, dependency inventory, approval, conformance results, deployment constraints, and rollback target.
Runtime	Workload, model, agent, policy, configuration, object, workflow, decision, and telemetry versions.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-06-001	Trace links SHALL be machine-readable, versioned, and queryable in both upstream and downstream directions.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-06-002	Every production artifact SHALL resolve to the source revision and requirement set from which it was built.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-06-003	Every material runtime output SHALL resolve to the exact service, schema, model, agent, policy, and configuration versions that produced it.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-06-004	A controlling change SHALL trigger deterministic impact analysis across dependent artifacts, tests, deployments, and decision-active products.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-06-005	Traceability data SHALL be protected as audit evidence and retained consistently with the governed records it explains.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Broken trace link	Mark the affected artifact or product non-conformant and prevent promotion or decision-active use.
Unresolved dependency impact	Hold release and route to accountable owners.

FAILURE CONDITION	REQUIRED BEHAVIOR
Runtime version mismatch	Quarantine output, preserve evidence, and reconcile deployment state.

Acceptance evidence

- Bidirectional traversal succeeds from each invariant to runtime evidence and back.
- Change-impact fixtures identify all seeded affected artifacts.
- Release manifests contain immutable source and dependency identifiers.
- Decision Replay resolves the complete execution and specification context.

CHAPTER ACCEPTANCE

Cross-Volume Traceability is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-005 · C-012 · C-025 · C-035 · C-036 · C-051 · C-052

PART II

Platform Construction Model

Deployable boundaries, runtime cells, state ownership, control enforcement, and parity.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

7 Runtime Topology

The runtime topology separates customer intelligence domains, scalable intelligence cells, shared control services, edge collection zones, controlled execution gateways, and recovery domains without changing canonical semantics.

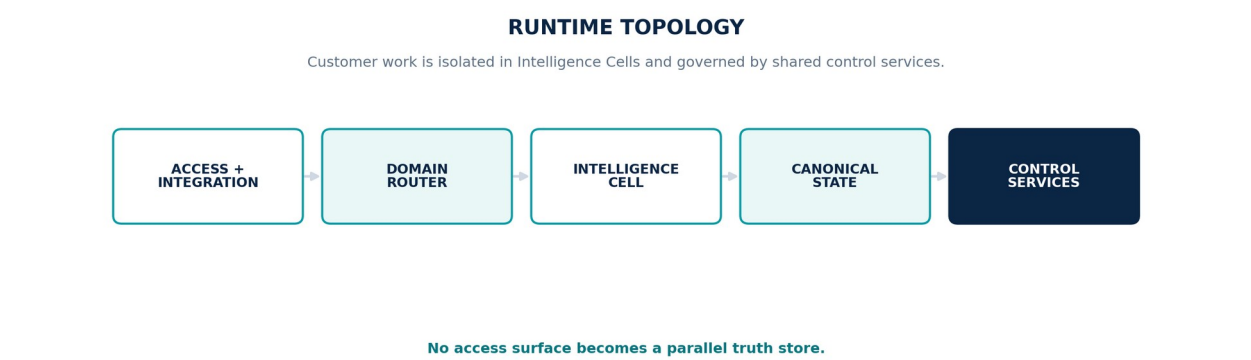


Figure 5. Runtime Topology. Customer work is isolated in Intelligence Cells and governed by shared control services.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Customer Intelligence Domain	Primary isolation and policy boundary for a customer's strategic system.
Intelligence Cell	Bounded deployment, scale, locality, and failure unit hosting canonical-layer workloads.
Shared Control Services	Policy-governed services whose shared operation never grants access to customer content by default.
Execution Gateway	Controlled boundary for notifications, exports, callbacks, enterprise actions, and other external side effects.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-07-001	Every request, event, object, workflow, agent execution, and telemetry record SHALL carry a resolved domain identity before owned behavior begins.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-07-002	Cell placement SHALL satisfy residency, classification, key ownership, model, network, capacity, and continuity policy.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-07-003	Cross-cell communication SHALL use authenticated contracts and SHALL preserve tenant, causation, trace, policy, and version context.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-07-004	Shared control services SHALL operate on the minimum metadata necessary and SHALL not become hidden canonical data owners.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.
ES-07-005	External side effects SHALL traverse an execution gateway with policy evaluation, approval context, idempotency, and audit evidence.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Cell unavailable	Route only to an approved compatible cell or expose bounded unavailability; never cross policy boundaries silently.
Domain identity unresolved	Reject before data access or side effect.

FAILURE CONDITION	REQUIRED BEHAVIOR
Shared control dependency degraded	Apply the declared fail-closed or bounded degraded policy and expose the control state.

Acceptance evidence

- Topology tests validate domain isolation under normal, failover, and recovery paths.
- Placement tests reject residency, key, model, network, and capacity conflicts.
- Gateway tests prove policy, approval, idempotency, and audit on every external effect class.
- Cell evacuation exercise preserves object, workflow, and replay continuity.

CHAPTER ACCEPTANCE

Runtime Topology is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-035 · C-037 · C-039 · C-041 · C-042 · C-043

8 Customer Intelligence Domain

A Customer Intelligence Domain is the stable logical boundary that binds customer identity, purpose, evidence, knowledge, models, agents, policies, decisions, audit, keys, residency, retention, and operational ownership.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Identity	Stable domain ID independent of deployment location, billing account, project, environment, or user interface.
Policy	Versioned domain policy set governing purpose, roles, data, models, agents, residency, retention, export, and execution.
State	Canonical objects and control evidence owned by the ten layers within the domain.
Administration	Explicit customer and platform authorities with separation of duties and no implicit content access.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-08-001	Domain creation SHALL be a governed workflow that binds legal entity, purpose, authorities, key policy, residency, retention, and recovery profile before activation.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-08-002	Domain identity SHALL be immutable; display names, organizational mappings, and deployment placements MAY change by version.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-08-003	All data access SHALL be evaluated against domain identity and purpose before object-level authorization.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-08-004	Cross-domain sharing SHALL create a governed, revocable reference or export with explicit source and recipient obligations; shared mutable state is prohibited.	Negative conformance test	Adversarial fixture proving the prohibited path is denied and audited.
ES-08-005	Domain suspension, export, and deletion SHALL preserve legal holds, audit continuity, customer proof, and declared residual-state evidence.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Policy incomplete	Keep the domain inactive and deny ingestion, processing, and execution.
Cross-domain reference revoked	Invalidate recipient access and propagate impact without rewriting source history.
Domain deletion partial	Remain in deleting state, expose residual inventory, and prohibit a false completion claim.

Acceptance evidence

- Isolation suite attempts cross-domain reads, writes, events, caches, search, graph traversal, model context, telemetry, and recovery access.
- Lifecycle tests cover create, activate, suspend, export, legal hold, delete, and verified closure.
- Administrative tests prove customer and platform role separation.
- Residual-state report reconciles canonical stores, projections, backups, logs, and external processors.

CHAPTER ACCEPTANCE

Customer Intelligence Domain is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-016 · C-037 · C-039 · C-040 · C-041

9 Intelligence Cells

An Intelligence Cell is the repeatable engineering unit for placement, scaling, failure containment, maintenance, and recovery. Cells host bounded workloads without owning a new semantic layer.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Cell identity	Immutable runtime identity with environment, region or site, trust zone, software baseline, and policy profile.
Admission	Compatibility and capacity decision for domain workloads, models, connectors, agents, and state replicas.
Failure boundary	Blast-radius unit with health, dependency, degradation, evacuation, and recovery state.
Portability	Same signed product artifacts and conformance fixtures across deployment modes.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-09-001	A cell SHALL publish supported artifact, schema, policy, model, security, and operational baseline versions before workload admission.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-09-002	Workloads SHALL be admitted only after compatibility, capacity, data locality, dependency, and recovery checks pass.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-09-003	Cell-local caches and projections SHALL be rebuildable and SHALL declare freshness, source revision, and invalidation state.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-09-004	Cell maintenance SHALL use drain, checkpoint, reconcile, and release transitions that preserve durable workflows and external-effect evidence.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-09-005	Cell health SHALL be computed from dependency and product signals, not process liveness alone.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Capacity breach	Throttle or reject according to priority and quality contracts; do not silently reduce correctness.
Baseline incompatibility	Deny admission or promotion and expose the unmet compatibility set.
Cell isolation event	Contain the cell, preserve evidence, pause unsafe work, and execute the recovery playbook.

Acceptance evidence

- Admission suite covers incompatible schemas, policies, models, dependencies, residency, and capacity.
- Fault injection proves cell containment and truthful degradation.
- Evacuation and restoration exercises reconcile workflows, objects, events, and external effects.
- Portability fixtures execute unchanged in SaaS, Private Cloud, and On-Premise cells.

CHAPTER ACCEPTANCE

Intelligence Cells is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-038 · C-041 · C-042 · C-043 · C-049

10 Shared Control Services

Shared Control Services distribute identity federation, policy templates, package metadata, licensing, support, and operational coordination while preserving customer control and domain isolation.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Allowed shared state	Product metadata, signed packages, public schemas, tenant directory references, entitlement, support routing, and aggregate service health.
Restricted state	Customer content, graph state, memory, prompts containing customer evidence, model context, decisions, and audit detail.
Trust	Every shared-service call is authenticated, purpose-bound, policy-evaluated, minimized, logged, and revocable.
Disconnected mode	Local authority packages and cached control state operate inside declared expiry and reconciliation rules.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-10-001	Shared services SHALL store only declared metadata fields and SHALL reject undeclared customer payloads.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-10-002	Administrative support access SHALL require explicit authorization, purpose, time bound, customer policy, and tamper-evident audit evidence.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-10-003	Control packages SHALL be signed, versioned, integrity-checked, compatible, and revocable before local activation.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-10-004	Disconnected operation SHALL expose control-state age and SHALL fail closed when a mandatory authority package expires.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.
ES-10-005	Reconnection SHALL reconcile changes, conflicts, revocations, and audit evidence before declaring healthy synchronization.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Shared service unavailable	Use an approved bounded cache or expose control-plane unavailability according to policy.
Package signature invalid	Quarantine the package and preserve verification evidence.
Support access exceeds scope	Terminate access, alert accountable authorities, and open an audit incident.

Acceptance evidence

- Data-minimization tests inspect shared stores and telemetry for prohibited content.
- Disconnected expiry tests prove mandatory fail-closed behavior.
- Support-access drills verify approval, session recording, revocation, and audit attribution.
- Package verification rejects tampering, rollback, incompatible dependencies, and revoked publishers.

CHAPTER ACCEPTANCE

Shared Control Services is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-035 · C-037 · C-040 · C-041 · C-042 · C-043

11 Component Ownership and Service Boundaries

Every logical component has one accountable owner, one declared state contract, one controlled interface surface, and explicit dependencies. Service count is an implementation variable; ownership is not.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Owner	Accountable team responsible for behavior, interfaces, state, quality, security, operations, and conformance.
Boundary	Declared inputs, outputs, owned state, dependencies, prohibited access, and failure behavior.
Service	Deployable realization with immutable artifact identity and runtime ownership metadata.
Library	Reusable stateless or local mechanics that cannot become an undeclared state or policy authority.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-11-001	Each component SHALL have an ownership record with primary and secondary operational contacts and a ratified transfer process.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.
ES-11-002	A service SHALL expose only contracts required by its owned behavior; convenience endpoints that bypass layer semantics are prohibited.	Negative conformance test	Adversarial fixture proving the prohibited path is denied and audited.
ES-11-003	State writes SHALL occur only through the owning component's transaction boundary or an explicitly delegated subcomponent contract.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-11-004	Dependency calls SHALL declare timeout, retry, idempotency, consistency, fallback, and degraded-product behavior.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-11-005	Ownership transfer SHALL include code, state, contracts, dashboards, alerts, runbooks, risks, exceptions, and unresolved incidents.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Owner unavailable	Escalate through the service registry; do not permit unowned changes or releases.
Boundary violation	Block the call or write, emit a security and conformance event, and reconcile affected state.
Undeclared dependency	Fail build or admission until registered and evaluated.

Acceptance evidence

- Catalog validation proves every component, service, endpoint, event, store, and dependency has an owner.
- Static and runtime tests detect forbidden cross-boundary calls and writes.
- Dependency failure tests exercise timeout, retry, fallback, and user-visible degradation.
- Ownership transfer exercise proves operational completeness without tribal knowledge.

CHAPTER ACCEPTANCE

Component Ownership and Service Boundaries is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-008 · C-035 · C-036 · C-043 · C-049

12 State Ownership and Persistence

Canonical state is owned by layer components and preserved as versioned records. Databases, graph engines, object stores, indexes, caches, and vector representations are implementation mechanisms with explicit authority classes.

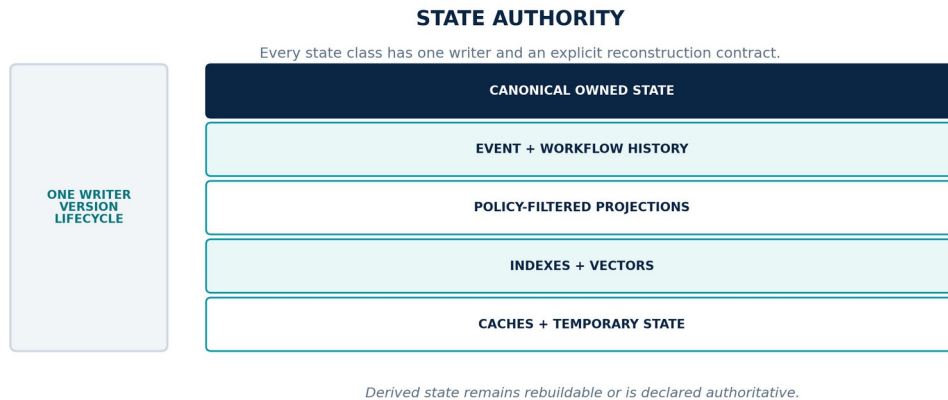


Figure 6. State Authority. Every state class has one writer and an explicit reconstruction contract.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Canonical store	Authoritative system of record for an owned object or transition.
Durable journal	Append-only transition or event history sufficient for recovery, audit, and replay.
Projection	Rebuildable query representation derived from canonical state.
Ephemeral state	Cache, scratch, session, or execution state whose loss has declared recovery semantics.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-12-001	Every persisted dataset SHALL declare owner, authority class, schema, temporal model, retention, encryption, backup, recovery, and deletion behavior.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-12-002	Canonical records SHALL be immutable by version; updates SHALL create a new version or an owned state transition with prior history preserved.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-12-003	Projection state SHALL store source revision and build version and SHALL never be accepted as canonical during recovery.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-12-004	Cross-store operations SHALL use a durable coordination pattern with causation, idempotency, reconciliation, and observable partial state.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-12-005	Backup and restore SHALL preserve domain isolation, key policy, temporal history, audit integrity, and deletion obligations.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Canonical/projection divergence	Prefer canonical state, mark the projection stale, rebuild, and report affected consumers.
Partial cross-store commit	Resume or compensate from the durable transition record; never infer completion from one store.
Unknown dataset authority	Prohibit decision-active use until ownership and authority are classified.

Acceptance evidence

- State catalog contains complete authority and lifecycle metadata.
- Projection rebuild tests start from canonical state and produce equivalent query results.
- Failure injection covers partial commits, duplicate delivery, reordering, and recovery.
- Restore tests verify history, isolation, keys, audit, retention, and deletion markers.

CHAPTER ACCEPTANCE

State Ownership and Persistence is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-010 · C-011 · C-015 · C-016 · C-025 · C-037 · C-043

13 Cross-Cutting Control Enforcement

Trust, security, privacy, identity, policy, workflow, evaluation, audit, sovereignty, and operations are in-band engineering behavior applied at every material boundary.

CONTROL ENFORCEMENT						
	READ	TRANSFORM	MODEL	AGENT	ACTION	EXPORT
IDENTITY	ENFORCE	VERIFY	ENFORCE	VERIFY	ENFORCE	VERIFY
POLICY	VERIFY	ENFORCE	VERIFY	ENFORCE	VERIFY	ENFORCE
PROVENANCE	ENFORCE	VERIFY	ENFORCE	VERIFY	ENFORCE	VERIFY
AUDIT	VERIFY	ENFORCE	VERIFY	ENFORCE	VERIFY	ENFORCE

Missing enforcement makes the operation non-conformant.

Figure 7. Control Enforcement. Controls operate at every material read, transformation, action, and export boundary.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Policy enforcement point	Interception boundary that evaluates a declared action against current authoritative policy.
Policy decision point	Versioned authority that returns allow, deny, obligations, review, or indeterminate.
Evidence point	Boundary that records material access, transformation, model or agent action, approval, decision, release, correction, or export.
Quality gate	Boundary that prevents unfit products, models, agents, packages, or data from entering a higher-trust state.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-13-001	Control evaluation SHALL occur before protected data access, model or agent invocation, state transition, external effect, export, and privileged administration.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-13-002	A control result SHALL carry policy version, subject, object, action, purpose, environment, obligations, time, and decision evidence.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-13-003	Indeterminate authorization SHALL be treated as deny unless an explicit bounded fail-open rule exists for a non-sensitive availability function.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-13-004	Obligations SHALL be executed and evidenced by the enforcing boundary; returning obligations without enforcement is non-conformant.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-13-005	Control bypass paths, including maintenance and break-glass access, SHALL be explicit, narrow, time-bound, monitored, and independently reviewed.	Operational exercise	Time-stamped exercise record with telemetry, operators, impact, recovery, and closure evidence.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Policy service unavailable	Use an approved unexpired decision cache only where authorized; otherwise deny and expose control unavailability.
Obligation failure	Abort or compensate the protected action and preserve partial evidence.
Bypass detected	Terminate the path, contain the actor or workload, and open a security and conformance incident.

Acceptance evidence

- Boundary inventory shows enforcement and evidence points for every protected action class.
- Policy tests cover allow, deny, obligations, review, indeterminate, cache, expiry, and revocation.
- Bypass testing covers administrative, recovery, batch, agent, and integration paths.
- Audit reconstruction proves policy and obligation execution for sampled material actions.

CHAPTER ACCEPTANCE

Cross-Cutting Control Enforcement is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-035 · C-036 · C-039 · C-040 · C-041 · C-043 · C-049

14 Deployment Semantic Parity

SaaS, Private Cloud, and On-Premise are operating envelopes for one product. Physical topology and operational responsibility may differ; object meaning, authority, contracts, workflow, evidence, and conformance do not.

DEPLOYMENT SEMANTIC PARITY					
	OBJECTS	WORKFLOW	POLICY	AUDIT	EXPORT
SAAS	ENFORCE	VERIFY	ENFORCE	VERIFY	ENFORCE
PRIVATE CLOUD	VERIFY	ENFORCE	VERIFY	ENFORCE	VERIFY
ON-PREMISE	ENFORCE	VERIFY	ENFORCE	VERIFY	ENFORCE

One canonical conformance suite governs every deployment profile.

Figure 8. Deployment Semantic Parity. Physical responsibility may move; product meaning does not.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Invariant core	Ten layers, canonical objects, truth and temporal model, provenance, human authority, workflow, audit, and external contracts.
Variable envelope	Capacity, managed services, connectivity, locality, key custody, administration, model availability, support, and update cadence.
Capability constraint	Declared unavailable or reduced function caused by an operating-envelope limit; never a hidden semantic substitute.
Portability proof	The same conformance fixtures and canonical artifacts execute across all supported modes.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-14-001	Product APIs, events, schemas, package manifests, policy contracts, export formats, and replay semantics SHALL remain portable across deployment modes.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-14-002	Mode-specific adapters SHALL terminate below canonical contracts and SHALL not leak provider semantics into object or workflow definitions.	Negative conformance test	Adversarial fixture proving the prohibited path is denied and audited.
ES-14-003	Disconnected operation SHALL declare authority-package expiry, local model and data boundaries, synchronization, conflict, and recovery behavior.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-14-004	Every mode SHALL expose capability constraints before request admission and in every affected product.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-14-005	Release qualification SHALL execute the same normative conformance suite against each supported deployment profile.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Mode-specific semantic drift	Classify the release as non-conformant and block promotion for that mode.
Unavailable dependency	Expose the constraint and use only an approved semantically equivalent adapter or bounded degraded behavior.

FAILURE CONDITION	REQUIRED BEHAVIOR
Disconnected authority expired	Stop protected actions that require current authority; preserve safe read or review functions only where policy permits.

Acceptance evidence

- Cross-mode golden fixtures produce equivalent canonical objects, decisions, lineage, and replay state.
- Contract tests run against each deployment profile without mode-specific test semantics.
- Disconnected exercises cover expiry, revocation, conflict, reconnect, and audit reconciliation.
- Capability matrix is machine-readable, versioned, customer-visible, and tested at admission.

CHAPTER ACCEPTANCE

Deployment Semantic Parity is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-007 · C-038 · C-041 · C-042 · C-043 · C-049

PART III

Contract System

Canonical APIs, commands, queries, events, envelopes, errors, and compatibility.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

15 Canonical Service Contract

Every service exposes a uniform engineering contract that makes identity, purpose, ownership, data access, time, semantics, quality, failure, and operations explicit before transport-specific details.

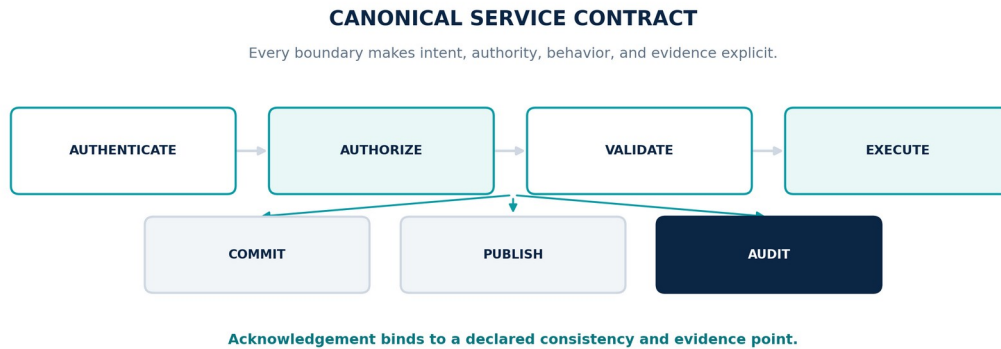


Figure 9. Canonical Service Contract. Every boundary makes intent, authority, behavior, and evidence explicit.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Identity	Service, component, layer, owner, artifact, workload, environment, domain, and caller assurance.
Behavior	Operations, inputs, outputs, side effects, state transitions, invariants, and prohibited actions.
Quality	Latency, availability, freshness, throughput, accuracy or fitness, capacity, and degradation contract.
Operations	Dependencies, timeouts, retries, idempotency, health, telemetry, runbook, recovery, and support.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-15-001	A service SHALL register its contract before accepting production traffic or events.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-15-002	Contract identity SHALL bind to the released artifact digest and SHALL not rely on mutable image tags or deployment names.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-15-003	Every operation SHALL declare authorization action, purpose, input and output object types, consistency, idempotency, deadlines, side-effect class, and error set.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-15-004	Service health SHALL include dependency, policy, state, quality, and product readiness rather than process liveness alone.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-15-005	Contract changes SHALL pass compatibility analysis and consumer verification before release.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Unregistered service	Deny admission to production routing, event subscriptions, and protected data.
Artifact/contract mismatch	Quarantine deployment and restore the last verified pair.
Undeclared operation	Reject at the service boundary and record a conformance event.

Acceptance evidence

- Registry validation checks complete metadata and artifact binding.
- Consumer-driven tests cover every published operation and error class.
- Dependency and policy fault tests verify health and degradation state.
- Compatibility checks evaluate additive, breaking, deprecated, and retired changes.

CHAPTER ACCEPTANCE

Canonical Service Contract is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-035 · C-038 · C-042 · C-043 · C-049

16 Synchronous API Profile

Synchronous APIs serve bounded commands and queries that require an immediate admission or result contract. They preserve the canonical envelope and never hide durable asynchronous work behind an ambiguous success response.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
External profile	Authenticated HTTPS resource and action APIs with explicit media type, schema version, deadline, idempotency, pagination, and error body.
Internal profile	Authenticated typed RPC or equivalent request-response transport using the same semantic envelope.
Long-running work	Returns an accepted operation reference whose durable state is queryable and event-producing.
Streaming	Bounded server or bidirectional stream with admission, sequence, checkpoint, cancellation, and backpressure semantics.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-16-001	Every request SHALL carry correlation, causation when applicable, domain, principal or workload, purpose, deadline, schema, and idempotency context.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-16-002	Success status SHALL mean the documented state transition or query completed; admission to long-running work SHALL return a distinct accepted result.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-16-003	Clients SHALL be able to distinguish validation, authorization, policy, conflict, stale state, capacity, dependency, timeout, and internal failure.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-16-004	Pagination SHALL use opaque stable cursors bound to query, sort, policy, and consistency context.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-16-005	APIs SHALL support cancellation or deadline propagation and SHALL prevent abandoned callers from creating uncontrolled work or side effects.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Deadline exceeded	Stop safe work, preserve durable state, and return an unambiguous timeout or accepted-operation reference.
Client retry	Deduplicate through the operation's idempotency contract and return the prior result or current durable state.
Backpressure	Reject or throttle with retry guidance and visible capacity state; do not queue without a declared bound.

Acceptance evidence

- Protocol conformance verifies headers, envelope, schemas, statuses, media types, and error bodies.
- Retry and timeout tests prove no duplicate material side effects.
- Pagination tests cover concurrent change, policy change, expiry, and cursor tampering.
- Load tests verify documented admission and backpressure thresholds.

CHAPTER ACCEPTANCE

Synchronous API Profile is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-035 · C-038 · C-039 · C-043 · C-049

17 Command Profile

Commands request owned behavior and may create durable state transitions. A command is directed to exactly one authority and returns an acceptance or rejection that is distinct from downstream completion.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Target	Owning component and operation.
Preconditions	Expected object version, workflow state, policy context, deadline, and required approval.
Identity	Command ID, idempotency key, correlation, causation, domain, actor, workload, and delegation.
Result	Rejected, completed synchronously, or accepted with durable operation and eventual event references.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-17-001	A command SHALL name one state-owning target and SHALL NOT broadcast an instruction to several competing authorities.	Negative conformance test	Adversarial fixture proving the prohibited path is denied and audited.
ES-17-002	Material commands SHALL include an idempotency key scoped to domain, operation, caller, and declared validity window.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-17-003	Precondition failures SHALL return current state references without applying a partial transition.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-17-004	Accepted commands SHALL create a durable operation record before acknowledging acceptance.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-17-005	Command handlers SHALL emit committed domain events through the same transaction boundary or a recoverable outbox pattern.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Duplicate command	Return the recorded result or operation state without repeating the material effect.
Concurrent version conflict	Reject with current version and require explicit caller reconciliation.
Partial downstream work	Resume or compensate from the durable operation state; do not report completion.

Acceptance evidence

- Command tests cover duplicate, reordered, concurrent, expired, cancelled, and unauthorized requests.
- Crash tests prove durable acceptance before acknowledgment.
- Event publication tests prove no committed state transition is permanently missing its domain event.
- Audit tests resolve actor, authority, preconditions, transition, and result.

CHAPTER ACCEPTANCE

Command Profile is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-011 · C-012 · C-025 · C-036 · C-043

18 Query and Consistency Profile

Queries retrieve policy-filtered state without changing canonical ownership. Callers declare the freshness, revision, temporal, and completeness guarantees required for their purpose.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Current	Latest committed version visible under the selected consistency and policy context.
As-of	State reconstructable at declared valid and system times without later contamination.
Revision-bound	Result fixed to named object, graph, ontology, memory, model, policy, or projection revisions.
Coverage	Declared completeness, freshness, source health, and known blind spots for the result set.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-18-001	A query SHALL declare current, as-of, or revision-bound semantics and SHALL return the resolved revision and temporal context.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-18-002	Query results SHALL include policy-filtering evidence and SHALL not reveal the existence of unauthorized objects through counts, errors, timing, or cursors.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-18-003	Projection-backed results SHALL declare projection build version, source revision, freshness, and known lag.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-18-004	Search, graph, vector, and analytical queries SHALL distinguish exact, approximate, partial, and truncated results.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-18-005	Decision-active queries SHALL fail or visibly degrade when declared freshness, completeness, consistency, or policy requirements cannot be met.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Projection stale	Return a visible stale or insufficient result according to the caller's declared contract; never label it current.
Temporal state unavailable	Return an explicit reconstruction gap with affected objects and evidence.
Policy changes mid-query	Apply the stronger valid policy before delivery or restart under the new context.

Acceptance evidence

- Temporal fixtures reconstruct known historical states before and after correction.
- Authorization tests detect enumeration and side-channel disclosure.
- Projection lag tests verify declared freshness and decision-active refusal.
- Approximation tests expose method, bounds, truncation, and reproducibility context.

CHAPTER ACCEPTANCE

Query and Consistency Profile is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

19 Event Profile

Domain events announce committed authoritative transitions. They are immutable facts about the platform state, not mutable messages, commands disguised as broadcasts, or unversioned integration payloads.



Figure 10. Command-To-Event Commit. State, event intent, and audit causation form one recoverable boundary.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Identity	Globally unique event ID, event type, schema version, producer component, producer artifact, and domain.
Causality	Correlation, causation, command, workflow, object, and trace references.
Time	Occurrence, commit, publication, observation where applicable, and expiry.
Delivery	Partition key, sequence, deduplication, retention, replay, and consumer checkpoint contract.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-19-001	An event SHALL be published only after the announced state transition is durable in the owning component.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-19-002	Event payloads SHALL contain stable references and the minimum transition data required by consumers; they SHALL not become uncontrolled data replicas.	Negative conformance test	Adversarial fixture proving the prohibited path is denied and audited.
ES-19-003	Consumers SHALL be idempotent and SHALL persist checkpoints only after their owned effects are durable.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-19-004	Ordering SHALL be guaranteed only within a declared partition and sequence scope; consumers SHALL handle cross-partition concurrency explicitly.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-19-005	Corrections SHALL be new events linked to affected prior events and objects; historical events SHALL not be mutated or deleted except under governed retention evidence.	Negative conformance test	Adversarial fixture proving the prohibited path is denied and audited.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Duplicate delivery	Deduplicate by event identity and consumer effect key.
Out-of-order delivery	Buffer, reconcile, or mark incomplete according to sequence contract; never silently apply invalid order.
Poison event	Quarantine with full evidence, continue unaffected partitions where safe, and surface consumer lag and impact.

Acceptance evidence

- Producer tests prove commit-before-publish and crash recovery from the outbox or journal.
- Consumer tests cover duplicate, reorder, gap, replay, poison, schema evolution, and checkpoint recovery.
- Event catalog validation checks ownership, schema, partition, retention, sensitivity, and consumer set.
- Replay tests reproduce owned projections without duplicate effects.

CHAPTER ACCEPTANCE

Event Profile is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-011 · C-012 · C-025 · C-036 · C-038 · C-043

20 Canonical Contract Envelope

The canonical envelope carries identity, intent, time, semantics, trust, control, and operational correlation across human, service, model, agent, workflow, and deployment boundaries.

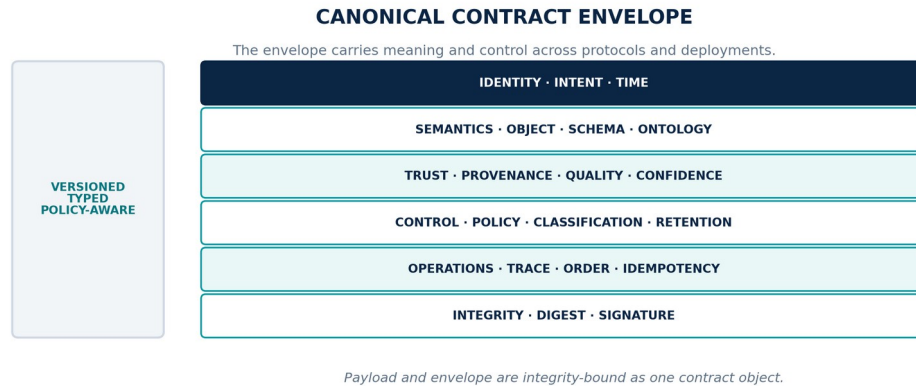


Figure 11. Canonical Contract Envelope. The envelope carries meaning and control across protocols and deployments.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Identity	domain, principal, actor assurance, workload, service, agent, delegation, and session.
Intent	purpose, operation, object scope, expected product, side-effect class, and approval context.
Time and semantics	event, observation, valid, system, deadline, expiry, object type, schema, ontology, and units.
Trust and operations	source, provenance, quality, truth state, policy, classification, correlation, causation, trace, idempotency, and checksum.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-20-001	Envelope fields SHALL have one canonical meaning across transports, languages, services, agents, models, and deployment modes.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-20-002	Mandatory fields SHALL be validated before payload processing or protected resource access.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-20-003	Envelope propagation SHALL preserve original actor and causation while adding the current workload and transition evidence.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-20-004	Sensitive envelope fields SHALL be minimized, classified, encrypted, access-controlled, and excluded from uncontrolled telemetry.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-20-005	Synthetic, inferred, stale, partial, and corrected states SHALL remain explicit and SHALL not be lost during translation or serialization.	Negative conformance test	Adversarial fixture proving the prohibited path is denied and audited.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Missing mandatory field	Reject before owned behavior and return a validation error with no protected disclosure.
Invalid delegation chain	Deny, preserve verification evidence, and alert according to risk policy.
Semantic translation loss	Reject the adapter output and mark the integration non-conformant.

Acceptance evidence

- Cross-language golden fixtures produce semantically equivalent envelopes.
- Propagation tests traverse service, event, agent, model, workflow, and human-gate boundaries.
- Security tests inspect logs, traces, and errors for sensitive-field leakage.
- Adapter conformance rejects lost truth, time, policy, provenance, or causation fields.

CHAPTER ACCEPTANCE

Canonical Contract Envelope is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-005 · C-010 · C-011 · C-012 · C-035 · C-038 · C-039 · C-042

21 Error, Abstention, and Partial-Result Model

Errors are typed engineering products with safe disclosure, causation, retry guidance, user impact, and audit evidence. Analytical abstention and partial results are first-class outcomes, not generic failures.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Validation	Input, schema, semantic, precondition, or compatibility failure before owned behavior.
Control	Authentication, authorization, policy, approval, classification, sovereignty, or safety decision.
Operational	Capacity, dependency, timeout, conflict, unavailable, degraded, or internal failure.
Intelligence	Insufficient evidence, low fitness, uncertainty breach, incoherent scenario, invalid simulation, or required human judgment.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-21-001	Every error SHALL include a stable code, category, safe message, correlation reference, retry class, owning component, and support or remediation path.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-21-002	Internal details, secrets, unauthorized object existence, stack traces, and sensitive policy logic SHALL not be exposed to untrusted callers.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-21-003	Abstention SHALL identify the unmet evidence, quality, uncertainty, policy, or authority condition and the path to review or additional information.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-21-004	Partial results SHALL declare missing scope, unavailable dependencies, freshness, confidence, affected conclusions, and whether decision-active use is prohibited.	Negative conformance test	Adversarial fixture proving the prohibited path is denied and audited.
ES-21-005	Retry guidance SHALL distinguish never, after caller correction, after policy or approval, after a stated delay, and safe idempotent retry.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Unknown error	Map to a non-disclosing internal error, preserve detailed evidence privately, and open defect analysis.
Abstention converted to success	Fail conformance and invalidate downstream products that relied on the false success.
Partial result used as complete	Mark affected decisions and briefings, propagate impact, and require review.

Acceptance evidence

- Error catalog tests cover every public and internal code and declared retry class.
- Security tests verify non-disclosure and correlation-based supportability.
- Analytical fixtures force abstention and partial-result paths.
- Downstream tests prove partial and abstained state cannot be silently promoted.

CHAPTER ACCEPTANCE

Error, Abstention, and Partial-Result Model is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

22 Idempotency, Ordering, Retry, and Compensation

Distributed execution assumes duplication, reordering, delay, partial failure, and replay. Correctness comes from explicit semantics and durable state, not from optimistic exactly-once claims.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Idempotency scope	Domain, operation, actor or workload, target, key, and validity window.
Ordering scope	Named aggregate or partition with monotonic sequence and gap behavior.
Retry policy	Attempt limit, backoff, jitter, deadline, retryable codes, budget, and ownership.
Compensation	Owned reversal or reconciliation action for an already committed external or cross-boundary effect.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-22-001	Every material side effect SHALL have a stable deduplication or reconciliation strategy before release.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-22-002	Retry loops SHALL be bounded by deadline and budget and SHALL not amplify load, bypass policy, or duplicate external effects.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-22-003	Ordering assumptions SHALL be local and declared; global ordering SHALL not be inferred from timestamps or transport position.	Negative conformance test	Adversarial fixture proving the prohibited path is denied and audited.
ES-22-004	Compensation SHALL be a new evidenced transition and SHALL not delete the original action or pretend it never occurred.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-22-005	Workflow and event replay SHALL preserve original causation while recording replay identity, initiator, reason, and result.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Idempotency record unavailable	Reject or pause material effects unless an approved alternate reconciliation path exists.
Sequence gap	Mark state incomplete and trigger bounded wait, fetch, or reconciliation.
Compensation failure	Escalate with original and compensation evidence; do not declare rollback complete.

Acceptance evidence

- Chaos fixtures inject duplicates, reorder, delay, retry storms, crash-after-effect, and crash-before-record.
- External-effect tests prove one material effect per idempotency scope.
- Sequence recovery tests reconcile gaps and late arrivals.
- Compensation tests preserve original evidence and expose residual state.

CHAPTER ACCEPTANCE

Idempotency, Ordering, Retry, and Compensation is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

23 Versioning, Compatibility, and Contract Registry

Schemas, APIs, events, services, policies, models, agents, packages, and configuration profiles evolve through explicit compatibility classes and a governed registry.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Additive	New optional meaning that existing conformant consumers safely ignore.
Behavioral	Same shape with changed semantics, thresholds, defaults, policy, or side effects; treated as compatibility-significant.
Breaking	Removes, renames, retypes, reinterprets, or changes required ordering, authority, truth, time, or error semantics.
Retirement	Time-bound removal after inventory, notice, migration, verification, and residual-consumer closure.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-23-001	Every contract SHALL have a stable identity, semantic version, owner, status, compatibility class, effective time, and machine-readable schema or definition.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-23-002	Behavioral changes SHALL be versioned even when wire shape remains unchanged.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-23-003	Breaking changes SHALL use parallel compatibility, explicit migration, consumer verification, and governed retirement; in-place reinterpretation is prohibited.	Negative conformance test	Adversarial fixture proving the prohibited path is denied and audited.
ES-23-004	Consumers SHALL declare supported version ranges and SHALL fail visibly when no compatible version exists.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-23-005	The registry SHALL prevent publication of duplicate authority, incompatible reuse of identity, invalid supersession, or unsigned production contracts.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
No compatible contract	Reject admission or interaction and expose the resolved compatibility conflict.
Retired consumer remains	Hold retirement or isolate the non-conformant consumer under an authorized time-bound exception.
Registry unavailable	Use only a verified unexpired local snapshot and prohibit unregistered change.

Acceptance evidence

- Compatibility suite checks wire, semantic, policy, error, ordering, and side-effect behavior.
- Consumer inventory reconciles declared and observed version use.
- Migration tests cover dual-read or dual-publish, cutover, rollback, and historical replay.
- Registry integrity tests reject identity reuse, invalid signatures, and supersession cycles.

CHAPTER ACCEPTANCE

Versioning, Compatibility, and Contract Registry is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

PART IV

Object and State Engineering

Identity, time, truth, provenance, policy, correction, and projections.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

24 Canonical Intelligence Object Header

Every material intelligence product inherits one universal header so identity, version, time, truth, evidence, control, quality, and operational lineage survive all service, agent, model, human, and deployment boundaries.

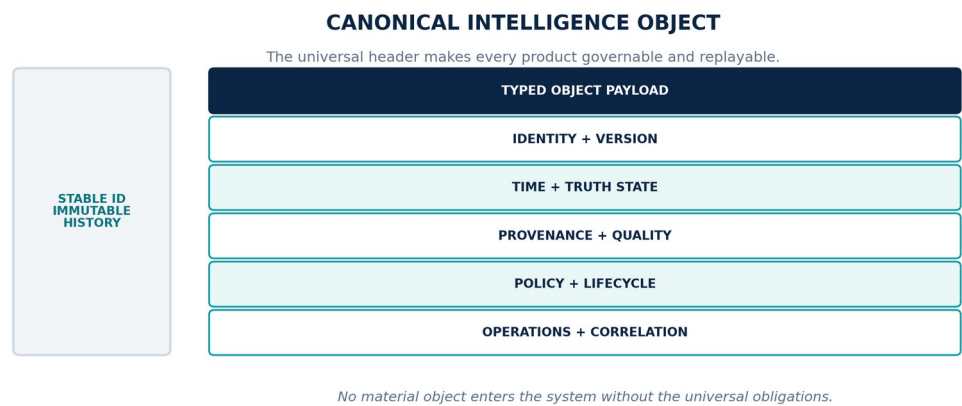


Figure 12. Canonical Intelligence Object. The universal header makes every product governable and replayable.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Identity	object_id, object_type, domain_id, schema_id, schema_version, and owning component.
Version	object_version, lifecycle_state, supersedes, correction, and immutable content digest.
Time and truth	event, observation, valid and system time; truth state, confidence, uncertainty, and synthetic marker.
Trust and control	source, evidence, transformation, model, agent, human, policy, classification, owner, quality, expiry, correlation, causation, and audit references.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-24-001	Object identity SHALL remain stable across versions, stores, indexes, exports, and deployment movement.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-24-002	Object versions SHALL be immutable and content-addressable; corrections and lifecycle changes SHALL create new versions or linked control records.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-24-003	Mandatory header fields SHALL be validated by the owning component before the object becomes decision-active.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-24-004	Derived objects SHALL reference exact input versions and producing transformation, model, agent, policy, and configuration versions.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-24-005	Serialization profiles MAY omit redundant fields only when the containing signed context supplies identical immutable values and reconstruction is lossless.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Header incomplete	Keep the object in proposed or quarantined state and prohibit decision-active use.
Digest mismatch	Quarantine as integrity failure and investigate all dependent products.
Owner unresolved	Reject commit to canonical state.

Acceptance evidence

- Schema fixtures validate mandatory, conditional, and prohibited field combinations.
- Round-trip tests preserve header meaning across encodings and deployment modes.
- Lineage tests resolve exact inputs and producing versions.
- Tamper tests detect payload, header, reference, and digest manipulation.

CHAPTER ACCEPTANCE

Canonical Intelligence Object Header is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-010 · C-011 · C-012 · C-016 · C-035 · C-042

25 Identity, Naming, and Tenancy

Stable identity separates what an object, actor, workload, model, agent, contract, or domain is from mutable names, locations, ownership assignments, and deployment instances.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Opaque ID	Globally unique, non-semantic, immutable identifier used for durable references.
Natural key	Governed domain identifier with authority, validity, namespace, collision, and correction rules.
Display name	Localized and mutable label never used as authorization or referential identity.
Domain scope	Every customer-controlled identity resolves inside one Customer Intelligence Domain unless a governed global authority explicitly owns it.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-25-001	Identifiers SHALL not embed mutable organizational names, regions, classifications, environments, or deployment locations.	Negative conformance test	Adversarial fixture proving the prohibited path is denied and audited.
ES-25-002	Every identity SHALL declare issuing authority, namespace, creation time, status, aliases, merges, splits, and supersession where applicable.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-25-003	Tenant and domain identity SHALL be explicit in storage keys, messages, caches, search indexes, telemetry, backups, and recovery operations.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-25-004	Identity resolution SHALL return evidence, confidence, competing candidates, stewardship state, and the ontology revision used.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-25-005	Merge and split operations SHALL preserve prior identifiers and forward impact links rather than rewriting history.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Identity collision	Quarantine affected commits and route to identity stewardship.
Domain mismatch	Deny before access, persistence, routing, or telemetry emission.
Ambiguous resolution	Return candidates or abstain; do not silently choose a canonical entity.

Acceptance evidence

- Collision fixtures cover concurrent creation, aliases, transliteration, and external identifier reuse.
- Isolation tests inspect every persisted and transmitted identity context.
- Resolution evaluation measures precision, recall, abstention, and stewardship workload.
- Merge and split replay verifies historical and forward references.

CHAPTER ACCEPTANCE

Identity, Naming, and Tenancy is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

26 Temporal Truth Engineering

The platform distinguishes when something occurred, when it was observed, when an assertion applies, and when The Eye recorded each version. This temporal model is mandatory for reconstruction and Decision Replay.

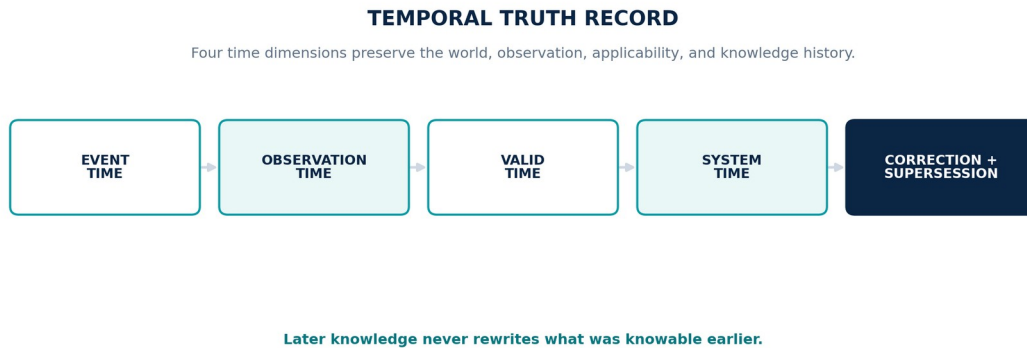


Figure 13. Temporal Truth Record. Four time dimensions preserve the world, observation, applicability, and knowledge history.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Event time	When the represented occurrence happened in the world.
Observation time	When a source or The Eye observed or received it.
Valid time	Interval in the world for which the assertion or state applies.
System time	Interval during which a specific version was recorded as current in The Eye.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-26-001	Material facts and relationships SHALL preserve all applicable time dimensions with timezone, precision, source, and uncertainty.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-26-002	System time SHALL be assigned by the authoritative committing component and SHALL be monotonic within the object's version history.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-26-003	As-of queries SHALL accept explicit valid and system time and SHALL not contaminate results with later corrections or knowledge.	Negative conformance test	Adversarial fixture proving the prohibited path is denied and audited.
ES-26-004	Unknown, approximate, bounded, recurring, and conflicting time SHALL be represented explicitly rather than forced into false precision.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-26-005	Clock quality, skew, and synchronization state SHALL be observable where ordering or expiry depends on wall time.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Invalid temporal interval	Reject canonical commit and retain the proposal for correction.
Clock uncertainty exceeds contract	Use sequence or authoritative commit order and expose temporal uncertainty.
Historical reconstruction gap	Return explicit missing range and affected objects; never substitute current state.

Acceptance evidence

- Bitemporal fixtures cover late arrival, correction, withdrawal, overlap, gap, and backdated validity.
- As-of replay reproduces known decision-time views.
- Clock-fault tests verify ordering and expiry behavior under skew and loss of synchronization.
- Temporal schema tests validate precision, timezone, interval, and uncertainty fields.

CHAPTER ACCEPTANCE

Temporal Truth Engineering is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-005 · C-011 · C-019 · C-025 · C-049

27 Truth State, Confidence, and Uncertainty

Truth state identifies what kind of knowledge an object represents; confidence and uncertainty characterize support and limits. Neither is a substitute for evidence or human authority.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Observed	Directly preserved from an identified source or instrument.
Claimed / inferred / assessed	Attributed assertion, derived result, or accountable analytical judgment.
Decided	Proposition or option incorporated into a recorded human decision within its scope.
Synthetic / withdrawn / superseded	Scenario or simulation state, or information no longer decision-active because history changed.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-27-001	Every decision-active object SHALL have an explicit truth state assigned through an owned transition.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-27-002	Confidence SHALL declare method, scale, calibration population, version, and interpretation; arbitrary global scores are prohibited.	Negative conformance test	Adversarial fixture proving the prohibited path is denied and audited.
ES-27-003	Uncertainty SHALL identify relevant distributions, intervals, alternatives, missing evidence, model limits, and irreducible unknowns.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-27-004	Promotion between truth states SHALL require the evidence, evaluation, authority, and audit defined for that transition.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.
ES-27-005	Synthetic state SHALL remain isolated from observed state until an explicit reconciliation establishes a new observed version.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Truth state missing	Prevent decision-active use and route to the owning component or human review.
Confidence outside calibration envelope	Mark unfit or abstain and propagate impact to dependent products.
Synthetic contamination	Quarantine affected state, reconstruct from observed history, and open an integrity incident.

Acceptance evidence

- Transition tests prove only authorized promotion paths are possible.
- Calibration tests validate confidence interpretation across declared populations and horizons.
- Downstream tests preserve truth state and uncertainty through summaries, briefings, exports, and APIs.
- Contamination fixtures attempt synthetic-to-observed writes through every integration path.

CHAPTER ACCEPTANCE

Truth State, Confidence, and Uncertainty is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-005 · C-010 · C-018 · C-021 · C-022 · C-023 · C-049

28 Provenance and Lineage Engineering

Provenance forms a traversable graph from source evidence through transformations, graph and memory revisions, twins, models, agents, forecasts, scenarios, simulations, recommendations, human decisions, commitments, and outcomes.

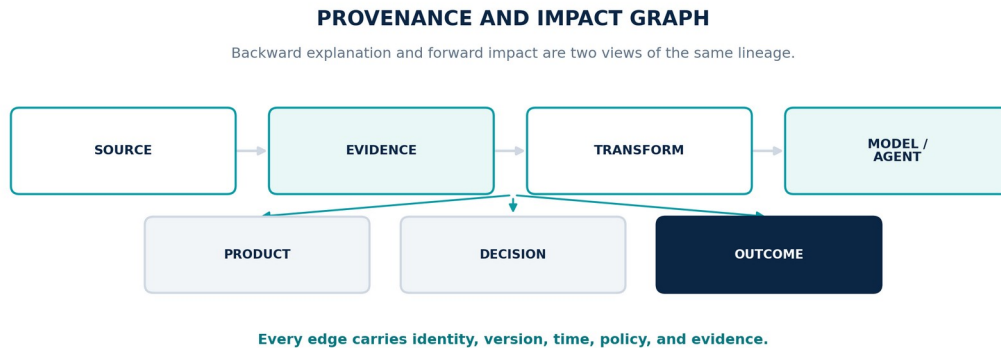


Figure 14. Provenance And Impact Graph. Backward explanation and forward impact are two views of the same lineage.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Source lineage	Authority, rights, source identity, acquisition, custody, checksum, and preserved evidence.
Transformation lineage	Exact inputs, code or prompt, model or agent, parameters, environment, policy, and outputs.
Decision lineage	Information boundary, alternatives, uncertainty, rationale, dissent, approval, obligations, and commitment.
Forward impact	Decision-active products and workflows that depend on a changed, corrected, withdrawn, or unfit object.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-28-001	Every material object and transition SHALL emit provenance links atomically or through a recoverable lineage journal.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-28-002	Lineage references SHALL identify exact versions and SHALL not rely on mutable latest aliases.	Negative conformance test	Adversarial fixture proving the prohibited path is denied and audited.
ES-28-003	Backward explanation SHALL resolve producing inputs and accountable actors; forward impact SHALL resolve active dependents and owners.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.
ES-28-004	Provenance SHALL preserve denied, abstained, challenged, overridden, corrected, and failed paths where they materially explain state.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-28-005	Lineage retention and deletion SHALL follow the strictest applicable obligation while preserving lawful audit proof and deletion evidence.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Lineage write failure	Do not promote the object to decision-active state; retain recoverable partial work.
Dangling reference	Mark lineage incomplete, identify affected products, and initiate repair.
Forward impact unavailable	Treat corrections and withdrawals as unresolved and escalate before closure.

Acceptance evidence

- Golden traces reconstruct source-to-outcome and outcome-to-source paths.
- Fault tests interrupt lineage writes at every boundary and verify no false completion.
- Correction tests identify all seeded downstream dependencies.
- Access tests ensure lineage reveals no unauthorized content while preserving accountable proof.

CHAPTER ACCEPTANCE

Provenance and Lineage Engineering is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-005 · C-012 · C-015 · C-025 · C-035 · C-036

29 Policy, Classification, Retention, and Rights

Control metadata travels with canonical objects and governs collection, access, processing, model and agent use, sharing, export, retention, correction, and deletion throughout the lifecycle.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Classification	Sensitivity and handling class with authority, markings, permitted environments, and downgrade rules.
Purpose	Declared allowed use linked to legal, contractual, customer, or institutional authority.
Rights	Acquisition, storage, transformation, analysis, model use, sharing, publication, export, and deletion rights.
Retention	Trigger, duration, hold, archive, correction, deletion, verification, and residual-state obligations.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-29-001	Objects SHALL receive domain, purpose, classification, rights, owner, retention, and residency metadata at or before canonical admission.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.
ES-29-002	Derived objects SHALL inherit the most restrictive applicable obligations unless an authorized transformation rule records a valid change.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-29-003	Policy evaluation SHALL consider subject, object, action, purpose, environment, time, location, model, agent, and requested side effect.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-29-004	Retention actions SHALL be durable workflows with scope resolution, holds, approvals, execution evidence, residual inventory, and verification.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.
ES-29-005	Exports SHALL carry enforceable recipient obligations, provenance, classification, expiry, and revocation context where applicable.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Policy metadata missing	Quarantine or apply an explicitly defined restrictive default; never infer permissive use.
Conflicting obligations	Apply the most restrictive valid obligation and route conflict for governance resolution.
Deletion incomplete	Keep the workflow open, expose residual state, and prohibit verified-deletion status.

Acceptance evidence

- Policy matrix tests cover roles, purposes, environments, models, agents, exports, time, and classifications.
- Inheritance tests verify restrictive propagation through derived objects and summaries.
- Retention tests cover hold, archive, delete, backup expiry, projection rebuild, and residual evidence.
- Export tests verify package obligations, revocation, and recipient scope.

CHAPTER ACCEPTANCE

Policy, Classification, Retention, and Rights is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-016 · C-035 · C-037 · C-039 · C-040 · C-041

30 Correction, Supersession, and Projection Rebuild

Corrections change what is decision-active without erasing what was previously recorded or knowable. Supersession and withdrawal propagate through durable impact analysis, projection invalidation, review, and closure.

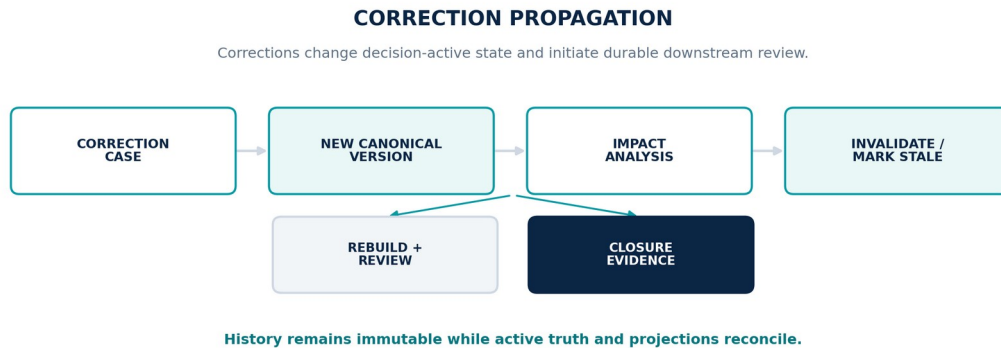


Figure 15. Correction Propagation. Corrections change decision-active state and initiate durable downstream review.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Correction	New version or linked object that replaces, adjudicates, merges, splits, or qualifies prior information.
Withdrawal	Authority-backed removal from decision-active use while preserving history and reason.
Supersession	Ordered relationship identifying the version now applicable under stated time and policy scope.
Projection rebuild	Deterministic regeneration of derived query state from corrected canonical history.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-30-001	Correction intake SHALL authenticate authority, identify target versions, preserve supplied evidence, and create a durable case before changing active state.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.
ES-30-002	Correction commit SHALL produce a new version and propagation event with reason, authority, time, scope, and affected truth state.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.
ES-30-003	Impact analysis SHALL identify dependent graph state, memory, twins, forecasts, scenarios, simulations, warnings, decisions, briefings, commitments, and learning products.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-30-004	Dependent products SHALL be marked affected, stale, withdrawn, or requiring review according to their owning contract.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-30-005	Closure SHALL require projection reconciliation, consumer acknowledgement or bounded exception, and complete audit evidence.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Target version missing	Hold the case and return an explicit unresolved target state.
Propagation incomplete	Keep correction open, expose affected consumers, and prohibit false closure.
Projection rebuild fails	Serve canonical or last-known projection only with visible degraded status and decision-use constraints.

Acceptance evidence

- End-to-end fixtures correct, withdraw, merge, and split seeded objects and verify all dependent products.
- Decision Replay proves prior knowledge remains unchanged while later correction is visible separately.
- Projection rebuild tests are deterministic and reconcile counts, versions, and digests.
- Closure tests reject unresolved consumers and missing audit evidence.

CHAPTER ACCEPTANCE

Correction, Supersession, and Projection Rebuild is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-011 · C-012 · C-016 · C-025 · C-033 · C-043

PART V

Canonical Layer Implementations

Build-directive specifications for the ten frozen architecture layers.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

31 World Observation Layer

Maintain a measurable, governed field of view across authorized external, internal, physical, and networked sources.

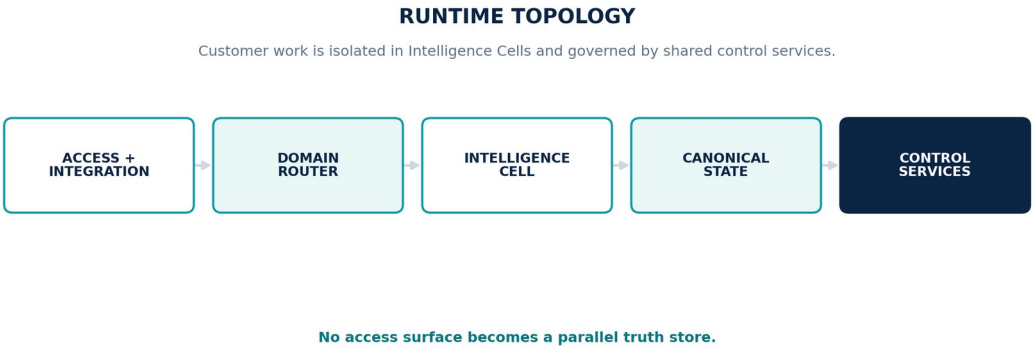


Figure 16. Runtime Topology. Customer work is isolated in Intelligence Cells and governed by shared control services.

Layer implementation contract

CONTRACT	ENGINEERING DEFINITION
Layer authority	Maintain a measurable, governed field of view across authorized external, internal, physical, and networked sources.
Boundary	Owns source contracts, acquisition, raw preservation, source health, coverage, and correction intake. It does not infer institutional truth or publish recommendations.
Authoritative state	Source contracts and credential references; Connector configuration and checkpoints; Immutable raw evidence; Acquisition and source-health history; Coverage map and blind spots.
Primary inputs	Authorized source definitions; Credentials and collection authority; Schedules, subscriptions, and stream endpoints; Customer purpose and coverage requirements.
Canonical outputs	Observation Records; Evidence Objects; Source health and freshness state; Coverage gaps; Correction and withdrawal events.
Quality dimensions	coverage, freshness, latency, completeness, authenticity, collection success, correction propagation time.

Logical component realization

ID	COMPONENT	IMPLEMENTATION DUTY	RUNTIME PATTERN
L1-C01	Source Registry and Contract Service	Owns versioned source registry and contract definitions, resolution, compatibility, stewardship, and lifecycle events.	Registry + resolver
L1-C02	Connector Runtime and Adapter SDK	Mediates connector runtime and adapter sdk through authenticated, policy-evaluated, typed contracts and isolates external semantics.	Stateless gateway + durable checkpoints
L1-C03	Collection Scheduler and Subscription Manager	Coordinates collection scheduler and subscription admission, ordering, budgets, retries, escalation, completion, and observable state.	Coordinator + durable queue
L1-C04	Streaming and Batch Receivers	Implements the streaming and batch receivers capability within World Observation Layer using versioned contracts, bounded state, controls, telemetry, and explicit failure behavior.	Owned service
L1-C05	Raw Preservation and Evidence Vault	Owns durable raw preservation and evidence vault state with version, policy, integrity, retention, correction, and recovery.	Canonical state service
L1-C06	Authenticity and Chain-of-Custody Service	Implements the authenticity and chain-of-custody service capability within World Observation Layer using versioned contracts, bounded state, controls, telemetry, and explicit failure behavior.	Owned service
L1-C07	Quarantine and Malware Inspection	Implements the quarantine and malware inspection capability within World Observation Layer using versioned contracts, bounded state, controls, telemetry, and explicit failure behavior.	Owned service

ID	COMPONENT	IMPLEMENTATION DUTY	RUNTIME PATTERN
L1-C08	Source Health, Freshness, and Coverage Monitor	Measures source health, freshness, and coverage monitor fitness, publishes product quality state, and initiates constraint or remediation workflow.	Stream + evaluation worker
L1-C09	Correction and Withdrawal Intake	Implements the correction and withdrawal intake capability within World Observation Layer using versioned contracts, bounded state, controls, telemetry, and explicit failure behavior.	Owned service
ID	STATE CLASS	PRIMARY SCALE KEY	FAILURE AND RECOVERY
L1-C01	Canonical control state	Domain / namespace	Last verified read; block unsafe writes
L1-C02	Checkpoint / quarantine	Source / destination / tenant	Backpressure, quarantine, bounded retry
L1-C03	Workflow / scheduling state	Tenant / workload class	Pause admission; drain or resume safely
L1-C04	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L1-C05	Authoritative	Tenant / object partition	Restore, reconcile, then reopen writes
L1-C06	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L1-C07	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L1-C08	Metrics + fitness decisions	Artifact / product / window	Expose indeterminate quality; constrain use
L1-C09	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract

Interface implementation profile

ID	INTERFACE	STYLE	TRANSPORT PROFILE	RELIABILITY CONTRACT
L1-I01	RegisterSource	Command	Typed asynchronous or bounded synchronous command with preconditions	Idempotent acceptance and one authoritative effect
L1-I02	Acquire	Command / stream	Flow-controlled stream with sequence, checkpoint, and bounded replay	At-least-once segments under stable partition key
L1-I03	ObservationRecorded	Domain event	Immutable versioned domain event through ordered partitioned transport	At-least-once delivery; consumer deduplication and checkpoint
L1-I04	SourceHealthChanged	Domain event	Immutable versioned domain event through ordered partitioned transport	At-least-once delivery; consumer deduplication and checkpoint
L1-I05	CorrectionReceived	Domain event	Immutable versioned domain event through ordered partitioned transport	At-least-once delivery; consumer deduplication and checkpoint

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-31-001	World Observation Layer SHALL own and version only its declared authoritative state: Source contracts and credential references; Connector configuration and checkpoints; Immutable raw evidence; Acquisition and source-health history; Coverage map and blind spots.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-31-002	Every logical component in World Observation Layer SHALL publish a service manifest with component ID, owner, contracts, state class, dependencies, controls, SLOs, failure domains, and recovery procedure.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-31-003	Deployable units implementing World Observation Layer SHALL preserve component-level telemetry, policy enforcement, state access, resource use, and failure isolation even when components share a runtime.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-31-004	All World Observation Layer interfaces SHALL implement the canonical envelope, error model, compatibility rules, idempotency or ordering profile, audit causation, and consumer conformance fixtures.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-31-005	Inputs to World Observation Layer SHALL be authenticated, authorized, schema-valid, temporally explicit, provenance-bearing, quality-marked, and admitted before decision-active use.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-31-006	Outputs from World Observation Layer SHALL be immutable versioned canonical products or explicit proposals and SHALL preserve truth state, uncertainty, lineage, policy, quality, and lifecycle.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-31-007	World Observation Layer SHALL enforce its declared trust, security, privacy, sovereignty, workflow, audit, evaluation, and human-authority controls at every material boundary.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-31-008	World Observation Layer SHALL measure coverage, freshness, latency, completeness, authenticity, collection success, correction propagation time through governed indicators whose failed or indeterminate state constrains downstream use.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-31-009	World Observation Layer SHALL implement and continuously test every declared degraded behavior without fabricating success, freshness, completeness, confidence, authorization, or approval.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-31-010	World Observation Layer SHALL pass the same object, contract, control, failure, recovery, export, and replay conformance fixtures in SaaS, Private Cloud, and On-Premise deployments.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.

Control enforcement

- Purpose and legal authority before collection
- Tenant, classification, and residency assignment at ingress
- Rate, robots, licence, and contractual limits
- Authenticity and malware checks
- Immutable evidence identifiers and custody events

Degraded behavior and acceptance

DECLARED DEGRADED BEHAVIOR	REQUIRED ENGINEERING PROOF
Stale source state remains visible	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.
Derived products receive source-health impact markers	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.
Unsafe content stays quarantined	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.
No silent substitution of an unavailable source	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.
Coverage gaps enter warning and decision packages	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.

LAYER ACCEPTANCE

World Observation Layer is accepted only when its 9 logical components, 5 canonical interfaces, authoritative state, controls, quality dimensions, degraded modes, and deployment-parity fixtures remain independently testable and operationally owned.

32 Intelligence Layer

Transform heterogeneous evidence into structured, reviewable understanding while preserving uncertainty, method, and source attribution.

	CONTROL ENFORCEMENT					
	READ	TRANSFORM	MODEL	AGENT	ACTION	EXPORT
IDENTITY	ENFORCE	VERIFY	ENFORCE	VERIFY	ENFORCE	VERIFY
POLICY	VERIFY	ENFORCE	VERIFY	ENFORCE	VERIFY	ENFORCE
PROVENANCE	ENFORCE	VERIFY	ENFORCE	VERIFY	ENFORCE	VERIFY
AUDIT	VERIFY	ENFORCE	VERIFY	ENFORCE	VERIFY	ENFORCE

Missing enforcement makes the operation non-conformant.

Figure 17. Control Enforcement. Controls operate at every material read, transformation, action, and export boundary.

Layer implementation contract

CONTRACT	ENGINEERING DEFINITION
Layer authority	Transform heterogeneous evidence into structured, reviewable understanding while preserving uncertainty, method, and source attribution.
Boundary	Owns deterministic and model-assisted transformation, extraction, classification, linking candidates, contradiction detection, assessments, and human review. It does not overwrite evidence or declare unreviewed model output as fact.
Authoritative state	Transformation definitions and versions; Candidate intelligence objects; Confidence and review state; Method and model lineage; Evaluation and adjudication history.
Primary inputs	Evidence Objects; Schemas and ontologies; Approved transformation and model versions; Domain policies and review thresholds.
Canonical outputs	Claims and Events; Entity and relationship candidates; Assessments and summaries; Contradictions and uncertainty; Transformation lineage and evaluation results.
Quality dimensions	precision, recall, entity-link accuracy, contradiction detection, calibration, review yield, latency, cost, abstention quality.

Logical component realization

ID	COMPONENT	IMPLEMENTATION DUTY	RUNTIME PATTERN
L2-C01	Format Detection, OCR, Transcription, and Parsing	Implements the format detection, ocr, transcription, and parsing capability within Intelligence Layer using versioned contracts, bounded state, controls, telemetry, and explicit failure behavior.	Owned service
L2-C02	Cleaning, Normalization, and Language Services	Implements the cleaning, normalization, and language services capability within Intelligence Layer using versioned contracts, bounded state, controls, telemetry, and explicit failure behavior.	Owned service
L2-C03	Classification and Routing	Implements the classification and routing capability within Intelligence Layer using versioned contracts, bounded state, controls, telemetry, and explicit failure behavior.	Owned service
L2-C04	Summarization and Structured Extraction	Implements the summarization and structured extraction capability within Intelligence Layer using versioned contracts, bounded state, controls, telemetry, and explicit failure behavior.	Owned service
L2-C05	Named-Entity Recognition and Candidate Linking	Implements the named-entity recognition and candidate linking capability within Intelligence Layer using versioned contracts, bounded state, controls, telemetry, and explicit failure behavior.	Owned service
L2-C06	Relationship, Event, and Claim Extraction	Implements the relationship, event, and claim extraction capability within Intelligence Layer using versioned contracts, bounded state, controls, telemetry, and explicit failure behavior.	Owned service

ID	COMPONENT	IMPLEMENTATION DUTY	RUNTIME PATTERN
L2-C07	Contradiction and Corroboration Detection	Implements the contradiction and corroboration detection capability within Intelligence Layer using versioned contracts, bounded state, controls, telemetry, and explicit failure behavior.	Owned service
L2-C08	Confidence and Truth-State Assessor	Executes the governed confidence and truth-state assessor method against exact inputs and versions and returns uncertainty, quality, and lineage.	Horizontally scaled worker
L2-C09	Human Review Queue and Adjudication	Coordinates durable human review queue and adjudication transitions, human tasks, deadlines, separation, compensation, and audit evidence.	Durable workflow
L2-C10	Transformation Registry and Evaluation Harness	Owns versioned transformation registry and evaluation harness definitions, resolution, compatibility, stewardship, and lifecycle events.	Registry + resolver
ID	STATE CLASS	PRIMARY SCALE KEY	FAILURE AND RECOVERY
L2-C01	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L2-C02	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L2-C03	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L2-C04	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L2-C05	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L2-C06	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L2-C07	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L2-C08	Immutable result + checkpoint	Tenant / object / job	Abstain or partial result; never fabricate completion
L2-C09	Workflow history	Instance / tenant	Resume from last commit; escalate residual effect
L2-C10	Canonical control state	Domain / namespace	Last verified read; block unsafe writes

Interface implementation profile

ID	INTERFACE	STYLE	TRANSPORT PROFILE	RELIABILITY CONTRACT
L2-I01	TransformEvidence	Command	Typed asynchronous or bounded synchronous command with preconditions	Idempotent acceptance and one authoritative effect
L2-I02	IntelligenceObjectProposed	Domain event	Immutable versioned domain event through ordered partitioned transport	At-least-once delivery; consumer deduplication and checkpoint
L2-I03	ContradictionDetected	Domain event	Immutable versioned domain event through ordered partitioned transport	At-least-once delivery; consumer deduplication and checkpoint
L2-I04	ReviewRequested	Workflow event	Immutable versioned domain event through ordered partitioned transport	At-least-once delivery; consumer deduplication and checkpoint
L2-I05	TransformationEvaluated	Domain event	Immutable versioned domain event through ordered partitioned transport	At-least-once delivery; consumer deduplication and checkpoint

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-32-001	Intelligence Layer SHALL own and version only its declared authoritative state: Transformation definitions and versions; Candidate intelligence objects; Confidence and review state; Method and model lineage; Evaluation and adjudication history.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-32-002	Every logical component in Intelligence Layer SHALL publish a service manifest with component ID, owner, contracts, state class, dependencies, controls, SLOs, failure domains, and recovery procedure.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-32-003	Deployable units implementing Intelligence Layer SHALL preserve component-level telemetry, policy enforcement, state access, resource use, and failure isolation even when components share a runtime.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-32-004	All Intelligence Layer interfaces SHALL implement the canonical envelope, error model, compatibility rules, idempotency or ordering profile, audit causation, and consumer conformance fixtures.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-32-005	Inputs to Intelligence Layer SHALL be authenticated, authorized, schema-valid, temporally explicit, provenance-bearing, quality-marked, and admitted before decision-active use.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-32-006	Outputs from Intelligence Layer SHALL be immutable versioned canonical products or explicit proposals and SHALL preserve truth state, uncertainty, lineage, policy, quality, and lifecycle.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-32-007	Intelligence Layer SHALL enforce its declared trust, security, privacy, sovereignty, workflow, audit, evaluation, and human-authority controls at every material boundary.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.
ES-32-008	Intelligence Layer SHALL measure precision, recall, entity-link accuracy, contradiction detection, calibration, review yield, latency, cost, abstention quality through governed indicators whose failed or indeterminate state constrains downstream use.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-32-009	Intelligence Layer SHALL implement and continuously test every declared degraded behavior without fabricating success, freshness, completeness, confidence, authorization, or approval.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-32-010	Intelligence Layer SHALL pass the same object, contract, control, failure, recovery, export, and replay conformance fixtures in SaaS, Private Cloud, and On-Premise deployments.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.

Control enforcement

- Truth-state labels on every output
- Evidence references remain mandatory
- Model and prompt versions are explicit
- Human review thresholds follow risk and impact
- Derived objects never replace source evidence

Degraded behavior and acceptance

DECLARED DEGRADED BEHAVIOR	REQUIRED ENGINEERING PROOF
Abstain when evidence or model fitness is insufficient	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.
Route to deterministic fallback or human review	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.
Expose reduced confidence and unavailable methods	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.
Quarantine prompt-injection or malicious content	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.
Prevent downstream promotion when required review is missing	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.

LAYER ACCEPTANCE

Intelligence Layer is accepted only when its 10 logical components, 5 canonical interfaces, authoritative state, controls, quality dimensions, degraded modes, and deployment-parity fixtures remain independently testable and operationally owned.

CONSTITUTIONAL INHERITANCE / C-005 · C-010 · C-011 · C-012 · C-030 · C-045

33 Enterprise Memory

Preserve durable, permission-aware organizational and world memory across evidence, analyses, decisions, actions, outcomes, and lessons.

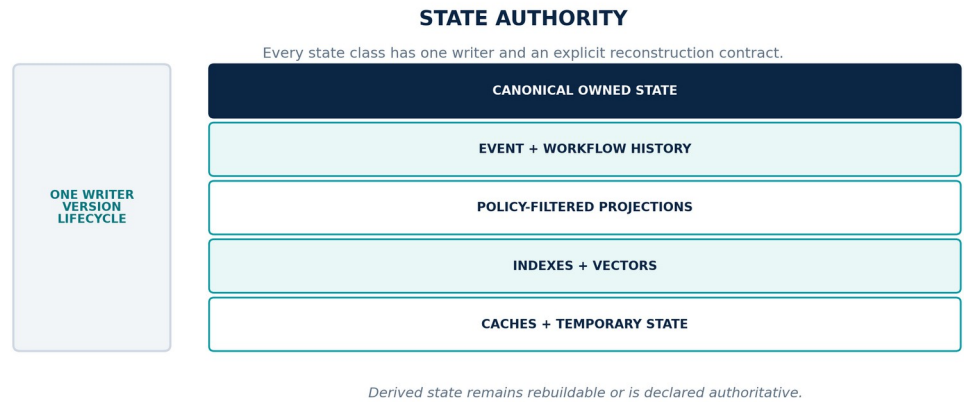


Figure 18. State Authority. Every state class has one writer and an explicit reconstruction contract.

Layer implementation contract

CONTRACT	ENGINEERING DEFINITION
Layer authority	Preserve durable, permission-aware organizational and world memory across evidence, analyses, decisions, actions, outcomes, and lessons.
Boundary	Owns persistence, version history, retention, legal hold, correction, retrieval indexes, and memory access policy. It does not convert retrieval frequency or model output into accepted organizational truth.
Authoritative state	Memory objects and versions; Retention and legal-hold state; Retrieval indexes; Correction and tombstone history; Archive, export, and deletion evidence.
Primary inputs	Evidence and intelligence objects; Documents and communications; Graph, twin, decision, and outcome artifacts; Retention, deletion, and legal-hold policy.
Canonical outputs	Versioned memory entries; Permission-aware retrieval context; Historical reconstruction; Retention and deletion evidence; Strategic record packages.
Quality dimensions	durability, retrieval relevance, permission correctness, temporal completeness, reconstruction success, retention compliance, deletion verification.

Logical component realization

ID	COMPONENT	IMPLEMENTATION DUTY	RUNTIME PATTERN
L3-C01	Memory Object Service	Owns durable memory object state with version, policy, integrity, retention, correction, and recovery.	Canonical state service
L3-C02	Document and Artifact Repository	Owns durable document and artifact repository state with version, policy, integrity, retention, correction, and recovery.	Canonical state service
L3-C03	Temporal Record Store	Owns durable temporal record store state with version, policy, integrity, retention, correction, and recovery.	Canonical state service
L3-C04	Version and Correction Ledger	Implements the version and correction ledger capability within Enterprise Memory using versioned contracts, bounded state, controls, telemetry, and explicit failure behavior.	Owned service
L3-C05	Retention, Legal Hold, and Deletion Service	Implements the retention, legal hold, and deletion service capability within Enterprise Memory using versioned contracts, bounded state, controls, telemetry, and explicit failure behavior.	Owned service
L3-C06	Permission-Aware Semantic Index	Implements the permission-aware semantic index capability within Enterprise Memory using versioned contracts, bounded state, controls, telemetry, and explicit failure behavior.	Owned service
L3-C07	Memory Retrieval and Context Assembly	Owns durable memory retrieval and context assembly state with version, policy, integrity, retention, correction, and recovery.	Canonical state service

ID	COMPONENT	IMPLEMENTATION DUTY	RUNTIME PATTERN
L3-C08	Archive and Cold-Tier Manager	Coordinates archive and cold-tier admission, ordering, budgets, retries, escalation, completion, and observable state.	Coordinator + durable queue
L3-C09	Memory Integrity and Orphan Detection	Owns durable memory integrity and orphan detection state with version, policy, integrity, retention, correction, and recovery.	Canonical state service
ID	STATE CLASS	PRIMARY SCALE KEY	FAILURE AND RECOVERY
L3-C01	Authoritative	Tenant / object partition	Restore, reconcile, then reopen writes
L3-C02	Authoritative	Tenant / object partition	Restore, reconcile, then reopen writes
L3-C03	Authoritative	Tenant / object partition	Restore, reconcile, then reopen writes
L3-C04	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L3-C05	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L3-C06	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L3-C07	Authoritative	Tenant / object partition	Restore, reconcile, then reopen writes
L3-C08	Workflow / scheduling state	Tenant / workload class	Pause admission; drain or resume safely
L3-C09	Authoritative	Tenant / object partition	Restore, reconcile, then reopen writes

Interface implementation profile

ID	INTERFACE	STYLE	TRANSPORT PROFILE	RELIABILITY CONTRACT
L3-I01	CommitMemory	Command	Typed asynchronous or bounded synchronous command with preconditions	Idempotent acceptance and one authoritative effect
L3-I02	RetrieveContext	Query	Synchronous typed query with as-of, projection watermark, and policy context	No state change; consistency and staleness explicit
L3-I03	MemoryCorrected	Domain event	Immutable versioned domain event through ordered partitioned transport	At-least-once delivery; consumer deduplication and checkpoint
L3-I04	RetentionActionDue	Domain event	Immutable versioned domain event through ordered partitioned transport	At-least-once delivery; consumer deduplication and checkpoint
L3-I05	DeletionVerified	Audit event	Immutable versioned domain event through ordered partitioned transport	At-least-once delivery; consumer deduplication and checkpoint

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-33-001	Enterprise Memory SHALL own and version only its declared authoritative state: Memory objects and versions; Retention and legal-hold state; Retrieval indexes; Correction and tombstone history; Archive, export, and deletion evidence.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-33-002	Every logical component in Enterprise Memory SHALL publish a service manifest with component ID, owner, contracts, state class, dependencies, controls, SLOs, failure domains, and recovery procedure.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-33-003	Deployable units implementing Enterprise Memory SHALL preserve component-level telemetry, policy enforcement, state access, resource use, and failure isolation even when components share a runtime.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-33-004	All Enterprise Memory interfaces SHALL implement the canonical envelope, error model, compatibility rules, idempotency or ordering profile, audit causation, and consumer conformance fixtures.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-33-005	Inputs to Enterprise Memory SHALL be authenticated, authorized, schema-valid, temporally explicit, provenance-bearing, quality-marked, and admitted before decision-active use.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-33-006	Outputs from Enterprise Memory SHALL be immutable versioned canonical products or explicit proposals and SHALL preserve truth state, uncertainty, lineage, policy, quality, and lifecycle.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-33-007	Enterprise Memory SHALL enforce its declared trust, security, privacy, sovereignty, workflow, audit, evaluation, and human-authority controls at every material boundary.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-33-008	Enterprise Memory SHALL measure durability, retrieval relevance, permission correctness, temporal completeness, reconstruction success, retention compliance, deletion verification through governed indicators whose failed or indeterminate state constrains downstream use.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-33-009	Enterprise Memory SHALL implement and continuously test every declared degraded behavior without fabricating success, freshness, completeness, confidence, authorization, or approval.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-33-010	Enterprise Memory SHALL pass the same object, contract, control, failure, recovery, export, and replay conformance fixtures in SaaS, Private Cloud, and On-Premise deployments.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.

Control enforcement

- Tenant and object-level access
- Purpose-aware retrieval
- Versioned corrections rather than destructive overwrite
- Lawful retention and verifiable deletion
- Isolation of unverified synthetic and model-generated content

Degraded behavior and acceptance

DECLARED DEGRADED BEHAVIOR	REQUIRED ENGINEERING PROOF
Fallback to metadata-only retrieval when content tier is unavailable	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.
Prevent stale index results from appearing current	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.
Expose partial reconstruction	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.
Pause deletion when referential scope cannot be proven	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.
Recover indexes from canonical records, not vice versa	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.

LAYER ACCEPTANCE

Enterprise Memory is accepted only when its 9 logical components, 5 canonical interfaces, authoritative state, controls, quality dimensions, degraded modes, and deployment-parity fixtures remain independently testable and operationally owned.

34 Knowledge Graph

Provide shared identity, semantics, temporal relationships, and provenance connecting the world, the enterprise, intelligence objects, and strategic intent.

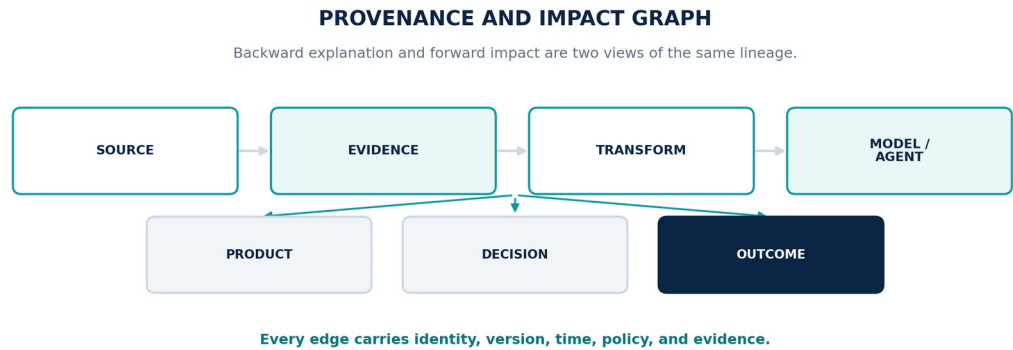


Figure 19. Provenance And Impact Graph. Backward explanation and forward impact are two views of the same lineage.

Layer implementation contract

CONTRACT	ENGINEERING DEFINITION
Layer authority	Provide shared identity, semantics, temporal relationships, and provenance connecting the world, the enterprise, intelligence objects, and strategic intent.
Boundary	Owns ontology governance, entity identity, relationship validation, temporal graph state, graph revisions, provenance edges, and graph subscriptions. It is the semantic heart, not the only physical store or computation engine.
Authoritative state	Ontology and semantic versions; Canonical entity identities; Temporal nodes and edges; Provenance links; Graph revision and stewardship history.
Primary inputs	Entity and relationship candidates; Claims, events, and assessments; Ontology and policy changes; Corrections, merges, splits, and withdrawals.
Canonical outputs	Canonical entity identities; Temporal relationships; Graph revisions and subscriptions; Impact paths and dependency views; Strategy Graph context.
Quality dimensions	identity precision, identity recall, edge validity, provenance completeness, temporal completeness, orphan rate, query latency, revision propagation.

Logical component realization

ID	COMPONENT	IMPLEMENTATION DUTY	RUNTIME PATTERN
L4-C01	Ontology Registry and Governance Workflow	Owns versioned ontology registry and governance workflow definitions, resolution, compatibility, stewardship, and lifecycle events.	Registry + resolver
L4-C02	Identity Resolution and Entity Stewardship	Implements the identity resolution and entity stewardship capability within Knowledge Graph using versioned contracts, bounded state, controls, telemetry, and explicit failure behavior.	Owned service
L4-C03	Graph Write Validator	Implements the graph write validator capability within Knowledge Graph using versioned contracts, bounded state, controls, telemetry, and explicit failure behavior.	Owned service
L4-C04	Temporal Graph Store	Owns durable temporal graph store state with version, policy, integrity, retention, correction, and recovery.	Canonical state service
L4-C05	Provenance and Evidence Edge Service	Implements the provenance and evidence edge service capability within Knowledge Graph using versioned contracts, bounded state, controls, telemetry, and explicit failure behavior.	Owned service
L4-C06	Graph Query and Traversal Service	Implements the graph query and traversal service capability within Knowledge Graph using versioned contracts, bounded state, controls, telemetry, and explicit failure behavior.	Owned service
L4-C07	Reasoning and Constraint Engine	Executes the governed reasoning and constraint method against exact inputs and versions and returns uncertainty, quality, and lineage.	Horizontally scaled worker

ID	COMPONENT	IMPLEMENTATION DUTY	RUNTIME PATTERN
L4-C08	Graph Change Subscription Service	Implements the graph change subscription service capability within Knowledge Graph using versioned contracts, bounded state, controls, telemetry, and explicit failure behavior.	Owned service
L4-C09	Merge, Split, and Reconciliation Manager	Coordinates merge, split, and reconciliation admission, ordering, budgets, retries, escalation, completion, and observable state.	Coordinator + durable queue
L4-C10	Graph Quality and Orphan Monitor	Measures graph quality and orphan monitor fitness, publishes product quality state, and initiates constraint or remediation workflow.	Stream + evaluation worker
ID	STATE CLASS	PRIMARY SCALE KEY	FAILURE AND RECOVERY
L4-C01	Canonical control state	Domain / namespace	Last verified read; block unsafe writes
L4-C02	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L4-C03	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L4-C04	Authoritative	Tenant / object partition	Restore, reconcile, then reopen writes
L4-C05	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L4-C06	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L4-C07	Immutable result + checkpoint	Tenant / object / job	Abstain or partial result; never fabricate completion
L4-C08	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L4-C09	Workflow / scheduling state	Tenant / workload class	Pause admission; drain or resume safely
L4-C10	Metrics + fitness decisions	Artifact / product / window	Expose indeterminate quality; constrain use

Interface implementation profile

ID	INTERFACE	STYLE	TRANSPORT PROFILE	RELIABILITY CONTRACT
L4-I01	ResolveIdentity	Command / query	Typed asynchronous or bounded synchronous command with preconditions	Idempotent acceptance and one authoritative effect
L4-I02	CommitGraphRevision	Command	Typed asynchronous or bounded synchronous command with preconditions	Idempotent acceptance and one authoritative effect
L4-I03	GraphChanged	Domain event	Immutable versioned domain event through ordered partitioned transport	At-least-once delivery; consumer deduplication and checkpoint
L4-I04	TraverseContext	Query	Synchronous typed query with as-of, projection watermark, and policy context	No state change; consistency and staleness explicit
L4-I05	OntologyChangeProposed	Workflow event	Immutable versioned domain event through ordered partitioned transport	At-least-once delivery; consumer deduplication and checkpoint

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-34-001	Knowledge Graph SHALL own and version only its declared authoritative state: Ontology and semantic versions; Canonical entity identities; Temporal nodes and edges; Provenance links; Graph revision and stewardship history.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-34-002	Every logical component in Knowledge Graph SHALL publish a service manifest with component ID, owner, contracts, state class, dependencies, controls, SLOs, failure domains, and recovery procedure.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-34-003	Deployable units implementing Knowledge Graph SHALL preserve component-level telemetry, policy enforcement, state access, resource use, and failure isolation even when components share a runtime.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-34-004	All Knowledge Graph interfaces SHALL implement the canonical envelope, error model, compatibility rules, idempotency or ordering profile, audit causation, and consumer conformance fixtures.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-34-005	Inputs to Knowledge Graph SHALL be authenticated, authorized, schema-valid, temporally explicit, provenance-bearing, quality-marked, and admitted before decision-active use.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-34-006	Outputs from Knowledge Graph SHALL be immutable versioned canonical products or explicit proposals and SHALL preserve truth state, uncertainty, lineage, policy, quality, and lifecycle.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-34-007	Knowledge Graph SHALL enforce its declared trust, security, privacy, sovereignty, workflow, audit, evaluation, and human-authority controls at every material boundary.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.
ES-34-008	Knowledge Graph SHALL measure identity precision, identity recall, edge validity, provenance completeness, temporal completeness, orphan rate, query latency, revision propagation through governed indicators whose failed or indeterminate state constrains downstream use.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-34-009	Knowledge Graph SHALL implement and continuously test every declared degraded behavior without fabricating success, freshness, completeness, confidence, authorization, or approval.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-34-010	Knowledge Graph SHALL pass the same object, contract, control, failure, recovery, export, and replay conformance fixtures in SaaS, Private Cloud, and On-Premise deployments.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.

Control enforcement

- No edge without type, time, provenance, and policy context
- Identity merges and splits are reversible
- Ontology changes require compatibility analysis
- Graph queries remain permission-aware
- Synthetic relationships are explicitly marked

Degraded behavior and acceptance

DECLARED DEGRADED BEHAVIOR	REQUIRED ENGINEERING PROOF
Serve last-known graph revision with staleness marker	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.
Queue writes when validation or consensus is unavailable	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.
Prevent partial multi-object revisions	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.
Expose unresolved identities rather than forcing links	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.
Rebuild projections from revision history	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.

LAYER ACCEPTANCE

Knowledge Graph is accepted only when its 10 logical components, 5 canonical interfaces, authoritative state, controls, quality dimensions, degraded modes, and deployment-parity fixtures remain independently testable and operationally owned.

CONSTITUTIONAL INHERITANCE / C-007 · C-009 · C-010 · C-011 · C-012 · C-017 · C-035

35 Digital Twins

Maintain living, versioned strategic representations of the customer, competitors, supply chain, products, markets, regulations, infrastructure, processes, and organization.

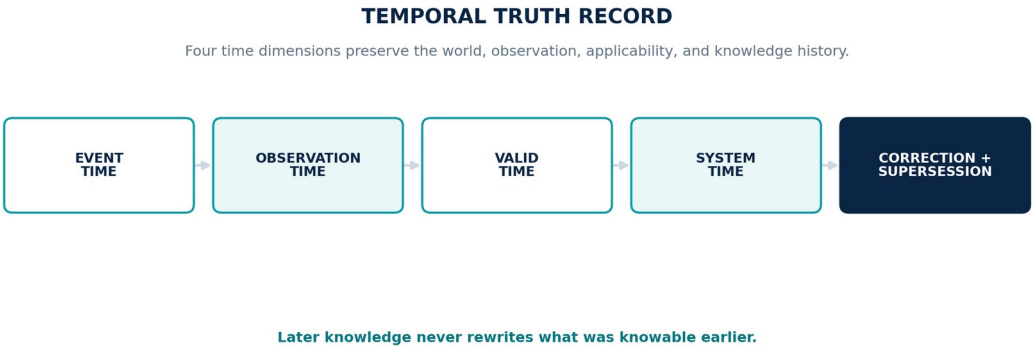


Figure 20. Temporal Truth Record. Four time dimensions preserve the world, observation, applicability, and knowledge history.

Layer implementation contract

CONTRACT	ENGINEERING DEFINITION
Layer authority	Maintain living, versioned strategic representations of the customer, competitors, supply chain, products, markets, regulations, infrastructure, processes, and organization.
Boundary	Owns twin definitions, evidence boundaries, state estimation, model bindings, snapshots, branches, reconciliation, validation, and dependency maps. It never presents simulated state as observed fact.
Authoritative state	Twin definitions and owners; Observed-state snapshots; Synthetic branches; Model and assumption bindings; Validation, confidence, and reconciliation history.
Primary inputs	Graph identities and relationships; Observed and operational state; Approved behavioral and causal models; Assumptions, constraints, and validation rules.
Canonical outputs	Twin snapshots; Observed and synthetic branches; Dependency and behavior state; Validation and confidence state; Scenario and simulation inputs.
Quality dimensions	state freshness, estimation error, model fitness, coverage, reconciliation success, branch isolation, dependency completeness.

Logical component realization

ID	COMPONENT	IMPLEMENTATION DUTY	RUNTIME PATTERN
L5-C01	Twin Registry and Boundary Service	Owns versioned twin registry and boundary definitions, resolution, compatibility, stewardship, and lifecycle events.	Registry + resolver
L5-C02	State Estimation and Reconciliation	Implements the state estimation and reconciliation capability within Digital Twins using versioned contracts, bounded state, controls, telemetry, and explicit failure behavior.	Owned service
L5-C03	Model Binding and Operating-Envelope Registry	Owns versioned model binding and operating-envelope registry definitions, resolution, compatibility, stewardship, and lifecycle events.	Registry + resolver
L5-C04	Snapshot and Version Manager	Coordinates snapshot and version admission, ordering, budgets, retries, escalation, completion, and observable state.	Coordinator + durable queue
L5-C05	Branch and Time-Travel Manager	Coordinates branch and time-travel admission, ordering, budgets, retries, escalation, completion, and observable state.	Coordinator + durable queue
L5-C06	Twin Dependency Mapper	Implements the twin dependency mapper capability within Digital Twins using versioned contracts, bounded state, controls, telemetry, and explicit failure behavior.	Owned service
L5-C07	Behavior and Constraint Runtime	Implements the behavior and constraint runtime capability within Digital Twins using versioned contracts, bounded state, controls, telemetry, and explicit failure behavior.	Owned service
L5-C08	Validation and Calibration Service	Measures validation and calibration fitness, publishes product quality state, and initiates constraint or remediation workflow.	Stream + evaluation worker

ID	COMPONENT	IMPLEMENTATION DUTY	RUNTIME PATTERN
L5-C09	Synthetic-State Labelling and Isolation	Implements the synthetic-state labelling and isolation capability within Digital Twins using versioned contracts, bounded state, controls, telemetry, and explicit failure behavior.	Owned service
ID	STATE CLASS	PRIMARY SCALE KEY	FAILURE AND RECOVERY
L5-C01	Canonical control state	Domain / namespace	Last verified read; block unsafe writes
L5-C02	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L5-C03	Canonical control state	Domain / namespace	Last verified read; block unsafe writes
L5-C04	Workflow / scheduling state	Tenant / workload class	Pause admission; drain or resume safely
L5-C05	Workflow / scheduling state	Tenant / workload class	Pause admission; drain or resume safely
L5-C06	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L5-C07	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L5-C08	Metrics + fitness decisions	Artifact / product / window	Expose indeterminate quality; constrain use
L5-C09	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract

Interface implementation profile

ID	INTERFACE	STYLE	TRANSPORT PROFILE	RELIABILITY CONTRACT
L5-I01	CreateTwinDefinition	Command	Typed asynchronous or bounded synchronous command with preconditions	Idempotent acceptance and one authoritative effect
L5-I02	ReconcileTwin	Command	Typed asynchronous or bounded synchronous command with preconditions	Idempotent acceptance and one authoritative effect
L5-I03	BranchTwin	Command	Typed asynchronous or bounded synchronous command with preconditions	Idempotent acceptance and one authoritative effect
L5-I04	TwinStateChanged	Domain event	Immutable versioned domain event through ordered partitioned transport	At-least-once delivery; consumer deduplication and checkpoint
L5-I05	ValidateTwin	Command / event	Immutable versioned domain event through ordered partitioned transport	At-least-once delivery; consumer deduplication and checkpoint

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-35-001	Digital Twins SHALL own and version only its declared authoritative state: Twin definitions and owners; Observed-state snapshots; Synthetic branches; Model and assumption bindings; Validation, confidence, and reconciliation history.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.
ES-35-002	Every logical component in Digital Twins SHALL publish a service manifest with component ID, owner, contracts, state class, dependencies, controls, SLOs, failure domains, and recovery procedure.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-35-003	Deployable units implementing Digital Twins SHALL preserve component-level telemetry, policy enforcement, state access, resource use, and failure isolation even when components share a runtime.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-35-004	All Digital Twins interfaces SHALL implement the canonical envelope, error model, compatibility rules, idempotency or ordering profile, audit causation, and consumer conformance fixtures.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-35-005	Inputs to Digital Twins SHALL be authenticated, authorized, schema-valid, temporally explicit, provenance-bearing, quality-marked, and admitted before decision-active use.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-35-006	Outputs from Digital Twins SHALL be immutable versioned canonical products or explicit proposals and SHALL preserve truth state, uncertainty, lineage, policy, quality, and lifecycle.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-35-007	Digital Twins SHALL enforce its declared trust, security, privacy, sovereignty, workflow, audit, evaluation, and human-authority controls at every material boundary.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.
ES-35-008	Digital Twins SHALL measure state freshness, estimation error, model fitness, coverage, reconciliation success, branch isolation, dependency completeness through governed indicators whose failed or indeterminate state constrains downstream use.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-35-009	Digital Twins SHALL implement and continuously test every declared degraded behavior without fabricating success, freshness, completeness, confidence, authorization, or approval.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-35-010	Digital Twins SHALL pass the same object, contract, control, failure, recovery, export, and replay conformance fixtures in SaaS, Private Cloud, and On-Premise deployments.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.

Control enforcement

- Observed and synthetic state separation
- Exact evidence and graph revision per snapshot
- Versioned model and assumption bindings
- Declared operating envelope
- Simulation preserves the exact twin version used

Degraded behavior and acceptance

DECLARED DEGRADED BEHAVIOR	REQUIRED ENGINEERING PROOF
Freeze last validated snapshot with freshness warning	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.
Disable behaviors outside calibrated envelope	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.
Require reconciliation before merging a branch	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.
Expose missing variables and dependency uncertainty	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.
Prevent synthetic branch leakage into observed state	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.

LAYER ACCEPTANCE

Digital Twins is accepted only when its 9 logical components, 5 canonical interfaces, authoritative state, controls, quality dimensions, degraded modes, and deployment-parity fixtures remain independently testable and operationally owned.

CONSTITUTIONAL INHERITANCE / C-010 · C-011 · C-018 · C-019 · C-023 · C-035 · C-043

36 Prediction Engine

Estimate future variables, events, and system states across the canonical 30-day, 90-day, 180-day, 1-year, 3-year, and 5-year horizons.

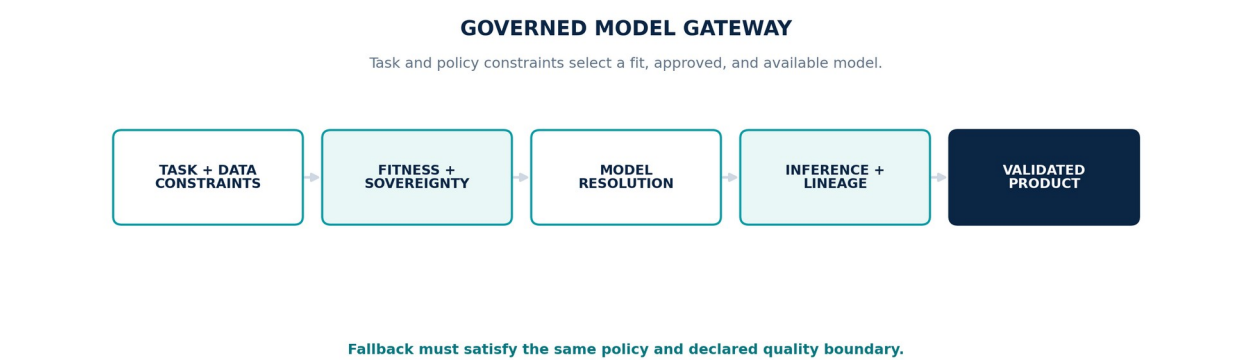


Figure 21. Governed Model Gateway. Task and policy constraints select a fit, approved, and available model.

Layer implementation contract

CONTRACT	ENGINEERING DEFINITION
Layer authority	Estimate future variables, events, and system states across the canonical 30-day, 90-day, 180-day, 1-year, 3-year, and 5-year horizons.
Boundary	Owns forecast definitions, horizon methods, feature context, model routing, ensembles, calibration, uncertainty, sensitivity, evaluation, drift, and forecast versions. It does not claim certainty or select institutional action.
Authoritative state	Forecast definitions; Feature and model bindings; Forecast versions; Calibration and backtest history; Drift, expiry, and withdrawal state.
Primary inputs	Governed evidence and graph context; Twin snapshots; Forecast definitions and horizons; Approved models, features, and evaluation thresholds.
Canonical outputs	Forecast Packages; Probability distributions and confidence; Drivers and sensitivity; Calibration and fitness state; Refresh triggers and failure conditions.
Quality dimensions	calibration, proper scoring rules, coverage, bias, sharpness, stability, drift, latency, cost, decision relevance.

Logical component realization

ID	COMPONENT	IMPLEMENTATION DUTY	RUNTIME PATTERN
L6-C01	Forecast Registry and Horizon Service	Owns versioned forecast registry and horizon definitions, resolution, compatibility, stewardship, and lifecycle events.	Registry + resolver
L6-C02	Feature and Context Assembler	Implements the feature and context assembler capability within Prediction Engine using versioned contracts, bounded state, controls, telemetry, and explicit failure behavior.	Owned service
L6-C03	Model Gateway and Candidate Router	Mediates model gateway and candidate router through authenticated, policy-evaluated, typed contracts and isolates external semantics.	Stateless gateway + durable checkpoints
L6-C04	Ensemble and Disagreement Manager	Coordinates ensemble and disagreement admission, ordering, budgets, retries, escalation, completion, and observable state.	Coordinator + durable queue
L6-C05	Calibration and Uncertainty Service	Measures calibration and uncertainty fitness, publishes product quality state, and initiates constraint or remediation workflow.	Stream + evaluation worker
L6-C06	Driver Attribution and Sensitivity Engine	Executes the governed driver attribution and sensitivity method against exact inputs and versions and returns uncertainty, quality, and lineage.	Horizontally scaled worker
L6-C07	Forecast Evaluation and Backtesting	Measures forecast evaluation and backtesting fitness, publishes product quality state, and initiates constraint or remediation workflow.	Stream + evaluation worker
L6-C08	Drift and Refresh Monitor	Measures drift and refresh monitor fitness, publishes product quality state, and initiates constraint or remediation workflow.	Stream + evaluation worker

ID	COMPONENT	IMPLEMENTATION DUTY	RUNTIME PATTERN
L6-C09	Forecast Version Store	Owens durable forecast version store state with version, policy, integrity, retention, correction, and recovery.	Canonical state service
ID	STATE CLASS	PRIMARY SCALE KEY	FAILURE AND RECOVERY
L6-C01	Canonical control state	Domain / namespace	Last verified read; block unsafe writes
L6-C02	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L6-C03	Checkpoint / quarantine	Source / destination / tenant	Backpressure, quarantine, bounded retry
L6-C04	Workflow / scheduling state	Tenant / workload class	Pause admission; drain or resume safely
L6-C05	Metrics + fitness decisions	Artifact / product / window	Expose indeterminate quality; constrain use
L6-C06	Immutable result + checkpoint	Tenant / object / job	Abstain or partial result; never fabricate completion
L6-C07	Metrics + fitness decisions	Artifact / product / window	Expose indeterminate quality; constrain use
L6-C08	Metrics + fitness decisions	Artifact / product / window	Expose indeterminate quality; constrain use
L6-C09	Authoritative	Tenant / object partition	Restore, reconcile, then reopen writes

Interface implementation profile

ID	INTERFACE	STYLE	TRANSPORT PROFILE	RELIABILITY CONTRACT
L6-I01	RequestForecast	Command	Typed asynchronous or bounded synchronous command with preconditions	Idempotent acceptance and one authoritative effect
L6-I02	ForecastProduced	Domain event	Immutable versioned domain event through ordered partitioned transport	At-least-once delivery; consumer deduplication and checkpoint
L6-I03	ForecastFitnessChanged	Domain event	Immutable versioned domain event through ordered partitioned transport	At-least-once delivery; consumer deduplication and checkpoint
L6-I04	BacktestForecast	Command	Typed asynchronous or bounded synchronous command with preconditions	Idempotent acceptance and one authoritative effect
L6-I05	ForecastWithdrawn	Domain event	Immutable versioned domain event through ordered partitioned transport	At-least-once delivery; consumer deduplication and checkpoint

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-36-001	Prediction Engine SHALL own and version only its declared authoritative state: Forecast definitions; Feature and model bindings; Forecast versions; Calibration and backtest history; Drift, expiry, and withdrawal state.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-36-002	Every logical component in Prediction Engine SHALL publish a service manifest with component ID, owner, contracts, state class, dependencies, controls, SLOs, failure domains, and recovery procedure.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-36-003	Deployable units implementing Prediction Engine SHALL preserve component-level telemetry, policy enforcement, state access, resource use, and failure isolation even when components share a runtime.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-36-004	All Prediction Engine interfaces SHALL implement the canonical envelope, error model, compatibility rules, idempotency or ordering profile, audit causation, and consumer conformance fixtures.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-36-005	Inputs to Prediction Engine SHALL be authenticated, authorized, schema-valid, temporally explicit, provenance-bearing, quality-marked, and admitted before decision-active use.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-36-006	Outputs from Prediction Engine SHALL be immutable versioned canonical products or explicit proposals and SHALL preserve truth state, uncertainty, lineage, policy, quality, and lifecycle.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-36-007	Prediction Engine SHALL enforce its declared trust, security, privacy, sovereignty, workflow, audit, evaluation, and human-authority controls at every material boundary.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.
ES-36-008	Prediction Engine SHALL measure calibration, proper scoring rules, coverage, bias, sharpness, stability, drift, latency, cost, decision relevance through governed indicators whose failed or indeterminate state constrains downstream use.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-36-009	Prediction Engine SHALL implement and continuously test every declared degraded behavior without fabricating success, freshness, completeness, confidence, authorization, or approval.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-36-010	Prediction Engine SHALL pass the same object, contract, control, failure, recovery, export, and replay conformance fixtures in SaaS, Private Cloud, and On-Premise deployments.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.

Control enforcement

- Horizon-specific method selection
- Distribution rather than point estimate by default
- Model and feature lineage
- Calibration and failure conditions
- Alternative models and disagreement remain inspectable

Degraded behavior and acceptance

DECLARED DEGRADED BEHAVIOR	REQUIRED ENGINEERING PROOF
Abstain or widen uncertainty when fitness fails	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.
Serve prior forecast only with explicit expiry state	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.
Exclude unavailable model paths and disclose loss	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.
Prevent warnings from treating withdrawn forecasts as current	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.
Route material failures to scenario and decision owners	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.

LAYER ACCEPTANCE

Prediction Engine is accepted only when its 9 logical components, 5 canonical interfaces, authoritative state, controls, quality dimensions, degraded modes, and deployment-parity fixtures remain independently testable and operationally owned.

CONSTITUTIONAL INHERITANCE / C-005 · C-020 · C-021 · C-030 · C-045 · C-049

37 Scenario Intelligence

Maintain a coherent, monitored portfolio of multiple plausible futures under uncertainty without allowing one forecast to masquerade as destiny.

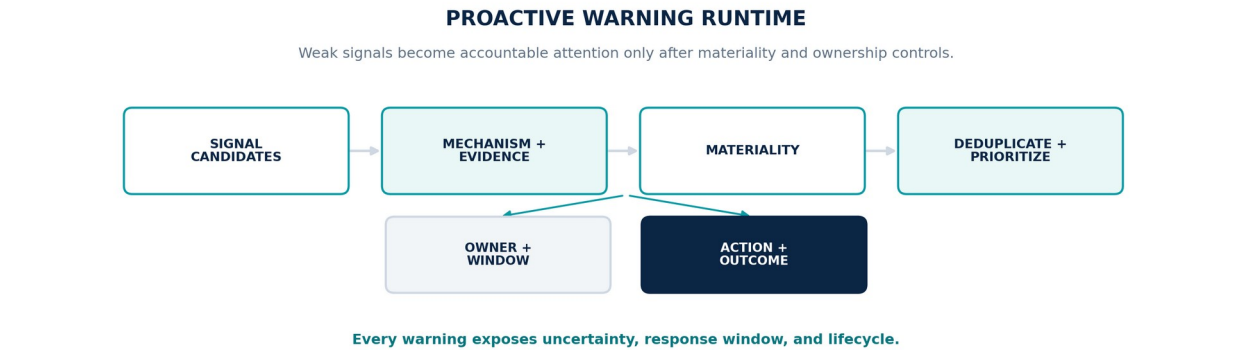


Figure 22. Proactive Warning Runtime. Weak signals become accountable attention only after materiality and ownership controls.

Layer implementation contract

CONTRACT	ENGINEERING DEFINITION
Layer authority	Maintain a coherent, monitored portfolio of multiple plausible futures under uncertainty without allowing one forecast to masquerade as destiny.
Boundary	Owns scenario definitions, assumptions, actors, drivers, branches, indicators, signposts, interventions, coherence, comparison, review cadence, and portfolio state. It does not execute simulations or approve decisions.
Authoritative state	Scenario definitions and branches; Assumptions, actors, drivers, and mechanisms; Indicators and signposts; Coherence and confidence state; Review and retirement history.
Primary inputs	Forecast Packages; Twin snapshots and graph context; Objectives, assumptions, risks, and opportunities; Human and agent hypotheses.
Canonical outputs	Scenario branches and portfolios; Assumption registers; Indicators and signposts; Branch confidence and coherence state; Simulation and decision context.
Quality dimensions	assumption coverage, internal coherence, branch diversity, indicator freshness, signpost discrimination, review timeliness, decision coverage.

Logical component realization

ID	COMPONENT	IMPLEMENTATION DUTY	RUNTIME PATTERN
L7-C01	Scenario Registry and Portfolio Manager	Owns versioned scenario registry and portfolio definitions, resolution, compatibility, stewardship, and lifecycle events.	Registry + resolver
L7-C02	Assumption and Hypothesis Register	Implements the assumption and hypothesis register capability within Scenario Intelligence using versioned contracts, bounded state, controls, telemetry, and explicit failure behavior.	Owned service
L7-C03	Branch and Path Engine	Executes the governed branch and path method against exact inputs and versions and returns uncertainty, quality, and lineage.	Horizontally scaled worker
L7-C04	Actor, Driver, and Mechanism Model	Implements the actor, driver, and mechanism model capability within Scenario Intelligence using versioned contracts, bounded state, controls, telemetry, and explicit failure behavior.	Owned service
L7-C05	Indicator and Signpost Service	Implements the indicator and signpost service capability within Scenario Intelligence using versioned contracts, bounded state, controls, telemetry, and explicit failure behavior.	Owned service
L7-C06	Scenario Coherence Validator	Implements the scenario coherence validator capability within Scenario Intelligence using versioned contracts, bounded state, controls, telemetry, and explicit failure behavior.	Owned service
L7-C07	Intervention and Impact Mapper	Implements the intervention and impact mapper capability within Scenario Intelligence using versioned contracts, bounded state, controls, telemetry, and explicit failure behavior.	Owned service

ID	COMPONENT	IMPLEMENTATION DUTY	RUNTIME PATTERN
L7-C08	Scenario Comparison and Review Workspace	Coordinates durable scenario comparison and review workspace transitions, human tasks, deadlines, separation, compensation, and audit evidence.	Durable workflow
L7-C09	Scenario Package Import and Localization	Implements the scenario package import and localization capability within Scenario Intelligence using versioned contracts, bounded state, controls, telemetry, and explicit failure behavior.	Owned service
ID	STATE CLASS	PRIMARY SCALE KEY	FAILURE AND RECOVERY
L7-C01	Canonical control state	Domain / namespace	Last verified read; block unsafe writes
L7-C02	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L7-C03	Immutable result + checkpoint	Tenant / object / job	Abstain or partial result; never fabricate completion
L7-C04	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L7-C05	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L7-C06	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L7-C07	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L7-C08	Workflow history	Instance / tenant	Resume from last commit; escalate residual effect
L7-C09	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract

Interface implementation profile

ID	INTERFACE	STYLE	TRANSPORT PROFILE	RELIABILITY CONTRACT
L7-I01	CreateScenario	Command	Typed asynchronous or bounded synchronous command with preconditions	Idempotent acceptance and one authoritative effect
L7-I02	BranchScenario	Command	Typed asynchronous or bounded synchronous command with preconditions	Idempotent acceptance and one authoritative effect
L7-I03	ScenarioIndicatorChanged	Domain event	Immutable versioned domain event through ordered partitioned transport	At-least-once delivery; consumer deduplication and checkpoint
L7-I04	ScenarioCoherenceFailed	Domain event	Immutable versioned domain event through ordered partitioned transport	At-least-once delivery; consumer deduplication and checkpoint
L7-I05	ScenarioReviewed	Workflow event	Immutable versioned domain event through ordered partitioned transport	At-least-once delivery; consumer deduplication and checkpoint

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-37-001	Scenario Intelligence SHALL own and version only its declared authoritative state: Scenario definitions and branches; Assumptions, actors, drivers, and mechanisms; Indicators and signposts; Coherence and confidence state; Review and retirement history.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-37-002	Every logical component in Scenario Intelligence SHALL publish a service manifest with component ID, owner, contracts, state class, dependencies, controls, SLOs, failure domains, and recovery procedure.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-37-003	Deployable units implementing Scenario Intelligence SHALL preserve component-level telemetry, policy enforcement, state access, resource use, and failure isolation even when components share a runtime.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-37-004	All Scenario Intelligence interfaces SHALL implement the canonical envelope, error model, compatibility rules, idempotency or ordering profile, audit causation, and consumer conformance fixtures.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-37-005	Inputs to Scenario Intelligence SHALL be authenticated, authorized, schema-valid, temporally explicit, provenance-bearing, quality-marked, and admitted before decision-active use.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-37-006	Outputs from Scenario Intelligence SHALL be immutable versioned canonical products or explicit proposals and SHALL preserve truth state, uncertainty, lineage, policy, quality, and lifecycle.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-37-007	Scenario Intelligence SHALL enforce its declared trust, security, privacy, sovereignty, workflow, audit, evaluation, and human-authority controls at every material boundary.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-37-008	Scenario Intelligence SHALL measure assumption coverage, internal coherence, branch diversity, indicator freshness, signpost discrimination, review timeliness, decision coverage through governed indicators whose failed or indeterminate state constrains downstream use.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-37-009	Scenario Intelligence SHALL implement and continuously test every declared degraded behavior without fabricating success, freshness, completeness, confidence, authorization, or approval.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-37-010	Scenario Intelligence SHALL pass the same object, contract, control, failure, recovery, export, and replay conformance fixtures in SaaS, Private Cloud, and On-Premise deployments.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.

Control enforcement

- Multiple branches remain visible
- Assumptions are explicit and versioned
- Evidence and forecast dependencies are linked
- Indicators have owners and thresholds
- Imported packages cannot access customer memory by default

Degraded behavior and acceptance

DECLARED DEGRADED BEHAVIOR	REQUIRED ENGINEERING PROOF
Suspend a branch when critical assumptions become invalid	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.
Expose missing indicators and stale dependencies	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.
Prevent incoherent branches from entering simulation	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.
Retain retired branches for Decision Replay	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.
Route material branch movement to accountable owners	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.

LAYER ACCEPTANCE

Scenario Intelligence is accepted only when its 9 logical components, 5 canonical interfaces, authoritative state, controls, quality dimensions, degraded modes, and deployment-parity fixtures remain independently testable and operationally owned.

CONSTITUTIONAL INHERITANCE / C-005 · C-022 · C-023 · C-032 · C-033 · C-046

38 Simulation Engine

Run reproducible experiments against declared twin versions, scenarios, assumptions, interventions, models, constraints, and seeds before institutional commitment.

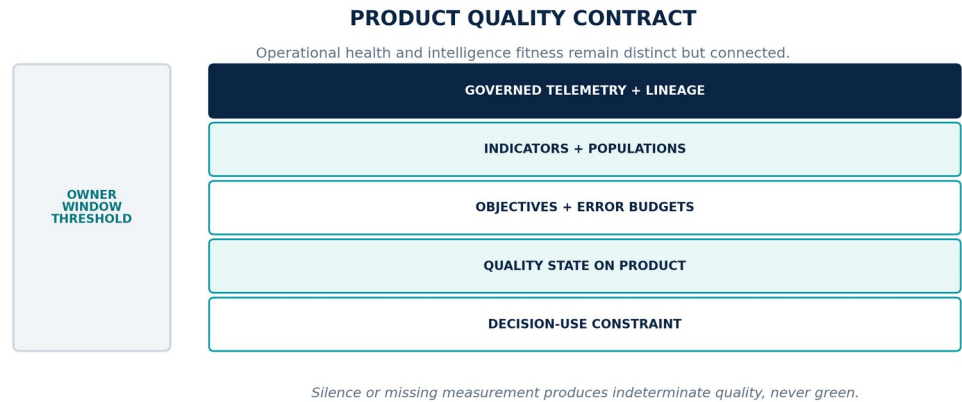


Figure 23. Product Quality Contract. Operational health and intelligence fitness remain distinct but connected.

Layer implementation contract

CONTRACT	ENGINEERING DEFINITION
Layer authority	Run reproducible experiments against declared twin versions, scenarios, assumptions, interventions, models, constraints, and seeds before institutional commitment.
Boundary	Owns experiment definitions, run orchestration, version resolution, constraints, solver adapters, impact analysis, sensitivity, validation, and reproducibility evidence. It does not silently alter production state or approve action.
Authoritative state	Experiment definitions; Resolved run manifests; Execution and resource state; Outputs and sensitivity; Validation, challenge, and invalidation history.
Primary inputs	Versioned twin and scenario branches; Interventions and constraints; Approved models and solver adapters; Run budgets, policies, and validation criteria.
Canonical outputs	Simulation Runs; Direct and second-order impacts; Sensitivity and uncertainty; Reproducibility package; Validation and challenge record.
Quality dimensions	reproducibility, numerical stability, constraint satisfaction, sensitivity coverage, validation status, run latency, cost, failure containment.

Logical component realization

ID	COMPONENT	IMPLEMENTATION DUTY	RUNTIME PATTERN
L8-C01	Experiment Registry and Run Planner	Owns versioned experiment registry and run planner definitions, resolution, compatibility, stewardship, and lifecycle events.	Registry + resolver
L8-C02	Twin, Scenario, and Model Version Resolver	Executes the governed twin, scenario, and model version resolver method against exact inputs and versions and returns uncertainty, quality, and lineage.	Horizontally scaled worker
L8-C03	Constraint and Intervention Engine	Executes the governed constraint and intervention method against exact inputs and versions and returns uncertainty, quality, and lineage.	Horizontally scaled worker
L8-C04	Seed and Determinism Manager	Coordinates seed and determinism admission, ordering, budgets, retries, escalation, completion, and observable state.	Coordinator + durable queue
L8-C05	Solver and Simulator Adapter Fabric	Mediates solver and simulator adapter fabric through authenticated, policy-evaluated, typed contracts and isolates external semantics.	Stateless gateway + durable checkpoints
L8-C06	Distributed Run Orchestrator	Coordinates distributed run orchestrator admission, ordering, budgets, retries, escalation, completion, and observable state.	Coordinator + durable queue
L8-C07	Impact and Second-Order Effect Analyzer	Executes the governed impact and second-order effect analyzer method against exact inputs and versions and returns uncertainty, quality, and lineage.	Horizontally scaled worker

ID	COMPONENT	IMPLEMENTATION DUTY	RUNTIME PATTERN
L8-C08	Sensitivity and Robustness Service	Implements the sensitivity and robustness service capability within Simulation Engine using versioned contracts, bounded state, controls, telemetry, and explicit failure behavior.	Owned service
L8-C09	Reproducibility and Validation Store	Owns durable reproducibility and validation store state with version, policy, integrity, retention, correction, and recovery.	Canonical state service
ID	STATE CLASS	PRIMARY SCALE KEY	FAILURE AND RECOVERY
L8-C01	Canonical control state	Domain / namespace	Last verified read; block unsafe writes
L8-C02	Immutable result + checkpoint	Tenant / object / job	Abstain or partial result; never fabricate completion
L8-C03	Immutable result + checkpoint	Tenant / object / job	Abstain or partial result; never fabricate completion
L8-C04	Workflow / scheduling state	Tenant / workload class	Pause admission; drain or resume safely
L8-C05	Checkpoint / quarantine	Source / destination / tenant	Backpressure, quarantine, bounded retry
L8-C06	Workflow / scheduling state	Tenant / workload class	Pause admission; drain or resume safely
L8-C07	Immutable result + checkpoint	Tenant / object / job	Abstain or partial result; never fabricate completion
L8-C08	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L8-C09	Authoritative	Tenant / object partition	Restore, reconcile, then reopen writes

Interface implementation profile

ID	INTERFACE	STYLE	TRANSPORT PROFILE	RELIABILITY CONTRACT
L8-I01	SubmitExperiment	Command	Typed asynchronous or bounded synchronous command with preconditions	Idempotent acceptance and one authoritative effect
L8-I02	SimulationStarted	Domain event	Immutable versioned domain event through ordered partitioned transport	At-least-once delivery; consumer deduplication and checkpoint
L8-I03	SimulationCompleted	Domain event	Immutable versioned domain event through ordered partitioned transport	At-least-once delivery; consumer deduplication and checkpoint
L8-I04	ChallengeSimulation	Workflow command	Typed asynchronous or bounded synchronous command with preconditions	Idempotent acceptance and one authoritative effect
L8-I05	SimulationInvalidated	Domain event	Immutable versioned domain event through ordered partitioned transport	At-least-once delivery; consumer deduplication and checkpoint

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-38-001	Simulation Engine SHALL own and version only its declared authoritative state: Experiment definitions; Resolved run manifests; Execution and resource state; Outputs and sensitivity; Validation, challenge, and invalidation history.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-38-002	Every logical component in Simulation Engine SHALL publish a service manifest with component ID, owner, contracts, state class, dependencies, controls, SLOs, failure domains, and recovery procedure.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-38-003	Deployable units implementing Simulation Engine SHALL preserve component-level telemetry, policy enforcement, state access, resource use, and failure isolation even when components share a runtime.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-38-004	All Simulation Engine interfaces SHALL implement the canonical envelope, error model, compatibility rules, idempotency or ordering profile, audit causation, and consumer conformance fixtures.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-38-005	Inputs to Simulation Engine SHALL be authenticated, authorized, schema-valid, temporally explicit, provenance-bearing, quality-marked, and admitted before decision-active use.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-38-006	Outputs from Simulation Engine SHALL be immutable versioned canonical products or explicit proposals and SHALL preserve truth state, uncertainty, lineage, policy, quality, and lifecycle.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-38-007	Simulation Engine SHALL enforce its declared trust, security, privacy, sovereignty, workflow, audit, evaluation, and human-authority controls at every material boundary.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-38-008	Simulation Engine SHALL measure reproducibility, numerical stability, constraint satisfaction, sensitivity coverage, validation status, run latency, cost, failure containment through governed indicators whose failed or indeterminate state constrains downstream use.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-38-009	Simulation Engine SHALL implement and continuously test every declared degraded behavior without fabricating success, freshness, completeness, confidence, authorization, or approval.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-38-010	Simulation Engine SHALL pass the same object, contract, control, failure, recovery, export, and replay conformance fixtures in SaaS, Private Cloud, and On-Premise deployments.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.

Control enforcement

- Exact model, twin, scenario, and policy versions
- Seed and environment capture
- Resource budgets and stop conditions
- No write-back to observed state
- Reproducibility package accompanies material results

Degraded behavior and acceptance

DECLARED DEGRADED BEHAVIOR	REQUIRED ENGINEERING PROOF
Fail closed when required versions cannot resolve	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.
Return partial runs only with declared missing outputs	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.
Contain unstable solver adapters	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.
Preserve checkpoints and audit continuity	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.
Prevent invalid results from remaining decision-active	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.

LAYER ACCEPTANCE

Simulation Engine is accepted only when its 9 logical components, 5 canonical interfaces, authoritative state, controls, quality dimensions, degraded modes, and deployment-parity fixtures remain independently testable and operationally owned.

CONSTITUTIONAL INHERITANCE / C-005 · C-018 · C-019 · C-023 · C-035 · C-043 · C-045

39 Decision Intelligence

Assemble complete, explainable choice packages and compare viable alternatives against objectives, constraints, obligations, risk, opportunity, uncertainty, reversibility, and information value.

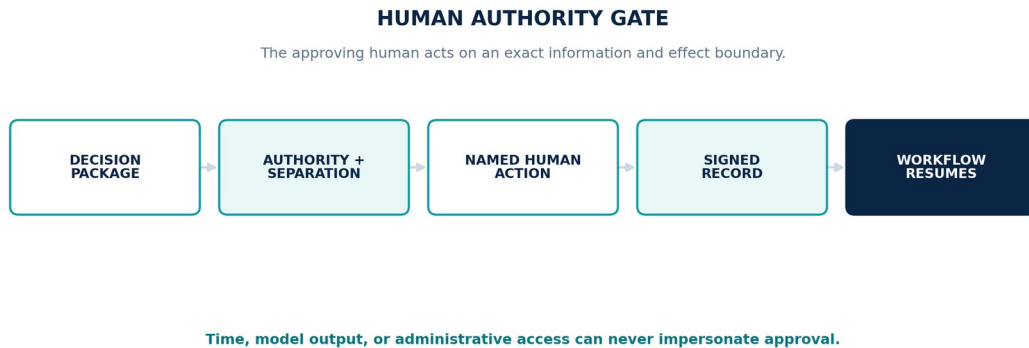


Figure 24. Human Authority Gate. The approving human acts on an exact information and effect boundary.

Layer implementation contract

CONTRACT	ENGINEERING DEFINITION
Layer authority	Assemble complete, explainable choice packages and compare viable alternatives against objectives, constraints, obligations, risk, opportunity, uncertainty, reversibility, and information value.
Boundary	Owns decision definitions, option context, evidence assembly, criteria, trade-offs, recommendation, dissent, approval routing, decision records, commitments, and monitoring contracts. A named human authority makes the final decision.
Authoritative state	Decision definitions and owners; Option and criteria versions; Decision package revisions; Approval, dissent, and override state; Final decisions, commitments, and monitoring contracts.
Primary inputs	Objectives and constraints; Evidence, assessments, forecasts, scenarios, and simulations; Options and stakeholder obligations; Risk appetite, policy, and approval rules.
Canonical outputs	Decision Packages; Recommendations and alternatives; Approvals, dissent, overrides, and final decision record; Commitments and monitoring contracts; Decision Replay state.
Quality dimensions	decision completeness, evidence coverage, alternative coverage, time-to-decision, reversibility, approval integrity, monitoring completeness, outcome linkage.

Logical component realization

ID	COMPONENT	IMPLEMENTATION DUTY	RUNTIME PATTERN
L9-C01	Decision Registry and Context Resolver	Owns versioned decision registry and context resolver definitions, resolution, compatibility, stewardship, and lifecycle events.	Registry + resolver
L9-C02	Objective, Constraint, and Obligation Service	Implements the objective, constraint, and obligation service capability within Decision Intelligence using versioned contracts, bounded state, controls, telemetry, and explicit failure behavior.	Owned service
L9-C03	Option and Alternative Manager	Coordinates option and alternative admission, ordering, budgets, retries, escalation, completion, and observable state.	Coordinator + durable queue
L9-C04	Evidence and Foresight Assembler	Implements the evidence and foresight assembler capability within Decision Intelligence using versioned contracts, bounded state, controls, telemetry, and explicit failure behavior.	Owned service
L9-C05	Criteria, Trade-off, and Information-Value Engine	Executes the governed criteria, trade-off, and information-value method against exact inputs and versions and returns uncertainty, quality, and lineage.	Horizontally scaled worker
L9-C06	Recommendation and Explanation Service	Implements the recommendation and explanation service capability within Decision Intelligence using versioned contracts, bounded state, controls, telemetry, and explicit failure behavior.	Owned service

ID	COMPONENT	IMPLEMENTATION DUTY	RUNTIME PATTERN
L9-C07	Approval, Dissent, and Override Workflow	Coordinates durable approval, dissent, and override workflow transitions, human tasks, deadlines, separation, compensation, and audit evidence.	Durable workflow
L9-C08	Decision Record and Commitment Service	Coordinates durable decision record and commitment transitions, human tasks, deadlines, separation, compensation, and audit evidence.	Durable workflow
L9-C09	Monitoring Contract and Review Scheduler	Coordinates durable monitoring contract and review scheduler transitions, human tasks, deadlines, separation, compensation, and audit evidence.	Durable workflow
ID	STATE CLASS	PRIMARY SCALE KEY	FAILURE AND RECOVERY
L9-C01	Canonical control state	Domain / namespace	Last verified read; block unsafe writes
L9-C02	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L9-C03	Workflow / scheduling state	Tenant / workload class	Pause admission; drain or resume safely
L9-C04	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L9-C05	Immutable result + checkpoint	Tenant / object / job	Abstain or partial result; never fabricate completion
L9-C06	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L9-C07	Workflow history	Instance / tenant	Resume from last commit; escalate residual effect
L9-C08	Workflow history	Instance / tenant	Resume from last commit; escalate residual effect
L9-C09	Workflow history	Instance / tenant	Resume from last commit; escalate residual effect

Interface implementation profile

ID	INTERFACE	STYLE	TRANSPORT PROFILE	RELIABILITY CONTRACT
L9-I01	OpenDecision	Command	Typed asynchronous or bounded synchronous command with preconditions	Idempotent acceptance and one authoritative effect
L9-I02	DecisionPackageReady	Domain event	Immutable versioned domain event through ordered partitioned transport	At-least-once delivery; consumer deduplication and checkpoint
L9-I03	ApproveDecision	Human command	Authenticated command bound to human gate and request digest	Exactly-once institutional effect through idempotency and durable workflow
L9-I04	DecisionCommitted	Domain event	Immutable versioned domain event through ordered partitioned transport	At-least-once delivery; consumer deduplication and checkpoint
L9-I05	DecisionReopened	Workflow event	Immutable versioned domain event through ordered partitioned transport	At-least-once delivery; consumer deduplication and checkpoint

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-39-001	Decision Intelligence SHALL own and version only its declared authoritative state: Decision definitions and owners; Option and criteria versions; Decision package revisions; Approval, dissent, and override state; Final decisions, commitments, and monitoring contracts.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.
ES-39-002	Every logical component in Decision Intelligence SHALL publish a service manifest with component ID, owner, contracts, state class, dependencies, controls, SLOs, failure domains, and recovery procedure.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-39-003	Deployable units implementing Decision Intelligence SHALL preserve component-level telemetry, policy enforcement, state access, resource use, and failure isolation even when components share a runtime.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-39-004	All Decision Intelligence interfaces SHALL implement the canonical envelope, error model, compatibility rules, idempotency or ordering profile, audit causation, and consumer conformance fixtures.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-39-005	Inputs to Decision Intelligence SHALL be authenticated, authorized, schema-valid, temporally explicit, provenance-bearing, quality-marked, and admitted before decision-active use.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-39-006	Outputs from Decision Intelligence SHALL be immutable versioned canonical products or explicit proposals and SHALL preserve truth state, uncertainty, lineage, policy, quality, and lifecycle.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-39-007	Decision Intelligence SHALL enforce its declared trust, security, privacy, sovereignty, workflow, audit, evaluation, and human-authority controls at every material boundary.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.
ES-39-008	Decision Intelligence SHALL measure decision completeness, evidence coverage, alternative coverage, time-to-decision, reversibility, approval integrity, monitoring completeness, outcome linkage through governed indicators whose failed or indeterminate state constrains downstream use.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.
ES-39-009	Decision Intelligence SHALL implement and continuously test every declared degraded behavior without fabricating success, freshness, completeness, confidence, authorization, or approval.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-39-010	Decision Intelligence SHALL pass the same object, contract, control, failure, recovery, export, and replay conformance fixtures in SaaS, Private Cloud, and On-Premise deployments.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.

Control enforcement

- Named human authority
- Separation of preparation and approval
- Alternatives and dissent retained
- Recommendation criteria remain inspectable
- State of knowledge at decision time is immutable for replay

Degraded behavior and acceptance

DECLARED DEGRADED BEHAVIOR	REQUIRED ENGINEERING PROOF
Block commitment when mandatory evidence or approval is missing	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.
Expose stale or withdrawn inputs	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.
Allow human deferral, rejection, or request for information	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.
Prevent agent self-approval	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.
Reopen rather than silently rewriting a committed decision	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.

LAYER ACCEPTANCE

Decision Intelligence is accepted only when its 9 logical components, 5 canonical interfaces, authoritative state, controls, quality dimensions, degraded modes, and deployment-parity fixtures remain independently testable and operationally owned.

CONSTITUTIONAL INHERITANCE / C-004 · C-005 · C-024 · C-025 · C-026 · C-036 · C-050

40 Executive Operating System

Govern strategic attention, responsibility, review, commitments, scenarios, decisions, material change, and outcomes across institutional cadence.

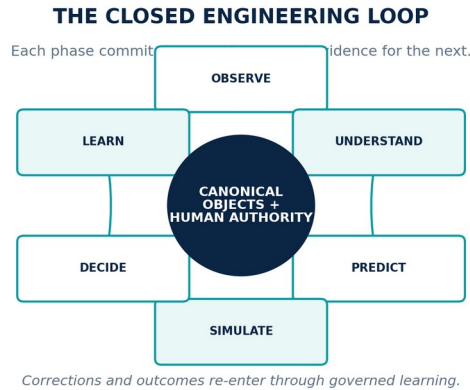


Figure 25. The Closed Engineering Loop. Each phase commits governed state and evidence for the next.

Layer implementation contract

CONTRACT	ENGINEERING DEFINITION
Layer authority	Govern strategic attention, responsibility, review, commitments, scenarios, decisions, material change, and outcomes across institutional cadence.
Boundary	Owns executive workspaces, attention policy, briefing products, decision rooms, responsibility views, strategic cadence, notification, and access modes. It is not a passive dashboard and does not create separate truths for different roles.
Authoritative state	Attention and materiality policy; Briefing definitions and versions; Workspace and responsibility configuration; Review cadence and agenda state; Notification, suppression, delegation, and acknowledgement history.
Primary inputs	Material changes and warnings; Objectives, risks, opportunities, and scenarios; Pending decisions and commitments; Monitoring, outcomes, and review cadence.
Canonical outputs	Executive Briefings; Decision and review rooms; Prioritized attention queues; Commitment and outcome reviews; Strategic cadence and institutional state.
Quality dimensions	time-to-understanding, decision latency, attention precision, warning acknowledgement, review completion, commitment visibility, accessibility, explanation use.

Logical component realization

ID	COMPONENT	IMPLEMENTATION DUTY	RUNTIME PATTERN
L10-C01	Executive Workspace and Responsibility Model	Presents policy-filtered executive workspace and responsibility model with evidence, uncertainty, authority, state, and required action intact.	Read model + command façade
L10-C02	Attention Policy and Materiality Engine	Executes the governed attention policy and materiality method against exact inputs and versions and returns uncertainty, quality, and lineage.	Horizontally scaled worker
L10-C03	Live Briefing Service	Presents policy-filtered live briefing with evidence, uncertainty, authority, state, and required action intact.	Read model + command façade
L10-C04	Decision and Scenario Rooms	Presents policy-filtered decision and scenario rooms with evidence, uncertainty, authority, state, and required action intact.	Read model + command façade
L10-C05	Objective, Commitment, and Outcome Review	Coordinates durable objective, commitment, and outcome review transitions, human tasks, deadlines, separation, compensation, and audit evidence.	Durable workflow
L10-C06	Strategic Cadence Scheduler	Coordinates strategic cadence scheduler admission, ordering, budgets, retries, escalation, completion, and observable state.	Coordinator + durable queue
L10-C07	Notification and Escalation Service	Implements the notification and escalation service capability within Executive Operating System using versioned contracts, bounded state, controls, telemetry, and explicit failure behavior.	Owned service

ID	COMPONENT	IMPLEMENTATION DUTY	RUNTIME PATTERN
L10-C08	Role-Appropriate Access Modes	Implements the role-appropriate access modes capability within Executive Operating System using versioned contracts, bounded state, controls, telemetry, and explicit failure behavior.	Owned service
L10-C09	Executive Search and Explanation	Implements the executive search and explanation capability within Executive Operating System using versioned contracts, bounded state, controls, telemetry, and explicit failure behavior.	Owned service
ID	STATE CLASS	PRIMARY SCALE KEY	FAILURE AND RECOVERY
L10-C01	Rebuildable projection	Tenant / audience / session	Stale-visible or read-only with watermark
L10-C02	Immutable result + checkpoint	Tenant / object / job	Abstain or partial result; never fabricate completion
L10-C03	Rebuildable projection	Tenant / audience / session	Stale-visible or read-only with watermark
L10-C04	Rebuildable projection	Tenant / audience / session	Stale-visible or read-only with watermark
L10-C05	Workflow history	Instance / tenant	Resume from last commit; escalate residual effect
L10-C06	Workflow / scheduling state	Tenant / workload class	Pause admission; drain or resume safely
L10-C07	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L10-C08	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L10-C09	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract

Interface implementation profile

ID	INTERFACE	STYLE	TRANSPORT PROFILE	RELIABILITY CONTRACT
L10-I01	PublishBriefing	Command	Typed asynchronous or bounded synchronous command with preconditions	Idempotent acceptance and one authoritative effect
L10-I02	MaterialChangeRaised	Domain event	Immutable versioned domain event through ordered partitioned transport	At-least-once delivery; consumer deduplication and checkpoint
L10-I03	ReviewConvened	Workflow event	Immutable versioned domain event through ordered partitioned transport	At-least-once delivery; consumer deduplication and checkpoint
L10-I04	ExecutiveActionRequest ed	Human command	Authenticated command bound to human gate and request digest	Exactly-once institutional effect through idempotency and durable workflow
L10-I05	AttentionPolicyChanged	Governance event	Immutable versioned domain event through ordered partitioned transport	At-least-once delivery; consumer deduplication and checkpoint

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-40-001	Executive Operating System SHALL own and version only its declared authoritative state: Attention and materiality policy; Briefing definitions and versions; Workspace and responsibility configuration; Review cadence and agenda state; Notification, suppression, delegation, and acknowledgement history.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-40-002	Every logical component in Executive Operating System SHALL publish a service manifest with component ID, owner, contracts, state class, dependencies, controls, SLOs, failure domains, and recovery procedure.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-40-003	Deployable units implementing Executive Operating System SHALL preserve component-level telemetry, policy enforcement, state access, resource use, and failure isolation even when components share a runtime.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-40-004	All Executive Operating System interfaces SHALL implement the canonical envelope, error model, compatibility rules, idempotency or ordering profile, audit causation, and consumer conformance fixtures.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-40-005	Inputs to Executive Operating System SHALL be authenticated, authorized, schema-valid, temporally explicit, provenance-bearing, quality-marked, and admitted before decision-active use.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-40-006	Outputs from Executive Operating System SHALL be immutable versioned canonical products or explicit proposals and SHALL preserve truth state, uncertainty, lineage, policy, quality, and lifecycle.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-40-007	Executive Operating System SHALL enforce its declared trust, security, privacy, sovereignty, workflow, audit, evaluation, and human-authority controls at every material boundary.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-40-008	Executive Operating System SHALL measure time-to-understanding, decision latency, attention precision, warning acknowledgement, review completion, commitment visibility, accessibility, explanation use through governed indicators whose failed or indeterminate state constrains downstream use.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-40-009	Executive Operating System SHALL implement and continuously test every declared degraded behavior without fabricating success, freshness, completeness, confidence, authorization, or approval.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-40-010	Executive Operating System SHALL pass the same object, contract, control, failure, recovery, export, and replay conformance fixtures in SaaS, Private Cloud, and On-Premise deployments.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.

Control enforcement

- Role and need-to-know access
- Materiality and notification policy
- Every displayed conclusion links to evidence and uncertainty
- Suppression and delegation remain visible
- Access mode never changes canonical meaning

Degraded behavior and acceptance

DECLARED DEGRADED BEHAVIOR	REQUIRED ENGINEERING PROOF
Expose incomplete briefings and unavailable dependencies	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.
Prefer silence over false certainty for unsafe products	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.
Retain urgent decision state during partial platform outage	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.
Preserve acknowledgement and escalation audit	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.
Allow controlled offline briefing packages where policy permits	Fault-injection trace, affected product watermark, consumer behavior, operator action, recovery, and reconciliation evidence.

LAYER ACCEPTANCE

Executive Operating System is accepted only when its 9 logical components, 5 canonical interfaces, authoritative state, controls, quality dimensions, degraded modes, and deployment-parity fixtures remain independently testable and operationally owned.

CONSTITUTIONAL INHERITANCE / C-003 · C-004 · C-027 · C-032 · C-033 · C-047 · C-048

PART VI

Runtime Intelligence Execution

Durable workflows, human gates, agents, models, evaluation, warnings, replay, and learning.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

41 Durable Workflow Runtime

Material intelligence work is executed as recoverable state, not as an in-memory chain of calls. The workflow runtime preserves intent, authority, progress, evidence, deadlines, compensation, and accountable completion across process, host, network, and deployment interruption.

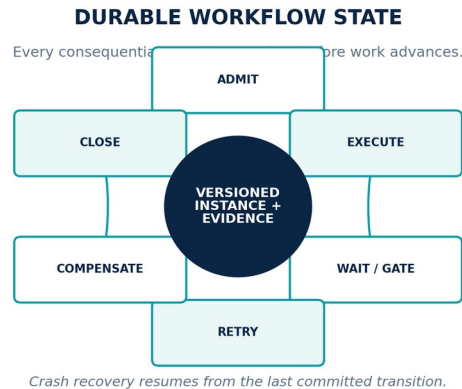


Figure 26. Durable Workflow State. Every consequential transition persists before work advances.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Workflow definition	Versioned states, transitions, guards, roles, timers, retries, compensations, evidence, and terminal conditions.
Workflow instance	Durable execution bound to tenant, purpose, initiating actor, definition version, business object, and correlation context.
Task	An idempotent unit of machine or human work with bounded inputs, side effects, deadlines, and completion evidence.
Terminal result	Completed, rejected, cancelled, compensated, expired, or failed-with-escalation; no silent or ambiguous completion state.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-41-001	Every consequential multi-step operation SHALL execute through a versioned durable workflow and persist each committed transition before acknowledging it.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-41-002	Workflow definitions SHALL identify owners, authorized transitions, timers, retry limits, compensation, human gates, cancellation semantics, and evidence required for closure.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-41-003	Task handlers SHALL be idempotent or expose an idempotency record and compensating action sufficient to prevent duplicate institutional effects.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-41-004	A running instance SHALL remain bound to its admitted definition version unless an explicit migration workflow validates and records the state transformation.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-41-005	Operational intervention SHALL use authenticated workflow commands and SHALL NOT mutate runtime persistence directly.	Operational exercise	Time-stamped exercise record with telemetry, operators, impact, recovery, and closure evidence.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Worker loss	Reassign from the last durable transition without repeating an acknowledged side effect.

FAILURE CONDITION	REQUIRED BEHAVIOR
Dependency unavailable	Enter a visible waiting or degraded state, preserve deadline policy, and escalate before consequence becomes hidden.
Compensation fails	Keep the instance non-terminal, identify residual effects, and route accountable remediation.

Acceptance evidence

- Crash-recovery tests interrupt every transition boundary and compare final state with an uninterrupted golden run.
- Duplicate-delivery tests prove task and effect idempotency under retry, timeout, and late acknowledgement.
- Definition-migration fixtures prove deterministic treatment of every active source state.
- Operator-access tests prove all interventions are authorized, audited, and reproducible from commands.

CHAPTER ACCEPTANCE

Durable Workflow Runtime is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-004 · C-011 · C-025 · C-029 · C-043 · C-049

42 Human Gates and Approvals

Human authority is implemented as a first-class runtime control. A human gate presents the exact decision context, verifies authority and separation of duties, records the accountable act, and resumes execution without allowing a model, agent, administrator, or timeout to impersonate consent.

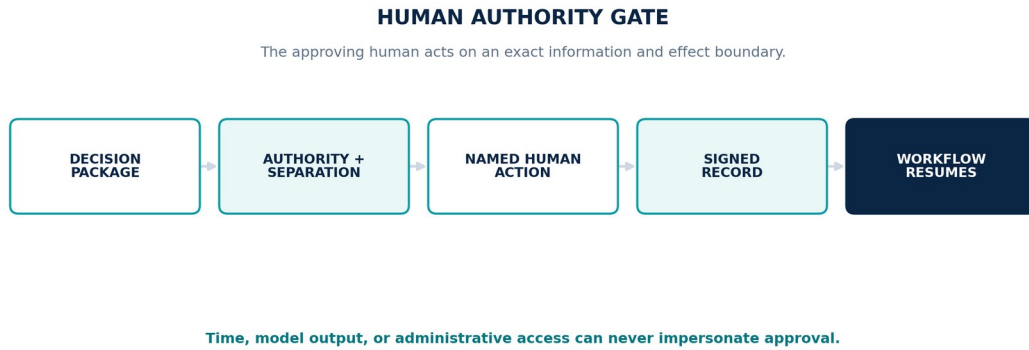


Figure 27. Human Authority Gate. The approving human acts on an exact information and effect boundary.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Gate definition	Required role, authority, quorum, separation, evidence, choices, deadline, escalation, revocation, and downstream effect.
Assignment	Resolved eligible principals and accountable owner at the moment the gate becomes actionable.
Decision record	Authenticated choice, rationale, information boundary, reservations, time, authority, and signature or equivalent proof.
Gate outcome	Approved, rejected, returned, abstained, expired, revoked, or escalated; each is distinct and policy governed.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-42-001	A consequential decision or irreversible institutional action SHALL require an explicit human gate owned by an authorized named role.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.
ES-42-002	The gate SHALL present evidence, uncertainty, alternatives, dissent, policy obligations, expected effects, reversibility, and missing information appropriate to the decision.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-42-003	Approval authority SHALL be evaluated at action time and SHALL NOT be inferred from task visibility, prior role, agent delegation, or administrative privilege.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.
ES-42-004	The approving principal SHALL be distinct from the proposing principal where separation-of-duties policy requires it.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-42-005	Expiry, delegation, reassignment, challenge, withdrawal, and emergency override SHALL be explicit transitions with independent audit evidence.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Approver unavailable	Escalate or reassign under policy; never convert elapsed time into approval.
Authority changed	Re-evaluate eligibility and invalidate an uncommitted action when authority no longer applies.

FAILURE CONDITION	REQUIRED BEHAVIOR
Decision context stale	Suspend the gate and require refresh or explicit acceptance of the changed information boundary.

Acceptance evidence

- Role and separation matrices are tested for allow, deny, delegation, expiry, and emergency cases.
- Context mutation tests prove stale gates cannot commit without acknowledged refresh.
- Replay tests reproduce exactly what the human could see and the authority applied at action time.
- Adversarial tests prove models, agents, service accounts, and administrators cannot synthesize approval.

CHAPTER ACCEPTANCE

Human Gates and Approvals is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-004 · C-005 · C-024 · C-025 · C-026 · C-029 · C-035

43 Multi-Agent Execution Runtime

Agents are bounded workers inside the operating system. The runtime converts a governed objective into a supervised execution graph whose plans, evidence, tools, budgets, uncertainty, intermediate products, human gates, and termination are observable and reproducible.

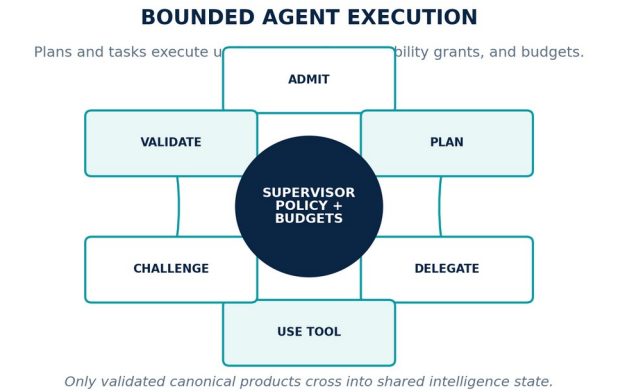


Figure 28. Bounded Agent Execution. Plans and tasks execute under a supervisor, capability grants, and budgets.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Execution	One immutable admission plus an append-only sequence of plans, tasks, observations, tool results, reviews, products, and terminal state.
Agent identity	Signed package, role, version, model policy, tool profile, evaluation status, and accountable owner.
Supervisor	Runtime authority that admits, delegates, monitors, challenges, suspends, reroutes, and terminates bounded work.
Product	Canonical output admitted only after schema, policy, provenance, quality, uncertainty, and ownership validation.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-43-001	Each execution SHALL declare purpose, initiating actor, data scope, agent versions, model policy, tools, budgets, deadlines, side-effect class, human gates, and stop conditions before work begins.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-43-002	Planner and supervisor responsibilities SHALL remain distinguishable; no agent may expand its own capability grant or approve its own consequential side effect.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-43-003	Inter-agent communication SHALL use versioned task and result contracts carrying causation, evidence references, confidence, policy context, and limits.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-43-004	The runtime SHALL checkpoint plans, tool invocations, intermediate products, challenges, and terminal reasons sufficient for audit and replay.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-43-005	Only validated canonical products SHALL enter shared memory, graph, twin, forecast, scenario, simulation, decision, or executive state.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Agent loops or stalls	Suspend at budget or progress thresholds and route evidence to the supervisor or human owner.
Conflicting agent results	Preserve alternatives and dissent; require adjudication rather than arbitrary overwrite.
Supervisor unavailable	Stop admission and side effects; retain safe read-only or checkpointed work only where policy permits.

Acceptance evidence

- Execution fixtures reproduce an admitted run from versioned manifests and preserved external results.
- Budget tests cover tokens, time, compute, tool calls, cost, data volume, retries, and side effects.
- Conflict tests prove alternative assessments remain visible through adjudication.
- Escape tests attempt prompt, tool, network, identity, policy, and capability escalation across agent boundaries.

CHAPTER ACCEPTANCE

Multi-Agent Execution Runtime is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-004 · C-005 · C-012 · C-027 · C-028 · C-029 · C-035

44 Agent Tool and Capability Control

Tools are controlled institutional capabilities, not model conveniences. Every read, computation, network action, mutation, communication, export, and physical or enterprise side effect is admitted through a least-privilege capability grant and an observable enforcement point.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Tool contract	Typed inputs and outputs, side-effect class, data handling, network destinations, timeout, idempotency, compensation, and audit profile.
Capability grant	Agent, task, tool, action, object scope, environment, budget, duration, purpose, obligations, and revocation.
Enforcement point	Gateway or sandbox that authenticates the caller, evaluates policy, constrains execution, and records the result.
Side-effect class	None, reversible, compensatable, approval-required, irreversible, or prohibited.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-44-001	An agent SHALL invoke external or privileged capability only through a registered tool contract and a valid task-specific capability grant.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-44-002	Tool inputs SHALL be schema validated, policy filtered, size bounded, classified, and protected against instruction or data boundary confusion.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-44-003	Tool outputs SHALL be treated as untrusted observations until integrity, schema, provenance, classification, and admissibility checks complete.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-44-004	Approval-required and irreversible effects SHALL require a human gate after the exact proposed effect is known and before dispatch.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.
ES-44-005	Revocation SHALL stop new calls immediately and SHALL define treatment of in-flight calls, credentials, cached results, and residual sessions.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Unknown tool or version	Deny invocation and emit a governed capability violation.
Ambiguous side effect	Classify at the more restrictive level and require owner review.
Tool result unverifiable	Preserve raw result as quarantined evidence and prohibit canonical admission.

Acceptance evidence

- Capability matrix tests enumerate every agent-tool-action-data-purpose combination.
- Sandbox tests cover filesystem, process, secret, network, tenant, and persistence isolation.
- Side-effect tests prove approval binds to the exact request digest and expires on mutation.
- Revocation drills measure termination and residual-access closure across all tool providers.

CHAPTER ACCEPTANCE

Agent Tool and Capability Control is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-004 · C-012 · C-028 · C-029 · C-035 · C-037 · C-040

45 Model Gateway and Inference Runtime

Models are replaceable governed dependencies behind one inference boundary. The gateway resolves an approved model and execution environment from task, data, policy, quality, sovereignty, cost, and availability constraints while preserving complete request and result lineage.

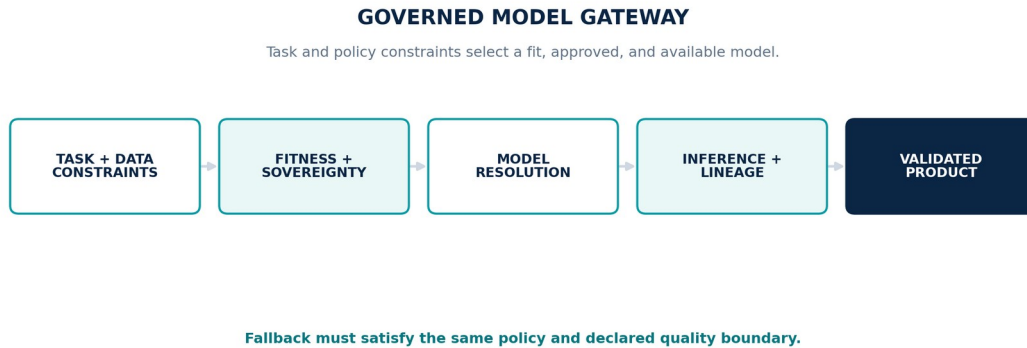


Figure 29. Governed Model Gateway. Task and policy constraints select a fit, approved, and available model.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Model profile	Identity, version, provider, task fitness, evaluation, data terms, context limits, sovereignty, cost, latency, and prohibited uses.
Inference policy	Allowed task, data class, tenant, deployment, region, purpose, model family, fallback, retention, and logging behavior.
Invocation record	Normalized request digest, template, parameters, context manifest, model resolution, result, usage, safety, and trace.
Result status	Complete, partial, abstained, refused, filtered, timed out, failed, or challenged; status remains distinct from content.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-45-001	All model inference SHALL pass through the governed model gateway, including embedded, local, external, fine-tuned, and marketplace models.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-45-002	Resolution SHALL select only model versions with current task-specific evaluation and data-handling approval for the requested deployment boundary.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.
ES-45-003	Prompt templates, retrieval context, tools, parameters, safety controls, and output schemas SHALL be immutable versioned dependencies of an invocation.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-45-004	Fallback SHALL preserve declared quality, data, sovereignty, and explanation constraints or return explicit unavailability; silent semantic downgrade is prohibited.	Negative conformance test	Adversarial fixture proving the prohibited path is denied and audited.
ES-45-005	Model output SHALL NOT become decision-active until the owning component validates schema, provenance, uncertainty, policy, and fitness for the product contract.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
No conforming model	Return unavailable or abstained with failed constraints; do not route to an unapproved provider.

FAILURE CONDITION	REQUIRED BEHAVIOR
Output schema failure	Retry only under bounded policy, then retain raw evidence and return an explicit failed result.
Model version withdrawn	Stop new resolution, identify affected products, and initiate review or recomputation policy.

Acceptance evidence

- Routing fixtures prove deterministic selection for task, classification, residency, quality, and failure cases.
- Canary tests compare new and incumbent versions against frozen evaluation populations and release thresholds.
- Lineage tests reconstruct every admitted model product from invocation dependencies and preserved evidence.
- Provider-failure tests prove fallback cannot cross policy, sovereignty, or quality boundaries.

CHAPTER ACCEPTANCE

Model Gateway and Inference Runtime is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-005 · C-012 · C-027 · C-030 · C-035 · C-041 · C-045

46 Evaluation and Fitness

Evaluation is a continuous control plane that determines whether a model, agent, workflow, source, analytical method, package, and composed product remains fit for a declared task and consequence class. Aggregate scores never conceal critical failure modes.

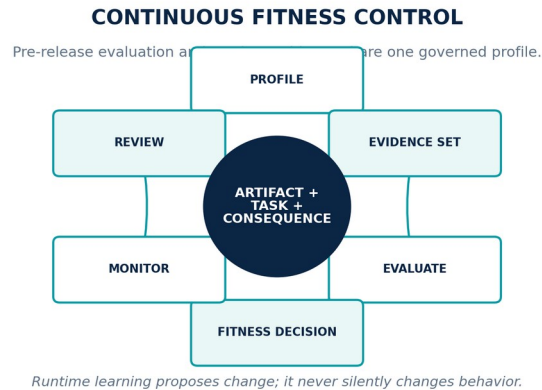


Figure 30. Continuous Fitness Control. Pre-release evaluation and runtime evidence share one governed profile.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Evaluation profile	Task, population, consequence, metrics, slices, thresholds, challenge set, judge method, owner, and expiry.
Evidence set	Versioned, access-controlled examples with provenance, rights, labels, uncertainty, and contamination status.
Fitness decision	Approved, constrained, shadow-only, suspended, withdrawn, or indeterminate for an exact artifact and use.
Runtime monitor	Signals for drift, error, calibration, abstention, bias, policy violation, operator override, and realized outcome.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-46-001	Every model, agent, workflow, and marketplace package SHALL have a current evaluation profile for each production task and consequence class.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-46-002	Evaluation SHALL measure correctness, calibration, evidence fidelity, uncertainty behavior, abstention, robustness, fairness, security, latency, cost, and operational failure where applicable.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-46-003	Critical slice failures SHALL block or constrain release even when an aggregate metric exceeds target.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-46-004	Evaluation data SHALL be separated from training and tuning data with explicit contamination controls and access evidence.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-46-005	Runtime outcomes, corrections, challenges, overrides, incidents, and drift SHALL trigger governed re-evaluation without silently changing production behavior.	Operational exercise	Time-stamped exercise record with telemetry, operators, impact, recovery, and closure evidence.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Insufficient evidence	Return indeterminate fitness and restrict production admission to an approved bounded mode.
Evaluation drift	Suspend affected resolution or apply declared constraint while investigation and revalidation proceed.
Judge disagreement	Preserve results by method and require adjudication for material fitness decisions.

Acceptance evidence

- Evaluation manifests pin artifact, data, metric, runner, judge, environment, and threshold versions.
- Reproduction runs match published results within declared tolerance.
- Slice and adversarial suites exercise high-consequence, sparse, novel, multilingual, and malicious conditions.
- Production monitors prove linkage from outcome signals to fitness review and rollback control.

CHAPTER ACCEPTANCE

Evaluation and Fitness is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-005 · C-021 · C-028 · C-030 · C-031 · C-035 · C-045 · C-049

47 Proactive Warning Runtime

The warning runtime converts material change into accountable attention. It joins weak signals, evidence, affected graph and twin state, forecasts, scenarios, strategic objectives, and time-to-act without overstating confidence or flooding operators with undifferentiated alerts.

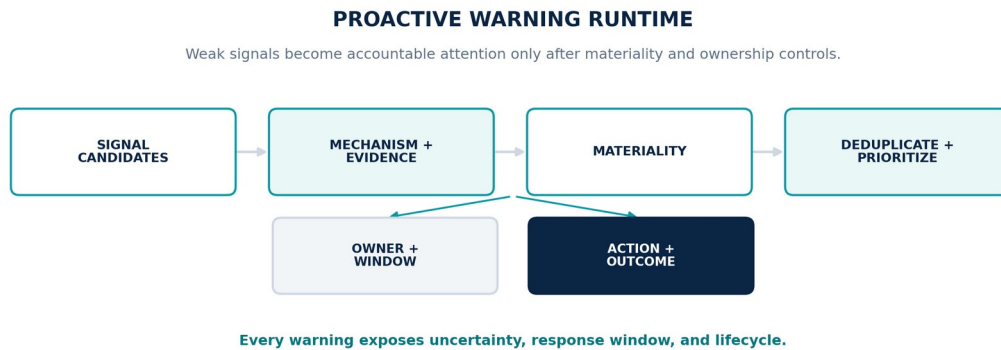


Figure 31. Proactive Warning Runtime. Weak signals become accountable attention only after materiality and ownership controls.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Candidate	A bounded change hypothesis with evidence, novelty, uncertainty, mechanism, affected entities, and monitoring state.
Materiality	Consequence, urgency, probability, reversibility, exposure, information value, and strategic relevance.
Warning	Deduplicated decision-active product with owner, response window, escalation, evidence, uncertainty, and lifecycle.
Lifecycle	Candidate, monitored, issued, acknowledged, investigating, acted, dismissed, expired, resolved, or superseded.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-47-001	Warning candidates SHALL retain supporting and contradicting evidence, uncertainty, and the mechanism connecting observed change to possible consequence.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-47-002	Materiality SHALL be computed from transparent dimensions and policy thresholds; no opaque global score may alone create executive urgency.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-47-003	Deduplication SHALL preserve distinct causes, affected objectives, geographies, time horizons, and policy domains even when presentation is consolidated.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-47-004	Each issued warning SHALL have an accountable recipient, acknowledgement expectation, response window, escalation path, and closure criteria.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-47-005	Feedback on false, late, missed, duplicated, and useful warnings SHALL enter evaluation and governed learning with Decision Replay context.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Signal confidence low	Maintain monitored candidate state and expose the information needed to raise or lower confidence.

FAILURE CONDITION	REQUIRED BEHAVIOR
Owner unresolved	Do not issue as actionable; escalate routing failure to an accountable operating owner.
Warning storm	Apply transparent grouping and prioritization while preserving each underlying candidate and urgent exception.

Acceptance evidence

- Historical replay measures precision, recall, timeliness, utility, acknowledgement, and outcome by materiality slice.
- Synthetic storms test grouping, suppression, escalation, and preservation of critical distinct signals.
- Closure tests require evidence, action or dismissal rationale, outcome, and learning disposition.
- Operator studies verify that presentation preserves uncertainty and action window without urgency inflation.

CHAPTER ACCEPTANCE

Proactive Warning Runtime is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-002 · C-005 · C-020 · C-022 · C-026 · C-032 · C-033 · C-034

48 Decision Replay and Governed Learning

Replay reconstructs what was knowable, believed, authorized, selected, committed, and later observed without hindsight contamination. Learning converts that evidence into change proposals that must be evaluated, approved, released, monitored, and reversible.

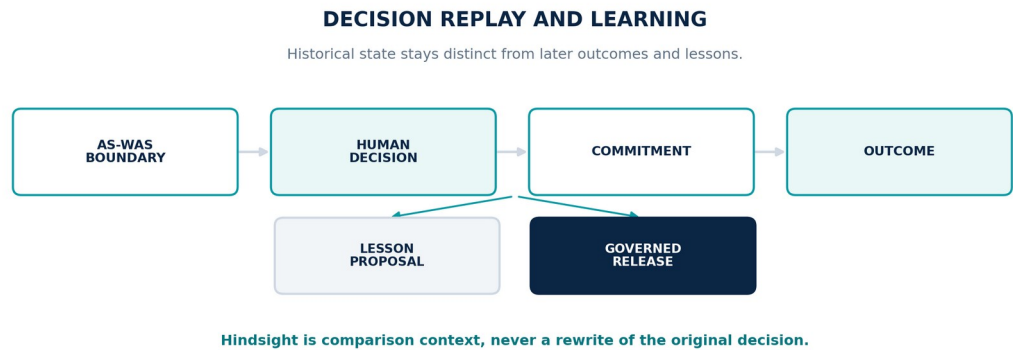


Figure 32. Decision Replay And Learning. Historical state stays distinct from later outcomes and lessons.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Replay boundary	Exact object, graph, memory, twin, forecast, scenario, simulation, model, agent, policy, and UI versions available at decision time.
Outcome record	Observed consequences, indicators, attribution limits, external changes, commitments, and accountable assessment.
Lesson proposal	Evidence-backed change to memory, ontology, model, agent, workflow, policy, control, product, or practice.
Learning release	Approved versioned change with evaluation, rollout, monitoring, rollback, affected products, and effective time.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-48-001	Decision Replay SHALL reconstruct the historical information boundary using system-time-correct state and SHALL display later knowledge only as a separate comparison.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-48-002	Replay SHALL include alternatives, dissent, uncertainty, approvals, rationale, commitments, execution state, policy, and presentation versions material to the decision.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-48-003	Outcome assessment SHALL distinguish observed result, inferred contribution, counterfactual claim, unresolved causality, and changed external conditions.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-48-004	A lesson SHALL remain a proposal until its owning authority approves a verified release; memory ingestion alone SHALL NOT change runtime behavior.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.
ES-48-005	Every learning release SHALL support impact analysis, staged deployment, runtime monitoring, rollback, and historical resolution of the prior behavior.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Historical dependency unavailable	Mark replay incomplete, enumerate missing evidence, and prohibit false completeness.

FAILURE CONDITION	REQUIRED BEHAVIOR
Outcome attribution uncertain	Preserve competing explanations and confidence; do not manufacture a single causal story.
Learning release degrades fitness	Rollback or constrain the release and preserve incident and evaluation evidence.

Acceptance evidence

- Golden decision cases reproduce historical state before and after seeded later corrections.
- Replay diff tests prove later evidence cannot contaminate the original boundary.
- Learning release tests trace proposal through approval, evaluation, deployment, monitoring, and rollback.
- Audit queries resolve the exact behavior and authority effective for any historical outcome.

CHAPTER ACCEPTANCE

Decision Replay and Governed Learning is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-004 · C-005 · C-011 · C-025 · C-031 · C-044 · C-045

PART VII

Security, Trust, and Sovereignty

Identity, authorization, isolation, cryptography, network control, privacy, audit, and supply chain.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

49 Identity and Authentication

Every human, workload, device, source, agent, model, package, and external organization acts through a verifiable identity with bounded assurance. Authentication establishes who or what is acting; it never establishes authority by itself.



Figure 33. Zero-Trust Request Path. Identity establishes the actor; policy establishes permitted action.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Principal	Stable identity for a human, workload, device, source, agent, model, package, or external organization.
Credential	Short-lived proof bound to principal, issuer, audience, assurance, key, validity, and revocation state.
Delegation	Explicit constrained authority chain from a principal to a workload or agent for a stated purpose and duration.
Session	Authenticated continuity with device, network, risk, assurance, purpose, and reauthentication context.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-49-001	All requests SHALL authenticate a principal or be explicitly admitted as a public source observation through a source-specific trust contract.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-49-002	Human authentication SHALL meet policy-defined assurance for the requested consequence, data class, administration plane, and deployment boundary.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-49-003	Workload, agent, model, package, and device identities SHALL be cryptographically bound to approved runtime or artifact state and rotated automatically.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-49-004	Delegation SHALL carry actor, delegate, purpose, actions, object scope, audience, duration, constraints, and revocation and SHALL remain visible downstream.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-49-005	Authentication failure, uncertainty, revocation, or assurance downgrade SHALL fail closed for protected actions and invalidate affected active sessions according to policy.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Identity provider unavailable	Use only pre-authorized bounded continuity for existing low-risk sessions; deny new privileged authentication.
Credential suspected compromised	Revoke, terminate affected sessions, rotate dependent credentials, and open incident impact analysis.

FAILURE CONDITION	REQUIRED BEHAVIOR
Delegation chain incomplete	Deny the action and retain the attempted context for audit.

Acceptance evidence

- Authentication suites cover assurance, audience, expiry, replay, key rotation, revocation, device, workload, and delegation.
- Chaos tests validate bounded identity-provider outage behavior by consequence class.
- Credential inventory proves no long-lived embedded secrets in production artifacts.
- Historical audit resolves principal, assurance, issuer, delegation, and session risk for each consequential action.

CHAPTER ACCEPTANCE

Identity and Authentication is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-004 · C-035 · C-036 · C-037 · C-038 · C-039 · C-040

50 Authorization and Policy Enforcement

Authorization evaluates whether a principal may perform an action on an object for a purpose in a particular environment and time. Decisions and obligations are enforced at every material read, transformation, model, agent, workflow, export, administration, and side-effect boundary.

POLICY ENFORCEMENT SURFACE					
	QUERY	WRITE	MODEL	AGENT	EXPORT
SUBJECT	ENFORCE	VERIFY	ENFORCE	VERIFY	ENFORCE
OBJECT	VERIFY	ENFORCE	VERIFY	ENFORCE	VERIFY
PURPOSE	ENFORCE	VERIFY	ENFORCE	VERIFY	ENFORCE
ENVIRONMENT	VERIFY	ENFORCE	VERIFY	ENFORCE	VERIFY

Authorization is evaluated at the owning boundary and evidenced at effect time.

Figure 34. Policy Enforcement Surface. The same decision and obligation semantics apply across product boundaries.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Policy input	Principal, delegation, action, object, purpose, classification, relationship, environment, time, location, risk, model, agent, and effect.
Decision	Allow, deny, indeterminate, or allow-with-obligations for one normalized request and policy version.
Obligation	Mask, filter, minimize, log, watermark, localize, approve, expire, encrypt, retain, notify, or prohibit onward use.
Enforcement proof	Decision ID, input digest, policy version, enforcement point, effect, and resulting object or error.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-50-001	Every protected action SHALL obtain and enforce a policy decision at the owning boundary; cached decisions SHALL be scope- and time-bounded and revocation aware.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-50-002	Default behavior SHALL be deny when identity, purpose, classification, ownership, policy, or enforcement state is missing or indeterminate.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-50-003	Object-level and relationship-level controls SHALL survive graph traversal, retrieval, summarization, export, model context assembly, and derived product creation.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-50-004	Administrative access SHALL use the same policy system with stronger assurance, explicit purpose, just-in-time scope, monitoring, and independent audit.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-50-005	Authorization semantics and conformance fixtures SHALL remain identical across SaaS, Private Cloud, and On-Premise deployments.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Policy service unavailable	Apply a declared restrictive continuity policy or deny; never assume previous broad access.

FAILURE CONDITION	REQUIRED BEHAVIOR
Obligation cannot be enforced	Deny the action and identify the unmet obligation.
Conflicting policies	Apply explicit precedence and the most restrictive applicable result pending governed resolution.

Acceptance evidence

- Policy matrices exhaustively cover principal, purpose, object, action, environment, time, and effect combinations.
- Graph and retrieval tests prove unauthorized relationships cannot be inferred through counts, paths, embeddings, or error differences.
- Revocation tests measure closure at caches, sessions, agents, models, exports, and derived views.
- Cross-deployment fixtures produce equivalent decisions and obligations for identical canonical inputs.

CHAPTER ACCEPTANCE

Authorization and Policy Enforcement is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-004 · C-035 · C-037 · C-038 · C-039 · C-040 · C-042 · C-047

51 Tenant and Domain Isolation

The Customer Intelligence Domain is the primary logical isolation boundary for evidence, knowledge, models, agents, policy, decisions, audit, administration, and export. Isolation is enforced through identity, authorization, cryptography, runtime placement, state partitioning, network controls, telemetry, and verification.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Domain identity	Stable tenant and domain identifiers present in every object, request, event, workflow, key, log, metric, trace, and package context.
Isolation cell	Bounded compute, state, queue, cache, model, agent, network, and operational context assigned to one or more policy-compatible domains.
Shared service	Control or utility service that processes domain-scoped data without owning cross-domain strategic state.
Cross-domain action	Explicit governed exchange with origin, recipient, purpose, package, obligations, approval, expiry, and audit.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-51-001	All canonical state and operational records SHALL carry authenticated domain context and SHALL reject absent, conflicting, or untrusted tenant identity.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-51-002	Domain context SHALL be propagated explicitly; process-local, connection-local, or user-interface context alone SHALL NOT authorize data access.	Negative conformance test	Adversarial fixture proving the prohibited path is denied and audited.
ES-51-003	Caches, indexes, vectors, queues, temporary files, model context, logs, traces, backups, and support artifacts SHALL enforce the same isolation as canonical stores.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-51-004	Cross-domain search, graph traversal, learning, benchmarking, sharing, or administration SHALL require an explicit product contract and policy authority.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-51-005	Isolation controls SHALL be continuously tested in every deployment mode and after changes to storage, routing, observability, model, agent, and package infrastructure.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Domain context mismatch	Reject, quarantine the attempted operation, and initiate security analysis when data exposure is possible.
Isolation control unavailable	Stop affected processing; do not fall back to an unpartitioned path.
Cross-domain package expired	Revoke recipient access and execute retention or deletion obligations with evidence.

Acceptance evidence

- Automated negative tests attempt cross-domain access through every API, event, cache, graph, vector, workflow, agent, model, and support path.
- Storage and backup inventories reconcile domain ownership and encryption context.
- Telemetry tests prove observability cannot leak customer content or identifiers outside approved boundaries.
- Red-team exercises validate confused-deputy, identifier substitution, cache poisoning, and cross-domain prompt attacks.

CHAPTER ACCEPTANCE

Tenant and Domain Isolation is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-016 · C-035 · C-037 · C-038 · C-039 · C-040 · C-041 · C-047

52 Cryptography, Keys, and Secrets

Cryptography protects identity, confidentiality, integrity, provenance, software supply, and non-repudiation across the full lifecycle. Key authority and custody are independently governed from data administration and may remain customer-controlled by deployment policy.

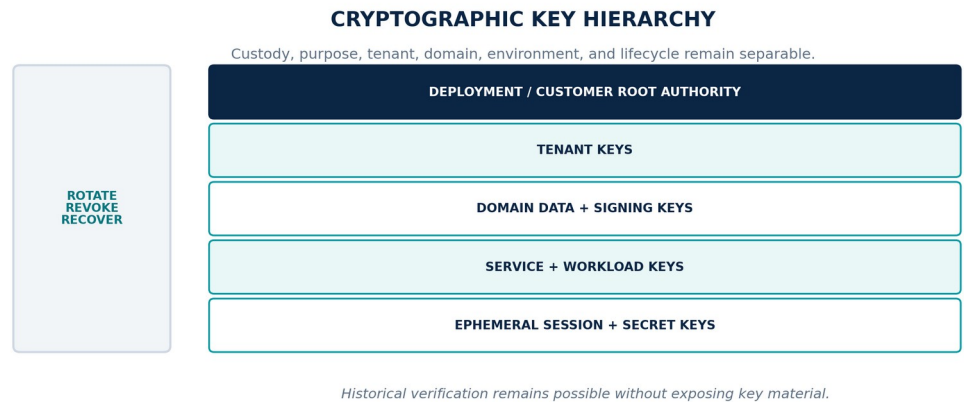


Figure 35. Cryptographic Key Hierarchy. Custody, purpose, tenant, domain, environment, and lifecycle remain separable.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Key hierarchy	Root, tenant, domain, data, service, signing, transport, backup, and ephemeral keys with declared owner and cryptoperiod.
Key custody	Provider-managed, customer-managed, customer-held, or external authority with explicit operations and continuity obligations.
Secret	Credential or sensitive value issued at runtime to an authenticated workload for a narrow audience and duration.
Crypto evidence	Algorithm suite, key reference, operation, time, principal, object, result, rotation, revocation, and verification status.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-52-001	Data SHALL be encrypted in transit and at rest using approved, versioned cryptographic profiles appropriate to classification and deployment policy.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-52-002	Keys SHALL be separated by tenant, domain, purpose, environment, and lifecycle where consequence requires; key identifiers SHALL remain resolvable historically without exposing key material.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-52-003	Secret material SHALL be issued dynamically, memory bounded, redacted from logs and artifacts, rotated automatically, and prohibited from source or build outputs.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-52-004	Signing keys for software, models, agents, policies, manifests, evidence, and exports SHALL be custody-separated from the identities that create those artifacts.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-52-005	Rotation, revocation, compromise, cryptographic migration, backup recovery, customer exit, and destruction SHALL be executable workflows with verified residual-state closure.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Key service unavailable	Apply classification-specific continuity; prohibit new sensitive writes or privileged operations when required key proof is absent.
Key compromise	Revoke, rotate, reissue trust, identify affected artifacts and data, and preserve incident evidence.
Algorithm withdrawn	Execute governed migration while maintaining historical verification and explicit unsupported state.

Acceptance evidence

- Cryptographic inventory maps every protected asset and operation to owner, profile, key, custody, rotation, and recovery evidence.
- Rotation and recovery drills cover active data, archives, backups, replicas, signatures, caches, and disconnected deployments.
- Secret scanning and runtime tests prove credentials do not enter source, images, logs, traces, prompts, support bundles, or crash dumps.
- Customer-held-key tests prove provider operators cannot decrypt protected content without customer-authorized key operation.

CHAPTER ACCEPTANCE

Cryptography, Keys, and Secrets is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-035 · C-036 · C-037 · C-038 · C-039 · C-041 · C-048

53 Network Trust and Controlled Egress

Network location is not authority. Every connection is authenticated, authorized, encrypted, destination-bound, observable, and constrained by data, purpose, deployment, and sovereignty policy. Egress is a governed capability with content-aware controls and explicit failure behavior.

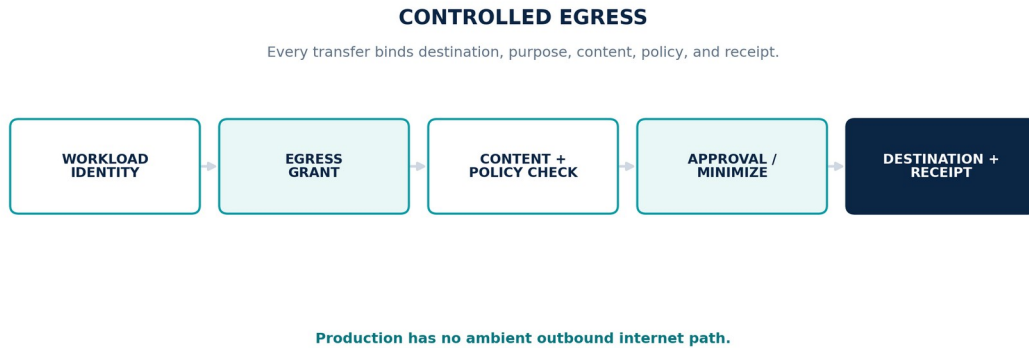


Figure 36. Controlled Egress. Every transfer binds destination, purpose, content, policy, and receipt.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Trust zone	Workload and data boundary with declared ingress, egress, identity, inspection, administration, and failure policy.
Service channel	Mutually authenticated, encrypted, authorized communication between registered workload identities.
Egress grant	Destination, protocol, data class, purpose, principal, workload, duration, volume, inspection, and approval.
Transfer record	Origin, destination, object or digest, policy, classification, bytes, time, result, and receipt where applicable.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-53-001	All service communication SHALL authenticate both endpoints and evaluate workload, action, destination, tenant, and purpose policy.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-53-002	Production workloads SHALL have no general outbound internet access; destinations and protocols SHALL be allow-listed through controlled egress points.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-53-003	Egress SHALL enforce classification, rights, minimization, residency, purpose, recipient, encryption, rate, inspection, and approval obligations before transfer.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-53-004	Disconnected and on-premise modes SHALL use signed import and export packages with equivalent policy, provenance, malware, integrity, and receipt checks.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-53-005	Network policy, service identity, certificate, destination, and egress-decision changes SHALL be versioned, tested, staged, monitored, and reversible.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Destination cannot be verified	Deny transfer and preserve the requested intent for accountable review.

FAILURE CONDITION	REQUIRED BEHAVIOR
Inspection unavailable	Apply destination and data-class failure policy; do not silently bypass mandatory inspection.
Partial transfer	Mark incomplete, prevent recipient admission where possible, reconcile receipts, and compensate or revoke.

Acceptance evidence

- Network tests enumerate every permitted workload-to-workload and workload-to-destination path and deny undeclared paths.
- Exfiltration exercises cover DNS, metadata, logs, traces, model providers, package managers, callbacks, and covert protocol paths.
- Disconnected-transfer fixtures validate signature, policy, provenance, malware, expiry, receipt, and revocation handling.
- Certificate and policy rotation drills demonstrate uninterrupted authorized traffic and prompt closure of revoked paths.

CHAPTER ACCEPTANCE

Network Trust and Controlled Egress is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-035 · C-036 · C-037 · C-038 · C-040 · C-041 · C-048

54 Privacy, Minimization, and Sovereignty

Privacy and sovereignty are executable policy dimensions spanning acquisition, identity, data, keys, models, agents, administration, telemetry, support, continuity, export, retention, deletion, and exit. They are not location labels or contractual footnotes.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Purpose record	Authorized institutional purpose, legal or contractual basis, scope, population, data classes, expiry, and owner.
Minimization plan	Fields, resolution, history, frequency, identifiers, recipients, retention, model use, and derived products strictly required.
Sovereignty profile	Independent constraints for data, keys, models, administration, telemetry, supply, support, backup, continuity, and exit.
Subject action	Access, correction, restriction, objection, portability, deletion, notification, or other applicable governed request.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-54-001	Collection and processing SHALL be admitted only for a declared purpose with approved authority, scope, minimization, retention, and downstream use.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.
ES-54-002	Sensitive identifiers SHALL be tokenized, pseudonymized, aggregated, masked, or excluded whenever the product contract does not require direct identity.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-54-003	Sovereignty policy SHALL be evaluated independently for data storage, processing, keys, models, administration, telemetry, support, supply, backup, and disaster recovery.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-54-004	Derived objects, embeddings, model context, evaluation data, logs, and exports SHALL inherit applicable privacy and sovereignty obligations.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-54-005	Subject requests, holds, legal conflicts, deletion, and customer exit SHALL use durable workflows with scope proof, residual inventory, completion evidence, and independent verification.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Purpose absent or expired	Stop new processing and route existing state through retention, restriction, or deletion policy.
Sovereignty constraints conflict	Deny the operation and require accountable legal, customer, and governance resolution.
Deletion residual remains	Keep the request open and expose residual location, reason, expiry, and accountable owner.

Acceptance evidence

- Data-flow inventories map purpose, fields, identities, systems, models, agents, regions, recipients, retention, and deletion behavior.
- Minimization tests compare actual payloads and context against approved field and history profiles.
- Residency and administration tests validate placement, routing, keys, telemetry, support, backup, and failover constraints.
- Lifecycle exercises prove subject and customer actions across canonical stores, projections, backups, exports, and model/evaluation artifacts.

CHAPTER ACCEPTANCE

Privacy, Minimization, and Sovereignty is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-016 · C-035 · C-037 · C-039 · C-040 · C-041 · C-048

55 Audit and Non-Repudiation

Audit is an independent, append-only account of consequential system and human action. It records who or what acted, under which authority and information boundary, what changed or was attempted, what controls decided, and what evidence proves the result.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Audit event	Actor, delegation, action, target, before/after reference, purpose, policy, result, time, location, trace, and evidence digest.
Audit stream	Ordered, integrity-protected domain and control-plane record with independent custody and retention.
Proof	Signature, digest chain, trusted timestamp, receipt, immutable storage evidence, or equivalent control appropriate to consequence.
Audit view	Policy-filtered query or export that preserves evidentiary meaning without exposing unrelated protected content.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-55-001	Consequential reads, writes, policy decisions, model and agent runs, human gates, administration, exports, corrections, releases, and security events SHALL emit audit records.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-55-002	Audit acceptance SHALL be durable before a consequential action is acknowledged complete, or the action SHALL enter an explicit unresolved evidence state.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-55-003	Audit records SHALL be append-only, integrity protected, time quality marked, access controlled, retained independently, and continuously reconciled against source systems.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-55-004	Audit content SHALL identify exact artifact, policy, schema, model, agent, workflow, object, and deployment versions material to the action.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-55-005	Audit search and export SHALL preserve chain of custody, query authority, redaction, completeness scope, and verification instructions.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Audit sink unavailable	Block or buffer according to consequence policy; never claim verified completion without required evidence.
Integrity gap detected	Seal affected range, preserve competing copies, alert security and governance, and prohibit unsupported non-repudiation claims.
Clock quality degraded	Record uncertainty and ordering evidence; do not fabricate precise event time.

Acceptance evidence

- Reconciliation compares authoritative state transitions, workflow histories, policy decisions, and external receipts with audit coverage.
- Tamper tests alter, remove, reorder, and duplicate records and verify detection and range isolation.
- Historical queries reproduce actor, authority, information boundary, effect, evidence, and governing versions.
- Export tests validate redaction, completeness declarations, signatures, receipt, and independent verification.

CHAPTER ACCEPTANCE

Audit and Non-Repudiation is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-004 · C-005 · C-012 · C-025 · C-035 · C-036 · C-049

56 Software Supply Chain and Marketplaces

Every executable, model, agent, workflow, connector, ontology extension, scenario, simulation, policy, and marketplace package is a governed supply-chain artifact. Admission depends on declared composition, provenance, evaluation, capabilities, compatibility, ownership, monitoring, and revocation.

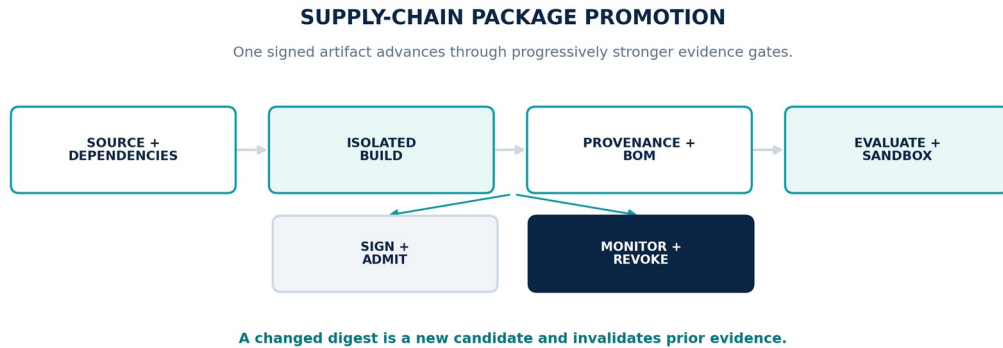


Figure 37. Supply-Chain Package Promotion. One signed artifact advances through progressively stronger evidence gates.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Package manifest	Identity, version, type, publisher, owner, dependencies, interfaces, permissions, data, network, models, resources, and compatibility.
Provenance	Source, builder, environment, inputs, tests, signatures, attestations, software or model bill of materials, and custody.
Admission decision	Approved scope, tenant or domain, deployment modes, capabilities, constraints, monitoring, expiry, and rollback.
Marketplace state	Draft, submitted, quarantined, evaluated, approved, restricted, published, deprecated, suspended, revoked, or retired.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-56-001	Production SHALL execute only signed artifacts whose provenance, composition, evaluation, vulnerabilities, licenses, policy, ownership, and compatibility pass current admission policy.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-56-002	Marketplace packages SHALL declare all data access, models, tools, network destinations, side effects, secrets, compute, storage, telemetry, and human gates before evaluation.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-56-003	Packages SHALL execute in capability-scoped isolation and SHALL NOT acquire ambient tenant, network, filesystem, credential, administration, or model access.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-56-004	Promotion SHALL preserve artifact identity across environments; rebuilds require a new digest and complete evidence chain.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-56-005	Revocation SHALL stop new use, identify active executions and derived products, apply rollback or review policy, and preserve historical replayability.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Signature or provenance invalid	Quarantine and prohibit execution or import.

FAILURE CONDITION	REQUIRED BEHAVIOR
Critical vulnerability or behavior discovered	Suspend resolution, revoke capabilities, identify impact, and execute bounded remediation.
Publisher unavailable	Apply ownership continuity, support, deprecation, and exit policy; no orphaned production authority.

Acceptance evidence

- Admission pipelines verify signatures, attestations, bills of materials, licenses, vulnerabilities, permissions, tests, and compatibility.
- Dynamic analysis observes file, process, network, model, secret, data, persistence, and side-effect behavior under adversarial inputs.
- Revocation drills trace package use to active runs, stored products, decisions, exports, and customer deployments.
- Offline package tests prove equivalent verification, policy, receipt, and revocation behavior for disconnected environments.

CHAPTER ACCEPTANCE

Software Supply Chain and Marketplaces is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-028 · C-035 · C-040 · C-041 · C-042 · C-045 · C-046 · C-048

PART VIII

Reliability and Operations

Quality contracts, performance, resilience, recovery, observability, incidents, and lifecycle operations.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

57 Quality Contracts and SLOs

Quality is part of product state. Every service and intelligence product declares what good means, how it is measured, which population and time window apply, what happens when evidence is weak, and which decisions remain safe when an objective is missed.

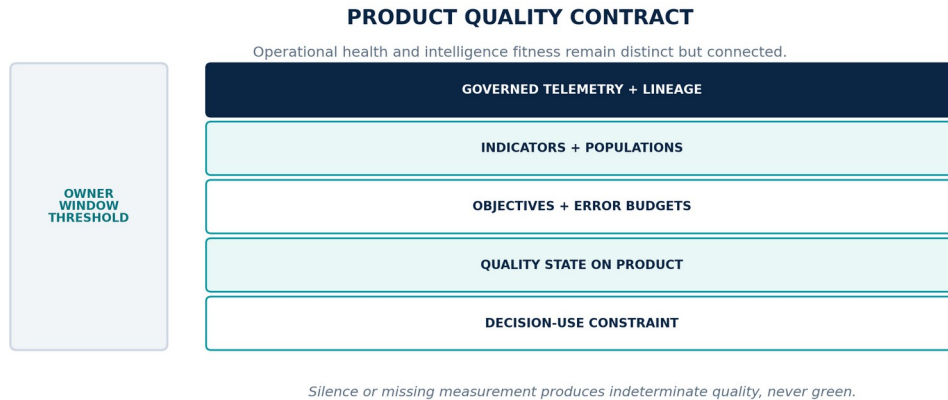


Figure 38. Product Quality Contract. Operational health and intelligence fitness remain distinct but connected.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Indicator	Measured signal with population, numerator, denominator, unit, aggregation, exclusions, source, freshness, and lineage.
Objective	Target, window, consequence class, confidence, owner, and permitted error budget for one indicator.
Product contract	Completeness, freshness, accuracy, calibration, provenance, coverage, latency, availability, and decision-use constraints.
Breach state	At risk, breached, exhausted, waived, recovering, or closed with impact, owner, actions, and verification.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-57-001	Every production service, workflow, source, model, agent, and canonical product SHALL publish owned quality indicators and objectives tied to user and institutional consequence.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-57-002	Indicators SHALL be computed from governed telemetry with declared populations, windows, exclusions, aggregation, sampling, delay, and missing-data behavior.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-57-003	Intelligence products SHALL expose decision-active quality and uncertainty on the object; operational dashboards alone SHALL NOT satisfy product transparency.	Operational exercise	Time-stamped exercise record with telemetry, operators, impact, recovery, and closure evidence.
ES-57-004	Error budgets SHALL govern change velocity and remediation priority but SHALL NOT authorize violation of security, privacy, human authority, or constitutional invariants.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.
ES-57-005	A breached objective SHALL trigger declared degradation, consumer notification, product invalidation or constraint, accountable remediation, and closure evidence.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Measurement unavailable	Mark quality indeterminate and apply the more restrictive product-use policy.

FAILURE CONDITION	REQUIRED BEHAVIOR
Objective breached	Expose affected population and decision impact, consume change budget, and initiate recovery workflow.
Metric disagreement	Preserve both computations, suspend unsupported closure, and reconcile definitions and lineage.

Acceptance evidence

- Indicator fixtures validate formulas, populations, windows, exclusions, late data, missing data, and time-zone behavior.
- SLO burn tests verify alert timing, change freeze, escalation, and consumer notification across consequence classes.
- Product tests prove quality state travels through APIs, exports, retrieval, briefings, and Decision Replay.
- Quarterly reviews correlate objectives with user harm, missed decisions, overrides, incidents, and realized outcomes.

CHAPTER ACCEPTANCE

Quality Contracts and SLOs is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-005 · C-020 · C-021 · C-032 · C-033 · C-043 · C-049

58 Performance and Capacity Engineering

Performance is engineered against strategic workloads, consequence, and time-to-decision rather than isolated endpoint speed. Capacity models cover ingestion bursts, graph growth, temporal history, twin and simulation portfolios, agent concurrency, model contention, human queues, and recovery demand.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Workload class	Interactive query, command, stream, batch, workflow, agent, model, graph, simulation, export, rebuild, or recovery.
Service envelope	Population, concurrency, data shape, latency distribution, throughput, freshness, resource, and deadline targets.
Capacity model	Demand drivers, unit costs, scaling key, headroom, saturation points, dependencies, and recovery reserve.
Admission control	Policy that accepts, queues, sheds, degrades, reschedules, or rejects work before unsafe saturation.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-58-001	Each component SHALL define workload classes, scaling keys, service envelopes, saturation signals, admission behavior, and resource budgets.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-58-002	Capacity planning SHALL model normal, forecast, campaign, source burst, geopolitical event, warning storm, model outage, recovery, and tenant-noise conditions.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-58-003	Interactive and consequential workloads SHALL be protected from analytical, training, simulation, export, and rebuild workloads through explicit priority and resource isolation.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-58-004	Performance tests SHALL preserve production-representative object sizes, graph density, temporal depth, policy evaluation, encryption, audit, and provenance overhead.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-58-005	Autoscaling or operator scaling SHALL respect tenancy, sovereignty, key locality, state movement, warm-up, model availability, and downstream capacity constraints.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Capacity forecast exceeded	Apply admission, priority, degradation, and expansion policy before uncontrolled queue growth.
Hot tenant or entity	Isolate the contention domain and preserve fairness and critical workload reserves.
Dependency saturation	Reduce or defer upstream work using propagated backpressure and visible product freshness state.

Acceptance evidence

- Load tests cover percentile latency, throughput, queue age, resource saturation, cost, fairness, and product freshness by workload class.
- Burst and soak tests expose leaks, compaction, cache churn, graph skew, workflow accumulation, and model contention.
- Backpressure tests prove bounded queues and propagated admission rather than hidden data loss.
- Capacity models are reconciled against production telemetry and revised under controlled change.

CHAPTER ACCEPTANCE

Performance and Capacity Engineering is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-032 · C-033 · C-042 · C-043 · C-049 · C-050

59 Availability and Resilience

Availability means the right capability remains trustworthy for its declared consequence. Resilience prevents a local dependency, tenant, source, model, region, administrator, or package failure from becoming an uncontrolled system-wide or cross-customer failure.

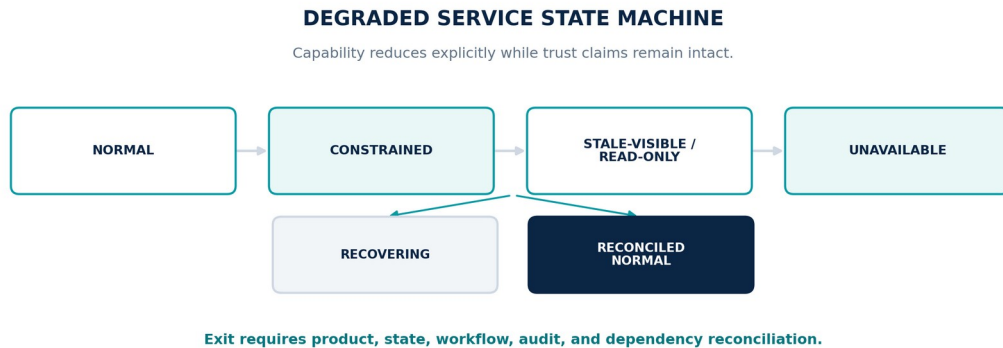


Figure 39. Degraded Service State Machine. Capability reduces explicitly while trust claims remain intact.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Failure domain	Process, host, zone, region, cell, tenant, store, queue, source, model, identity, key, control plane, or operator boundary.
Resilience pattern	Replication, partition, checkpoint, retry, circuit break, bulkhead, fallback, queue, reconstruction, or manual continuity.
Dependency policy	Criticality, timeout, retry, fallback, freshness, isolation, recovery, and customer-visible consequence.
Continuity state	Normal, constrained, read-only, stale-visible, queued, isolated, unavailable, recovering, or reconciled.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-59-001	Every component SHALL identify failure domains, critical dependencies, required availability, recovery objectives, isolation controls, and declared degraded states.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-59-002	Retries SHALL be bounded, jittered, idempotent, deadline-aware, and subordinate to circuit breaking and system-wide load protection.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-59-003	Intelligence Cells SHALL limit tenant and workload blast radius while shared control services SHALL degrade without creating cross-domain data or authority failure.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-59-004	Fallback SHALL preserve identity, policy, provenance, time, uncertainty, and decision-use semantics; a lower-quality path SHALL be declared and constrained.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-59-005	Recovery from partition or failover SHALL reconcile authoritative state, event order, workflow effects, audit, caches, and projections before normal status resumes.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Regional or cell failure	Fail over or isolate under placement policy, expose continuity state, and preserve sovereignty constraints.
Control-plane partition	Use only bounded pre-authorized continuity; deny state or authority changes that require current control.

FAILURE CONDITION	REQUIRED BEHAVIOR
Split-brain state	Fence writers, preserve both histories, adjudicate authority, and reconcile before reopening writes.

Acceptance evidence

- Fault injection removes each process, host, zone, cell, region, queue, store, model, key, identity, and policy dependency.
- Blast-radius tests prove tenant, cell, workload class, and control-plane isolation under saturation and corruption.
- Partition tests validate fencing, ordering, bounded continuity, reconciliation, and audit completeness.
- Resilience evidence reports customer-visible capability and truth state, not infrastructure uptime alone.

CHAPTER ACCEPTANCE

Availability and Resilience is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-037 · C-041 · C-042 · C-043 · C-049 · C-050

60 Degraded Service Semantics

Degradation is a named product state with precise guarantees and prohibitions. The Eye may reduce capability, currency, automation, or throughput, but it may never fabricate freshness, confidence, provenance, authorization, approval, or completeness.

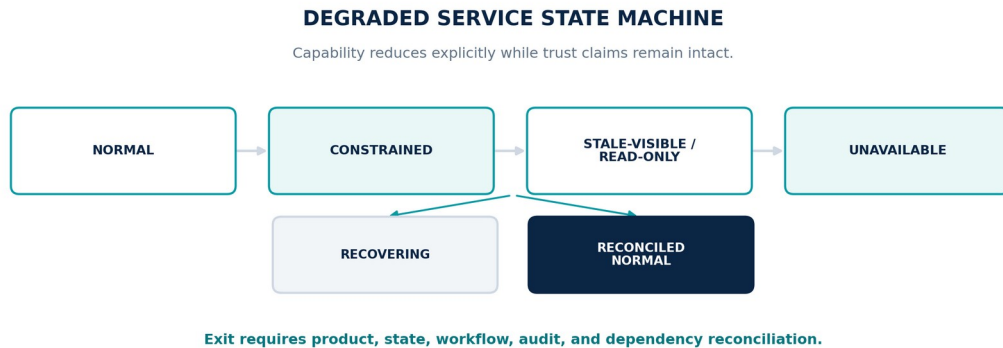


Figure 40. Degraded Service State Machine. Capability reduces explicitly while trust claims remain intact.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Mode	Normal, constrained, delayed, stale-visible, read-only, queued, manually supervised, unavailable, or recovering.
Entry condition	Measured dependency, quality, security, policy, capacity, data, model, or control failure that selects the mode.
Guarantee	Capabilities and object properties that remain trustworthy in the mode.
Prohibition	Actions and claims that are disabled until recovery and reconciliation complete.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-60-001	Every customer-visible component and canonical product SHALL publish degraded modes, entry and exit criteria, guarantees, prohibitions, operator actions, and consumer signals.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-60-002	Stale data SHALL carry last successful observation or computation time, cause, affected scope, expected review, and decision-use constraint.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-60-003	Read-only mode SHALL prevent all indirect writes, side effects, acknowledgements, learning releases, approvals, and hidden queueing not explicitly declared.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.
ES-60-004	Manual continuity SHALL preserve authenticated authority, evidence, version, policy, audit, and later reconciliation with system state.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-60-005	Exit to normal SHALL require dependency health, backlog treatment, state reconciliation, audit closure, product refresh or invalidation, and accountable confirmation.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Mode cannot be determined	Select unavailable or the most restrictive safe mode for the affected consequence.
Consumer ignores degraded state	Enforce at the owning boundary where use would be unsafe and record the attempted action.
Recovery incomplete	Remain recovering and expose unresolved state; do not declare normal from infrastructure health alone.

Acceptance evidence

- Mode-transition tests cover every trigger, guard, consumer signal, prohibited operation, recovery step, and rollback.
- UI, API, event, export, briefing, and Decision Replay fixtures present equivalent degraded semantics.
- Chaos tests verify no failure path emits false success, currency, confidence, completeness, approval, or authorization.
- Recovery exercises reconcile queues, workflows, canonical state, projections, products, and audit before normal status.

CHAPTER ACCEPTANCE

Degraded Service Semantics is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-004 · C-005 · C-033 · C-043 · C-049 · C-050

61 Backup, Recovery, and Disaster Recovery

Recovery restores an internally consistent intelligence system, not merely files or databases. The plan preserves identity, policy, keys, canonical history, graph and twin state, workflows, audit, model and agent artifacts, projections, exports, and customer sovereignty within declared loss and time objectives.

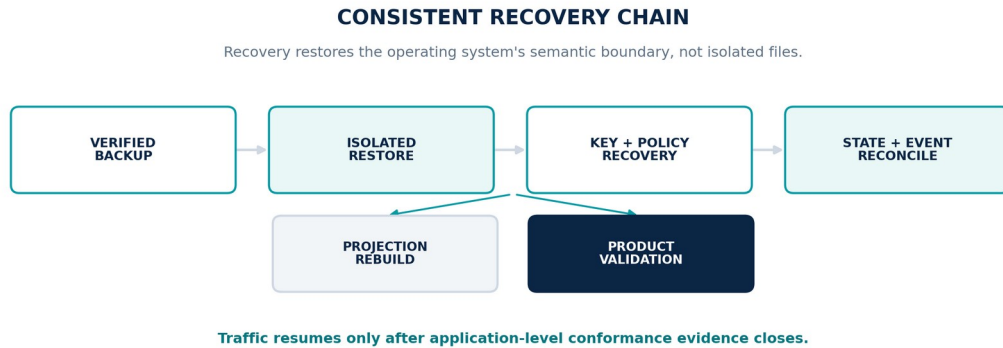


Figure 41. Consistent Recovery Chain. Recovery restores the operating system's semantic boundary, not isolated files.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Recovery unit	Customer domain, cell, service, canonical store, artifact registry, control plane, region, or full deployment.
Recovery point	Latest mutually consistent state across authoritative data, events, workflows, audit, policies, keys, and manifests.
Recovery time	Time from declared incident to verified capability state, including reconciliation and product validation.
Recovery evidence	Backup identity, integrity, custody, key, restore log, reconciliation, validation, exceptions, and approval.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-61-001	Each state owner SHALL define backup scope, frequency, consistency boundary, retention, encryption, custody, restoration order, RPO, RTO, and verification owner.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-61-002	Backups SHALL include or reference exact schema, policy, key, artifact, workflow, event, and dependency versions required for restoration.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-61-003	Restoration SHALL occur into isolated infrastructure, verify integrity and authority, reconcile cross-store state, rebuild projections, and validate product behavior before traffic resumes.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.
ES-61-004	Disaster recovery SHALL preserve sovereignty, tenant isolation, customer-held key authority, administration boundaries, and deployment semantic parity.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-61-005	Recovery plans SHALL cover corruption, deletion, ransomware, region loss, control-plane loss, key loss, supply-chain compromise, and operator error.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Backup integrity fails	Quarantine the set, select the latest verified recovery point, and expose increased loss boundary.

FAILURE CONDITION	REQUIRED BEHAVIOR
Key unavailable	Do not bypass custody; execute customer or authority continuity and report unrecoverable scope explicitly.
Cross-store reconciliation fails	Keep the deployment isolated, preserve evidence, and continue deterministic repair or accepted bounded loss workflow.

Acceptance evidence

- Automated restore tests run continuously for every state class and verify application-level semantics, not archive readability alone.
- Quarterly recovery exercises measure end-to-end RPO and RTO including keys, identity, policy, workflows, audit, projections, and validation.
- Corruption drills identify last-known-good recovery points and prove immutable evidence survives the compromised authority.
- Customer-domain export and exit restoration tests validate portability into an independently controlled environment.

CHAPTER ACCEPTANCE

Backup, Recovery, and Disaster Recovery is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-011 · C-025 · C-037 · C-041 · C-042 · C-043 · C-048 · C-050

62 Observability and Telemetry

Observability connects technical execution to intelligence product state and institutional consequence. Metrics, logs, traces, events, profiles, and synthetic checks carry common identity and causation without becoming an uncontrolled copy of customer evidence or sensitive content.

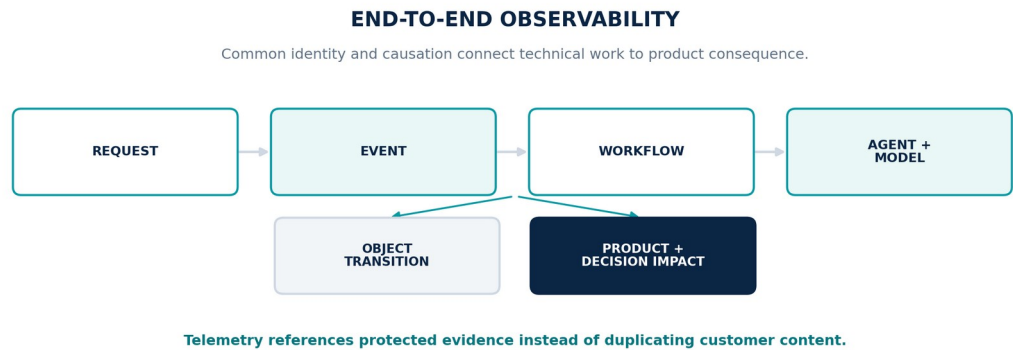


Figure 42. End-To-End Observability. Common identity and causation connect technical work to product consequence.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Correlation model	Tenant, domain, principal, workload, component, object, workflow, agent, model, policy, deployment, trace, and causation identifiers.
Signal	Metric, event, log, trace, profile, synthetic result, quality state, or audit reference with owner and retention.
Telemetry policy	Permitted fields, classification, redaction, sampling, region, access, retention, export, and support use.
Health model	Dependency, service, workflow, product, quality, security, and decision-impact states with explicit derivation.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-62-001	Every request, event, workflow, agent run, model call, object transition, and external effect SHALL propagate trace and causation context across synchronous and asynchronous boundaries.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-62-002	Telemetry SHALL include service, dependency, queue, workflow, quality, security, policy, model, agent, cost, and customer-impact signals owned by named teams.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-62-003	Logs and traces SHALL exclude secrets and minimize customer content; protected evidence SHALL remain referenced by governed identifiers rather than duplicated.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-62-004	Health SHALL distinguish infrastructure, control, data, product, quality, and decision-use state and SHALL NOT reduce them to a single green indicator.	Negative conformance test	Adversarial fixture proving the prohibited path is denied and audited.
ES-62-005	Telemetry pipelines SHALL have their own availability, integrity, loss, delay, sampling, retention, residency, access, and recovery contracts.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Telemetry delayed or lost	Expose observability uncertainty and apply consequence-specific operational caution; do not infer health from silence.

FAILURE CONDITION	REQUIRED BEHAVIOR
Sensitive content detected	Restrict access, purge under evidence-preserving workflow, correct instrumentation, and assess exposure.
Correlation broken	Mark traces incomplete and repair propagation before relying on them for conformance or closure.

Acceptance evidence

- Trace fixtures span API, event, workflow, agent, model, tool, store, policy, and external receipt boundaries.
- Telemetry schema tests enforce classification, redaction, field budgets, units, cardinality, and retention.
- Synthetic tests validate customer-visible product paths and degraded-mode representation from every deployment zone.
- Incident reviews prove signals can reconstruct scope, cause, product impact, authority, recovery, and unresolved uncertainty.

CHAPTER ACCEPTANCE

Observability and Telemetry is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-005 · C-012 · C-035 · C-036 · C-041 · C-043 · C-049 · C-050

63 Incident, Change, and Problem Management

Operational governance treats incidents, changes, and recurring problems as connected evidence. Response protects people, customer authority, strategic truth, service continuity, legal obligations, and institutional memory before optimizing restoration speed.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Incident	Unplanned loss or credible threat to confidentiality, integrity, availability, quality, authority, explainability, or decision safety.
Change	Authorized modification to code, configuration, policy, schema, model, agent, infrastructure, data, package, or operating procedure.
Problem	Underlying or systemic cause whose removal reduces recurrence or consequence across incidents and near misses.
Record	Timeline, scope, evidence, authority, actions, communications, decisions, customer impact, obligations, recovery, and learning.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-63-001	Incident classification SHALL consider strategic and decision impact, data and tenant scope, authority, quality, provenance, explainability, sovereignty, and legal obligation in addition to uptime.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-63-002	Responders SHALL use just-in-time controlled access, named roles, an incident workflow, protected evidence, decision logs, communication authority, and independent audit.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.
ES-63-003	Emergency changes SHALL remain versioned, reviewed, tested to the feasible extent, time-bounded, monitored, reversible, and subject to post-event ratification or removal.	Operational exercise	Time-stamped exercise record with telemetry, operators, impact, recovery, and closure evidence.
ES-63-004	Recovery declaration SHALL require technical, canonical state, workflow, audit, product, customer-impact, and control reconciliation appropriate to the incident.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-63-005	Post-incident learning SHALL generate owned corrective actions and lesson proposals with deadlines, verification, effectiveness monitoring, and recurrence linkage.	Operational exercise	Time-stamped exercise record with telemetry, operators, impact, recovery, and closure evidence.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Scope uncertain	Assume the credible wider boundary, preserve evidence, and narrow only through verified analysis.
Primary controls compromised	Move coordination and evidence to an independently protected continuity environment.
Recovery causes regression	Reopen the incident, restore the last verified state, and retain both change and failure evidence.

Acceptance evidence

- Exercises cover security breach, data corruption, model defect, warning failure, decision replay gap, region loss, and insider misuse.
- Change records prove proposal, risk, dependencies, tests, approvals, artifact identity, rollout, monitoring, and rollback.
- Incident timelines reconcile telemetry, audit, workflow, communication, external receipt, and human decision evidence.
- Problem reviews measure corrective-action effectiveness and recurrence across services, tenants, deployments, and consequence classes.

CHAPTER ACCEPTANCE

Incident, Change, and Problem Management is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-004 · C-005 · C-025 · C-031 · C-035 · C-043 · C-045 · C-049

64 Operational Data Lifecycle

Operational data has the same governed lifecycle discipline as strategic data. Queues, caches, indexes, vectors, temporary files, telemetry, support bundles, checkpoints, replicas, backups, evaluation sets, and derived artifacts have declared ownership, authority, retention, reconstruction, and deletion behavior.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
State class	Canonical, event, workflow, projection, cache, temporary, telemetry, audit, backup, artifact, evaluation, or support.
Lifecycle profile	Creation, owner, authority, policy, version, replication, compaction, archive, correction, deletion, and proof.
Rebuild contract	Canonical inputs, algorithm, version, ordering, checkpoint, reconciliation, target time, and validation.
Residual inventory	Known remaining copies after retention, deletion, customer exit, incident, or migration action.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-64-001	Every persistent and transient state class SHALL have an accountable owner, authority classification, policy, retention, recovery, export, correction, and deletion profile.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-64-002	Projections, caches, indexes, vectors, and summaries SHALL be reconstructable from declared canonical inputs or explicitly designated as authoritative state.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-64-003	Compaction and archival SHALL preserve identity, temporal truth, provenance, correction, audit, and Decision Replay requirements.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-64-004	Temporary state SHALL use bounded lifetime, encrypted storage, tenant isolation, crash cleanup, and verifiable deletion appropriate to classification.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-64-005	Lifecycle actions SHALL propagate to replicas, caches, exports, model context, evaluation data, support artifacts, backups, and disconnected deployments according to policy.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Orphaned state discovered	Quarantine, identify ownership and policy, assess exposure, and disposition through an audited workflow.
Projection cannot rebuild	Treat hidden state as authoritative drift, stop unsupported use, and repair the state contract.
Residual copy cannot be deleted	Record location, reason, access restriction, expiry, and accountable exception.

Acceptance evidence

- Automated inventories reconcile declared state classes with runtime stores, volumes, buckets, queues, caches, registries, and backups.
- Projection rebuilds compare counts, identities, versions, digests, query behavior, and policy outcomes.
- Retention and deletion fixtures track seeded objects through every derived and operational state class.
- Customer-exit tests produce complete export, deletion, residual, key-destruction, and independent verification evidence.

CHAPTER ACCEPTANCE

Operational Data Lifecycle is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-011 · C-012 · C-016 · C-025 · C-035 · C-039 · C-041 · C-048

PART IX

Delivery and Verification

Source organization, configuration, tests, release gates, promotion, packaging, acceptance, and governance.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

65 Repository and Module Standards

Source organization makes ownership, boundaries, contracts, controls, and verification visible. Repositories and modules may follow team and deployment needs, but no layout may conceal canonical responsibility, shared mutable state, ungoverned dependencies, or unauditable generation.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Repository	Version-controlled source and declarative assets with owner, purpose, classification, build, tests, release, and archival policy.
Module	Cohesive implementation unit with explicit public contract, dependencies, state access, configuration, telemetry, and tests.
Generated asset	Reproducible output with generator, inputs, version, review, digest, and update procedure.
Ownership file	Machine-readable mapping from paths and artifacts to accountable engineering, product, security, data, and operational owners.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-65-001	Each repository SHALL declare mapped layer and component IDs, owners, supported artifacts, build environment, dependencies, interfaces, state access, controls, tests, and release path.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.
ES-65-002	Modules SHALL expose only versioned public contracts and SHALL NOT bypass owners through internal imports, shared schemas, direct database access, or generated client modifications.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.
ES-65-003	Dependency direction SHALL follow declared architecture boundaries and SHALL be continuously checked for cycles, forbidden imports, and undeclared runtime links.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-65-004	Generated code, schemas, clients, policies, infrastructure, prompts, and documentation SHALL be reproducible and reviewed from authoritative sources.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-65-005	Secrets, customer content, production identifiers, unlicensed data, mutable binaries, and environment-specific authority SHALL be prohibited from source repositories.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Owner unavailable	Stop material release and invoke service ownership continuity.
Forbidden dependency introduced	Fail integration and require boundary correction or approved architecture change.
Generated output differs	Reject the artifact until generator, inputs, and reproducibility are reconciled.

Acceptance evidence

- Static architecture tests enforce module, dependency, database, client, and layer-boundary rules.
- Repository scanners validate ownership, secrets, licenses, provenance, generated assets, and required manifests.
- Clean-room builds reproduce signed outputs from pinned source and toolchain inputs.
- Ownership and support exercises verify escalation and stewardship for every production artifact.

CHAPTER ACCEPTANCE

Repository and Module Standards is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-008 · C-035 · C-036 · C-046 · C-048 · C-051 · C-052

66 Configuration and Feature Control

Configuration is versioned product behavior under the same authority as code. Flags, policies, schemas, thresholds, routing, models, agents, connectors, workflow definitions, and operational limits are typed, owned, validated, staged, observable, and historically resolvable.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Configuration item	Typed name, owner, authority, schema, default, scope, sensitivity, validation, rollout, expiry, and rollback.
Scope	Global, deployment, region, cell, tenant, domain, component, workload, user role, or controlled experiment.
Feature state	Disabled, shadow, internal, restricted, staged, generally enabled, suspended, deprecated, or removed.
Resolution record	Effective value, source layers, overrides, policy, actor, time, version, and runtime acknowledgement.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-66-001	All production configuration SHALL be schema validated, version controlled, reviewed, approved, signed, promoted, and audited through a controlled path.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-66-002	Defaults SHALL be safe and semantically identical across deployment modes unless the specification explicitly identifies a physical implementation difference.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-66-003	Feature flags SHALL have an owner, purpose, eligible scope, dependency, telemetry, rollout, expiry, removal, and degraded-state behavior.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-66-004	Configuration changes affecting security, policy, object meaning, model or agent behavior, human authority, data lifecycle, or deployment parity SHALL receive independent domain approval.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-66-005	Runtime components SHALL expose effective configuration version and reject invalid, stale, unsigned, conflicting, or authority-incompatible updates.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Configuration service unavailable	Continue with last verified bounded configuration or enter declared degraded mode; never invent defaults.
Conflicting scopes	Apply deterministic precedence and reject ambiguity for consequential behavior.
Flag expires	Disable or apply the declared terminal state and open removal verification.

Acceptance evidence

- Schema and property tests cover type, range, dependency, precedence, invalid input, downgrade, and migration.
- Promotion tests prove the same signed configuration object advances without manual re-entry.
- Kill-switch exercises measure propagation, enforcement, acknowledgement, and residual behavior.
- Historical queries resolve effective configuration for every consequential execution and decision replay.

CHAPTER ACCEPTANCE

Configuration and Feature Control is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-004 · C-035 · C-042 · C-043 · C-045 · C-049 · C-051

67 Test Architecture

Testing is a layered proof system. It validates local logic, contracts, canonical semantics, policy, security, data quality, AI fitness, resilience, recovery, deployment parity, human authority, and institutional outcomes using versioned evidence that can be reproduced independently.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Test level	Unit, property, component, contract, integration, system, end-to-end, operational, recovery, or acceptance.
Conformance fixture	Portable canonical input, environment assumptions, expected outputs and effects, tolerances, and evidence.
Test population	Curated, generated, historical, adversarial, synthetic, or production-shadow data with rights and contamination controls.
Evidence record	Artifact, environment, dependencies, seed, population, runner, result, logs, exceptions, reviewer, and signature.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-67-001	Every normative requirement SHALL map to one or more owned verification methods and release evidence records.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-67-002	Contract tests SHALL run for every provider and consumer version and include errors, abstention, retries, ordering, compatibility, policy, and degraded behavior.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-67-003	Canonical conformance suites SHALL validate identity, time, truth, provenance, correction, human authority, deployment parity, and replay across the complete strategic loop.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-67-004	AI and probabilistic tests SHALL use distributions, calibration, slices, uncertainty, abstention, drift, challenge, and outcome measures rather than deterministic exact-match alone.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-67-005	Production-like system tests SHALL include real control enforcement, encryption, audit, data scale, temporal depth, graph density, model and agent boundaries, and failure injection.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Flaky test	Quarantine only with owner, impact, expiry, monitoring, and equivalent release protection; never normalize nondeterminism.
Unrepresentative fixture	Invalidate unsupported conclusions and rebuild the population with provenance and coverage evidence.
Test environment drift	Stop promotion until environment identity and production equivalence are restored or explicitly bounded.

Acceptance evidence

- Traceability reports show complete requirement-to-test-to-evidence-to-artifact closure.
- Mutation and fault tests demonstrate suites detect semantic, security, policy, temporal, and recovery defects.
- Cross-deployment suites run the same canonical fixtures in SaaS, Private Cloud, and On-Premise reference environments.
- Independent assurance reproduces high-consequence evidence from immutable manifests and artifacts.

CHAPTER ACCEPTANCE

Test Architecture is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-004 · C-005 · C-011 · C-035 · C-042 · C-043 · C-045 · C-049 · C-052

68 CI/CD and Release Gates

The delivery system converts reviewed source into one immutable, attested release candidate and admits it only when architecture, contracts, quality, security, privacy, AI fitness, resilience, operations, documentation, and rollback evidence are complete for the intended scope.

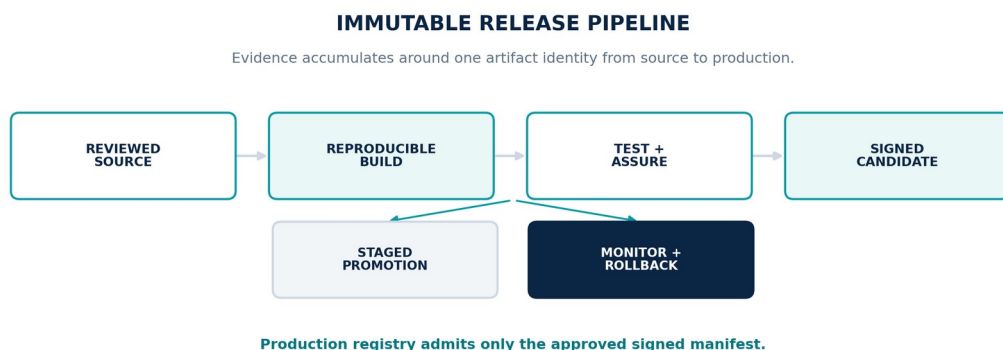


Figure 43. Immutable Release Pipeline. Evidence accumulates around one artifact identity from source to production.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Candidate	Immutable set of code, schemas, policies, models, agents, workflows, packages, infrastructure, configuration, migrations, and documentation.
Build provenance	Source revision, toolchain, dependencies, builder identity, environment, commands, tests, attestations, and artifact digests.
Release gate	Owned policy decision over required evidence, scope, consequence, exceptions, approvals, and expiry.
Release manifest	Candidate identity, compatibility, requirements, changes, migrations, deployment support, rollout, monitoring, rollback, and owners.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-68-001	Builds SHALL execute in isolated, ephemeral, least-privilege environments from pinned inputs and SHALL produce signed provenance and composition attestations.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-68-002	The same immutable artifact set SHALL be promoted across environments; production-specific rebuilds or manual binary replacement are prohibited.	Negative conformance test	Adversarial fixture proving the prohibited path is denied and audited.
ES-68-003	Release gates SHALL evaluate architecture conformance, tests, contracts, vulnerabilities, licenses, policy, privacy, data migration, AI fitness, performance, resilience, recovery, observability, runbooks, and documentation.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-68-004	Exceptions SHALL be explicit, risk-accepted by authorized independent owners, time-bounded, monitored, and included in the release and operational acceptance record.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.
ES-68-005	A release SHALL have staged rollout, stop conditions, product and technical monitoring, rollback or forward-repair path, and accountable command authority before promotion.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Evidence missing or stale	Fail the gate and preserve an indeterminate status rather than infer success.

FAILURE CONDITION	REQUIRED BEHAVIOR
Artifact identity changes	Invalidate downstream evidence and restart promotion for the new digest.
Gate system unavailable	Pause promotion; emergency release requires the separately governed continuity path and later ratification.

Acceptance evidence

- Clean-room rebuilds reproduce candidate digests and attestations from pinned source and toolchain inputs.
- Gate-policy tests seed failures, expiries, exceptions, incompatible artifacts, and tampered evidence.
- Release dry runs execute migrations, rollout, monitoring, rollback, reconciliation, and operator communication.
- Production registry rejects artifacts and configurations not present in the signed approved release manifest.

CHAPTER ACCEPTANCE

CI/CD and Release Gates is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-035 · C-042 · C-045 · C-046 · C-048 · C-049 · C-051 · C-052

69 Environment Promotion

Environments progressively increase consequence without changing product meaning. Development, verification, integration, pre-production, production, and recovery contexts differ in authority and data, but they resolve one release, contract, policy, and deployment-parity baseline.

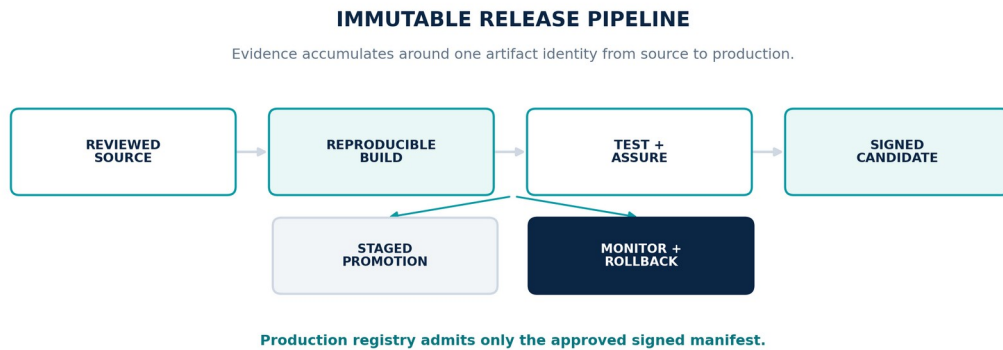


Figure 44. Immutable Release Pipeline. Evidence accumulates around one artifact identity from source to production.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Environment	Declared purpose, authority, data classes, identities, network, keys, models, integrations, observability, retention, and consequence.
Promotion	Authorized movement of an immutable release and configuration set after evidence closure for the target environment.
Data boundary	Synthetic, masked, generated, approved reference, controlled production copy, or live production with purpose and lifecycle.
Drift record	Observed divergence from declared environment state with owner, consequence, correction, and closure.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-69-001	Each environment SHALL have a machine-readable contract for authority, data, identity, network, keys, models, integrations, telemetry, support, retention, and permitted side effects.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-69-002	Promotion SHALL reference an approved immutable release manifest and target configuration package; manual recreation of environment state is prohibited.	Negative conformance test	Adversarial fixture proving the prohibited path is denied and audited.
ES-69-003	Production data SHALL NOT enter non-production without explicit purpose, minimization, masking, approval, isolation, retention, and verified deletion.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-69-004	Pre-production SHALL exercise production-equivalent control paths, policy, encryption, audit, scale characteristics, model and agent boundaries, migrations, failure modes, and rollback.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-69-005	Environment drift SHALL be detected continuously and corrected through the same controlled delivery path, with emergency intervention independently recorded and reconciled.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Target contract not met	Block promotion and identify the missing capability or control.

FAILURE CONDITION	REQUIRED BEHAVIOR
Unmanaged drift	Restrict changes, capture evidence, restore declared state, and re-run affected acceptance.
Data lifecycle violation	Quarantine the environment, stop processing, execute incident and deletion workflows, and verify residual closure.

Acceptance evidence

- Environment conformance scans compare runtime state with signed contracts and release manifests.
- Promotion rehearsals validate identity, policy, keys, integrations, telemetry, migrations, feature state, and rollback.
- Data-seeding tests trace provenance, masking, access, retention, and deletion across environment copies.
- Parity suites execute the same canonical scenarios across deployment and environment combinations.

CHAPTER ACCEPTANCE

Environment Promotion is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-035 · C-037 · C-041 · C-042 · C-045 · C-048 · C-049

70 Deployment Packaging

A deployment package is a complete, portable expression of The Eye for SaaS, Private Cloud, or On-Premise operation. It carries application, data, policy, model, agent, workflow, infrastructure, migration, observability, operations, verification, and exit contracts without creating deployment-specific semantics.

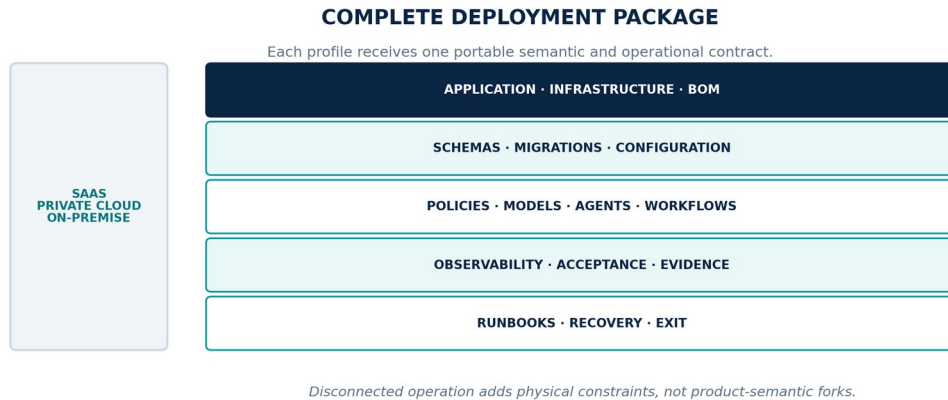


Figure 45. Complete Deployment Package. Each profile receives one portable semantic and operational contract.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Package	Signed release manifest plus all artifacts and declarative state required for the target deployment profile.
Deployment profile	SaaS, Private Cloud, On-Premise connected, restricted, or disconnected with declared physical and sovereignty boundaries.
Installer	Idempotent, resumable, preflighted workflow that realizes and verifies the signed target state.
Bill of materials	Application, infrastructure, software, model, agent, data, policy, schema, dependency, license, and vulnerability inventory.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-70-001	Deployment packages SHALL implement the same canonical objects, contracts, workflows, human authority, trust controls, quality semantics, audit, export, and replay across supported profiles.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.
ES-70-002	A package SHALL include signed artifacts, bills of materials, schemas, migrations, policies, models, agents, workflows, configuration, conformance fixtures, runbooks, recovery, and uninstall or exit procedures.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-70-003	Installation and upgrade SHALL preflight identity, keys, compute, storage, network, time, policy, dependencies, capacity, compatibility, backup, recovery, and rollback prerequisites.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-70-004	Disconnected packages SHALL support authenticated import, export, update, revocation, evidence return, time quality, and vulnerability handling without hidden external dependency.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-70-005	Installation SHALL be idempotent and resumable and SHALL not declare success until application-level conformance, state reconciliation, control, telemetry, and recovery verification complete.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Preflight fails	Make no consequential change and return exact unmet prerequisites and remediation evidence.
Upgrade partially applies	Resume or rollback from durable checkpoints, isolate affected state, and preserve the last verified service mode.
Offline revocation delayed	Apply declared validity and local deny policy and surface disconnected risk until signed update is admitted.

Acceptance evidence

- Fresh install, upgrade, rollback, scale, backup, restore, export, exit, and uninstall are tested for every supported deployment profile.
- Package contents reconcile exactly with signed manifests and bills of materials.
- Offline fixtures validate import, export, update, revocation, time uncertainty, evidence, and no-call-home operation.
- Canonical conformance suites produce equivalent semantic results across SaaS, Private Cloud, and On-Premise references.

CHAPTER ACCEPTANCE

Deployment Packaging is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-035 · C-041 · C-042 · C-043 · C-046 · C-048 · C-049 · C-050

71 Operational Acceptance

A capability becomes operational only when product, engineering, security, data, AI, privacy, reliability, support, customer, and governance authorities can operate, recover, explain, challenge, and retire it. A successful deployment is necessary but insufficient.

OPERATIONAL ACCEPTANCE						
	PRODUCT	SECURITY	DATA	AI	SRE	CUSTOMER
EVIDENCE	ENFORCE	VERIFY	ENFORCE	VERIFY	ENFORCE	VERIFY
AUTHORITY	VERIFY	ENFORCE	VERIFY	ENFORCE	VERIFY	ENFORCE
READINESS	ENFORCE	VERIFY	ENFORCE	VERIFY	ENFORCE	VERIFY
REVIEW	VERIFY	ENFORCE	VERIFY	ENFORCE	VERIFY	ENFORCE

Deployment success is necessary but never sufficient for acceptance.

Figure 46. Operational Acceptance. Independent authorities close evidence for an exact release and scope.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Acceptance scope	Exact release, configuration, tenant or domain, deployment, use cases, consequence classes, integrations, and exclusions.
Readiness evidence	Conformance, tests, security, privacy, data, AI fitness, SLOs, capacity, recovery, operations, support, training, and documentation.
Authority	Named accountable roles that accept owned risk and readiness without approving outside their domain.
Acceptance state	Pending, conditional, accepted, suspended, withdrawn, expired, or retired with effective time and obligations.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-71-001	Operational acceptance SHALL identify exact scope, owners, evidence versions, open risks, exceptions, constraints, support model, monitoring, review date, and withdrawal criteria.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.
ES-71-002	No single role SHALL accept architecture, security, privacy, AI fitness, data, operations, and business readiness on behalf of all independent authorities.	Automated conformance test	Versioned fixture and signed result for the exact provider, consumer, schema, and environment.
ES-71-003	Acceptance SHALL require verified runbooks, dashboards, alerts, on-call, incident roles, capacity, backup, recovery, degraded modes, customer communications, and escalation.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-71-004	Conditional acceptance SHALL be time-bounded, consequence-limited, monitored, reversible, and blocked from silent expansion to new tenants, data, decisions, or deployment profiles.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-71-005	Material change, incident, control failure, model or agent withdrawal, SLO exhaustion, or expired evidence SHALL trigger acceptance review or suspension according to policy.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Owner or evidence missing	Remain pending; production availability does not create acceptance.
Condition expires	Suspend affected scope unless closure or authorized renewal is complete before expiry.

FAILURE CONDITION	REQUIRED BEHAVIOR
Readiness degrades	Constrain or withdraw the capability and execute recovery and reacceptance workflow.

Acceptance evidence

- Acceptance checklist resolves every requirement to current signed evidence and accountable authority.
- Game days validate incident, degradation, recovery, customer communication, security, and decision-safety procedures.
- Support exercises reproduce diagnosis and remediation without uncontrolled customer-data or administrative access.
- Periodic reviews reconcile actual use, incidents, SLOs, overrides, outcomes, exceptions, and retirement obligations.

CHAPTER ACCEPTANCE

Operational Acceptance is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-004 · C-005 · C-035 · C-043 · C-045 · C-049 · C-051 · C-052

72 Engineering Governance and Exceptions

Engineering governance preserves a coherent operating system across teams, customers, deployments, and generations of technology. It controls standards, requirements, interfaces, object semantics, ownership, exceptions, technical debt, deprecation, and evidence without substituting committee opinion for explicit authority.

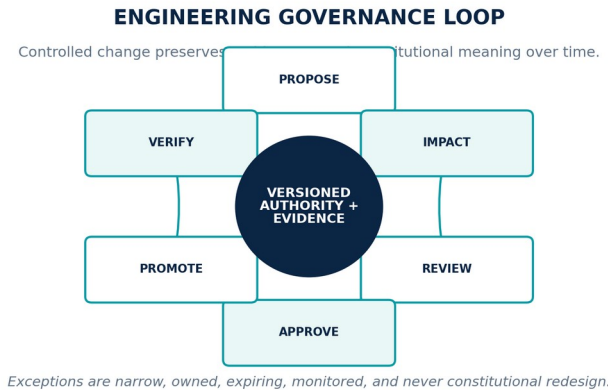


Figure 47. Engineering Governance Loop. Controlled change preserves architecture and constitutional meaning over time.

Engineering contract

CONTRACT ELEMENT	BINDING DEFINITION
Governed artifact	Requirement, contract, schema, ontology, policy, model, agent, workflow, package, configuration, SLO, test, runbook, or decision record.
Change proposal	Current and proposed state, authority, rationale, alternatives, impact, migration, evidence, rollout, rollback, and communication.
Exception	Narrow time-bounded deviation with owner, scope, risk acceptance, compensating control, telemetry, expiry, and closure.
Deprecation	Announced lifecycle with replacement, compatibility, migration, support, consumer acknowledgement, and final removal evidence.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
ES-72-001	Every governed artifact SHALL have an accountable owner, approving authority, lifecycle, version, effective time, dependencies, review cadence, and supersession path.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.
ES-72-002	Changes to constitutional meaning or frozen architecture SHALL follow their controlling amendment authority; Volume 4 governance cannot approve semantic redesign by exception.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.
ES-72-003	Interface, object, policy, model, agent, workflow, and deployment changes SHALL include consumer impact, compatibility, migration, replay, rollback, and evidence plans before approval.	Fault / recovery test	Fault trace, recovered state digest, reconciliation record, and declared customer-visible mode.
ES-72-004	Exceptions SHALL NOT remove human final authority, provenance, tenant isolation, audit, deployment semantic parity, or any constitutional invariant.	Security test	Signed policy/security suite result bound to artifact, configuration, deployment, and data class.
ES-72-005	Technical debt and deprecation SHALL be maintained as decision-active risk with affected objectives, customers, consequences, owner, due date, investment, and verified closure.	Test + accountable review	Executable result plus named independent authority, scope, reservations, and effective time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Authority disputed	Preserve the current controlled state and escalate through the specification hierarchy.
Exception expired or expanded	Mark non-conformant, block new promotion or use, and execute closure or amendment.
Consumer cannot migrate	Maintain declared compatibility or constrain retirement through an approved bounded plan; never silently break the contract.

Acceptance evidence

- Governance registry reconciles owners, versions, dependencies, reviews, exceptions, debt, deprecations, and closure evidence.
- Change-impact tests trace affected components, contracts, objects, policies, models, agents, workflows, deployments, customers, and documents.
- Exception audits verify scope, compensating controls, telemetry, expiry, non-expansion, and independent approval.
- Annual engineering-baseline review proves every constitutional invariant and architecture decision remains implemented and testable.

CHAPTER ACCEPTANCE

Engineering Governance and Exceptions is conformant only when every requirement is implemented for the declared scope, all failure states are observable and recoverable, and the listed evidence resolves to the exact released artifacts and governing versions.

CONSTITUTIONAL INHERITANCE / C-004 · C-008 · C-035 · C-042 · C-049 · C-051 · C-052

PART APP

Engineering Reference Appendices

Normative implementation, contract, object, quality, verification, traceability, and terminology catalogs.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

A Logical Component Implementation Catalog

The 92 stable logical component identifiers bind Volume 3 architecture to engineering ownership. Deployable decomposition may change only when duty, state authority, contract, controls, telemetry, and failure behavior remain independently resolvable.

ID	LAYER	COMPONENT	RUNTIME	STATE	SCALE KEY	FAILURE / RECOVERY
L1-C01	World Observation Layer	Source Registry and Contract Service	Registry + resolver	Canonical control state	Domain / namespace	Last verified read; block unsafe writes
L1-C02	World Observation Layer	Connector Runtime and Adapter SDK	Stateless gateway + durable checkpoints	Checkpoint / quarantine	Source / destination / tenant	Backpressure, quarantine, bounded retry
L1-C03	World Observation Layer	Collection Scheduler and Subscription Manager	Coordinator + durable queue	Workflow / scheduling state	Tenant / workload class	Pause admission; drain or resume safely
L1-C04	World Observation Layer	Streaming and Batch Receivers	Owned service	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L1-C05	World Observation Layer	Raw Preservation and Evidence Vault	Canonical state service	Authoritative	Tenant / object partition	Restore, reconcile, then reopen writes
L1-C06	World Observation Layer	Authenticity and Chain-of-Custody Service	Owned service	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L1-C07	World Observation Layer	Quarantine and Malware Inspection	Owned service	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L1-C08	World Observation Layer	Source Health, Freshness, and Coverage Monitor	Stream + evaluation worker	Metrics + fitness decisions	Artifact / product / window	Expose indeterminate quality; constrain use
L1-C09	World Observation Layer	Correction and Withdrawal Intake	Owned service	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L2-C01	Intelligence Layer	Format Detection, OCR, Transcription, and Parsing	Owned service	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L2-C02	Intelligence Layer	Cleaning, Normalization, and Language Services	Owned service	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L2-C03	Intelligence Layer	Classification and Routing	Owned service	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L2-C04	Intelligence Layer	Summarization and Structured Extraction	Owned service	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L2-C05	Intelligence Layer	Named-Entity Recognition and Candidate Linking	Owned service	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L2-C06	Intelligence Layer	Relationship, Event, and Claim Extraction	Owned service	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L2-C07	Intelligence Layer	Contradiction and Corroboration Detection	Owned service	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L2-C08	Intelligence Layer	Confidence and Truth-State Assessor	Horizontally scaled worker	Immutable result + checkpoint	Tenant / object / job	Abstain or partial result; never fabricate completion
L2-C09	Intelligence Layer	Human Review Queue and Adjudication	Durable workflow	Workflow history	Instance / tenant	Resume from last commit; escalate residual effect
L2-C10	Intelligence Layer	Transformation Registry and Evaluation Harness	Registry + resolver	Canonical control state	Domain / namespace	Last verified read; block unsafe writes
L3-C01	Enterprise Memory	Memory Object Service	Canonical state service	Authoritative	Tenant / object partition	Restore, reconcile, then reopen writes
L3-C02	Enterprise Memory	Document and Artifact Repository	Canonical state service	Authoritative	Tenant / object partition	Restore, reconcile, then reopen writes
L3-C03	Enterprise Memory	Temporal Record Store	Canonical state service	Authoritative	Tenant / object partition	Restore, reconcile, then reopen writes
L3-C04	Enterprise Memory	Version and Correction Ledger	Owned service	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract

ID	LAYER	COMPONENT	RUNTIME	STATE	SCALE KEY	FAILURE / RECOVERY
L3-C05	Enterprise Memory	Retention, Legal Hold, and Deletion Service	Owned service	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L3-C06	Enterprise Memory	Permission-Aware Semantic Index	Owned service	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L3-C07	Enterprise Memory	Memory Retrieval and Context Assembly	Canonical state service	Authoritative	Tenant / object partition	Restore, reconcile, then reopen writes
L3-C08	Enterprise Memory	Archive and Cold-Tier Manager	Coordinator + durable queue	Workflow / scheduling state	Tenant / workload class	Pause admission; drain or resume safely
L3-C09	Enterprise Memory	Memory Integrity and Orphan Detection	Canonical state service	Authoritative	Tenant / object partition	Restore, reconcile, then reopen writes
L4-C01	Knowledge Graph	Ontology Registry and Governance Workflow	Registry + resolver	Canonical control state	Domain / namespace	Last verified read; block unsafe writes
L4-C02	Knowledge Graph	Identity Resolution and Entity Stewardship	Owned service	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L4-C03	Knowledge Graph	Graph Write Validator	Owned service	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L4-C04	Knowledge Graph	Temporal Graph Store	Canonical state service	Authoritative	Tenant / object partition	Restore, reconcile, then reopen writes
L4-C05	Knowledge Graph	Provenance and Evidence Edge Service	Owned service	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L4-C06	Knowledge Graph	Graph Query and Traversal Service	Owned service	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L4-C07	Knowledge Graph	Reasoning and Constraint Engine	Horizontally scaled worker	Immutable result + checkpoint	Tenant / object / job	Abstain or partial result; never fabricate completion
L4-C08	Knowledge Graph	Graph Change Subscription Service	Owned service	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L4-C09	Knowledge Graph	Merge, Split, and Reconciliation Manager	Coordinator + durable queue	Workflow / scheduling state	Tenant / workload class	Pause admission; drain or resume safely
L4-C10	Knowledge Graph	Graph Quality and Orphan Monitor	Stream + evaluation worker	Metrics + fitness decisions	Artifact / product / window	Expose indeterminate quality; constrain use
L5-C01	Digital Twins	Twin Registry and Boundary Service	Registry + resolver	Canonical control state	Domain / namespace	Last verified read; block unsafe writes
L5-C02	Digital Twins	State Estimation and Reconciliation	Owned service	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L5-C03	Digital Twins	Model Binding and Operating-Envelope Registry	Registry + resolver	Canonical control state	Domain / namespace	Last verified read; block unsafe writes
L5-C04	Digital Twins	Snapshot and Version Manager	Coordinator + durable queue	Workflow / scheduling state	Tenant / workload class	Pause admission; drain or resume safely
L5-C05	Digital Twins	Branch and Time-Travel Manager	Coordinator + durable queue	Workflow / scheduling state	Tenant / workload class	Pause admission; drain or resume safely
L5-C06	Digital Twins	Twin Dependency Mapper	Owned service	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L5-C07	Digital Twins	Behavior and Constraint Runtime	Owned service	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L5-C08	Digital Twins	Validation and Calibration Service	Stream + evaluation worker	Metrics + fitness decisions	Artifact / product / window	Expose indeterminate quality; constrain use
L5-C09	Digital Twins	Synthetic-State Labelling and Isolation	Owned service	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L6-C01	Prediction Engine	Forecast Registry and Horizon Service	Registry + resolver	Canonical control state	Domain / namespace	Last verified read; block unsafe writes
L6-C02	Prediction Engine	Feature and Context Assembler	Owned service	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L6-C03	Prediction Engine	Model Gateway and Candidate Router	Stateless gateway + durable checkpoints	Checkpoint / quarantine	Source / destination / tenant	Backpressure, quarantine, bounded retry
L6-C04	Prediction Engine	Ensemble and Disagreement Manager	Coordinator + durable queue	Workflow / scheduling state	Tenant / workload class	Pause admission; drain or resume safely
L6-C05	Prediction Engine	Calibration and Uncertainty Service	Stream + evaluation worker	Metrics + fitness decisions	Artifact / product / window	Expose indeterminate quality; constrain use
L6-C06	Prediction Engine	Driver Attribution and Sensitivity Engine	Horizontally scaled worker	Immutable result + checkpoint	Tenant / object / job	Abstain or partial result; never fabricate completion

ID	LAYER	COMPONENT	RUNTIME	STATE	SCALE KEY	FAILURE / RECOVERY
L6-C07	Prediction Engine	Forecast Evaluation and Backtesting	Stream + evaluation worker	Metrics + fitness decisions	Artifact / product / window	Expose indeterminate quality; constrain use
L6-C08	Prediction Engine	Drift and Refresh Monitor	Stream + evaluation worker	Metrics + fitness decisions	Artifact / product / window	Expose indeterminate quality; constrain use
L6-C09	Prediction Engine	Forecast Version Store	Canonical state service	Authoritative	Tenant / object partition	Restore, reconcile, then reopen writes
L7-C01	Scenario Intelligence	Scenario Registry and Portfolio Manager	Registry + resolver	Canonical control state	Domain / namespace	Last verified read; block unsafe writes
L7-C02	Scenario Intelligence	Assumption and Hypothesis Register	Owned service	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L7-C03	Scenario Intelligence	Branch and Path Engine	Horizontally scaled worker	Immutable result + checkpoint	Tenant / object / job	Abstain or partial result; never fabricate completion
L7-C04	Scenario Intelligence	Actor, Driver, and Mechanism Model	Owned service	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L7-C05	Scenario Intelligence	Indicator and Signpost Service	Owned service	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L7-C06	Scenario Intelligence	Scenario Coherence Validator	Owned service	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L7-C07	Scenario Intelligence	Intervention and Impact Mapper	Owned service	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L7-C08	Scenario Intelligence	Scenario Comparison and Review Workspace	Durable workflow	Workflow history	Instance / tenant	Resume from last commit; escalate residual effect
L7-C09	Scenario Intelligence	Scenario Package Import and Localization	Owned service	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L8-C01	Simulation Engine	Experiment Registry and Run Planner	Registry + resolver	Canonical control state	Domain / namespace	Last verified read; block unsafe writes
L8-C02	Simulation Engine	Twin, Scenario, and Model Version Resolver	Horizontally scaled worker	Immutable result + checkpoint	Tenant / object / job	Abstain or partial result; never fabricate completion
L8-C03	Simulation Engine	Constraint and Intervention Engine	Horizontally scaled worker	Immutable result + checkpoint	Tenant / object / job	Abstain or partial result; never fabricate completion
L8-C04	Simulation Engine	Seed and Determinism Manager	Coordinator + durable queue	Workflow / scheduling state	Tenant / workload class	Pause admission; drain or resume safely
L8-C05	Simulation Engine	Solver and Simulator Adapter Fabric	Stateless gateway + durable checkpoints	Checkpoint / quarantine	Source / destination / tenant	Backpressure, quarantine, bounded retry
L8-C06	Simulation Engine	Distributed Run Orchestrator	Coordinator + durable queue	Workflow / scheduling state	Tenant / workload class	Pause admission; drain or resume safely
L8-C07	Simulation Engine	Impact and Second-Order Effect Analyzer	Horizontally scaled worker	Immutable result + checkpoint	Tenant / object / job	Abstain or partial result; never fabricate completion
L8-C08	Simulation Engine	Sensitivity and Robustness Service	Owned service	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L8-C09	Simulation Engine	Reproducibility and Validation Store	Canonical state service	Authoritative	Tenant / object partition	Restore, reconcile, then reopen writes
L9-C01	Decision Intelligence	Decision Registry and Context Resolver	Registry + resolver	Canonical control state	Domain / namespace	Last verified read; block unsafe writes
L9-C02	Decision Intelligence	Objective, Constraint, and Obligation Service	Owned service	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L9-C03	Decision Intelligence	Option and Alternative Manager	Coordinator + durable queue	Workflow / scheduling state	Tenant / workload class	Pause admission; drain or resume safely
L9-C04	Decision Intelligence	Evidence and Foresight Assembler	Owned service	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L9-C05	Decision Intelligence	Criteria, Trade-off, and Information-Value Engine	Horizontally scaled worker	Immutable result + checkpoint	Tenant / object / job	Abstain or partial result; never fabricate completion
L9-C06	Decision Intelligence	Recommendation and Explanation Service	Owned service	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L9-C07	Decision Intelligence	Approval, Dissent, and Override Workflow	Durable workflow	Workflow history	Instance / tenant	Resume from last commit; escalate residual effect
L9-C08	Decision Intelligence	Decision Record and Commitment Service	Durable workflow	Workflow history	Instance / tenant	Resume from last commit; escalate residual effect
L9-C09	Decision Intelligence	Monitoring Contract and Review Scheduler	Durable workflow	Workflow history	Instance / tenant	Resume from last commit; escalate residual effect

ID	LAYER	COMPONENT	RUNTIME	STATE	SCALE KEY	FAILURE / RECOVERY
L10-C01	Executive Operating System	Executive Workspace and Responsibility Model	Read model + command façade	Rebuildable projection	Tenant / audience / session	Stale-visible or read-only with watermark
L10-C02	Executive Operating System	Attention Policy and Materiality Engine	Horizontally scaled worker	Immutable result + checkpoint	Tenant / object / job	Abstain or partial result; never fabricate completion
L10-C03	Executive Operating System	Live Briefing Service	Read model + command façade	Rebuildable projection	Tenant / audience / session	Stale-visible or read-only with watermark
L10-C04	Executive Operating System	Decision and Scenario Rooms	Read model + command façade	Rebuildable projection	Tenant / audience / session	Stale-visible or read-only with watermark
L10-C05	Executive Operating System	Objective, Commitment, and Outcome Review	Durable workflow	Workflow history	Instance / tenant	Resume from last commit; escalate residual effect
L10-C06	Executive Operating System	Strategic Cadence Scheduler	Coordinator + durable queue	Workflow / scheduling state	Tenant / workload class	Pause admission; drain or resume safely
L10-C07	Executive Operating System	Notification and Escalation Service	Owned service	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L10-C08	Executive Operating System	Role-Appropriate Access Modes	Owned service	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract
L10-C09	Executive Operating System	Executive Search and Explanation	Owned service	Declared by component	Tenant / domain	Isolate, expose state, and recover by contract

B Interface, API, and Event Catalog

The 50 canonical layer interfaces are stable semantic boundaries. Protocol bindings may vary, but envelope, authorization, reliability, error, compatibility, audit, and deployment-parity behavior remains one contract.

ID	LAYER	INTERFACE	STYLE	SEMANTIC CONTRACT	RELIABILITY	FAILURE
L1-I01	World Observation Layer	RegisterSource	Command	Creates or versions a governed source contract after authorization and rights checks.	Idempotent acceptance and one authoritative effect	Reject conflict; retry only with same idempotency boundary
L1-I02	World Observation Layer	Acquire	Command / stream	Collects content or state under contract, purpose, rate, and residency constraints.	At-least-once segments under stable partition key	Backpressure, resume, or explicit incomplete range
L1-I03	World Observation Layer	ObservationRecorded	Domain event	Announces an immutable observation and evidence reference.	At-least-once delivery; consumer deduplication and checkpoint	Quarantine invalid event; reconcile committed publication
L1-I04	World Observation Layer	SourceHealthChanged	Domain event	Reports freshness, completeness, authenticity, latency, or coverage state changes.	At-least-once delivery; consumer deduplication and checkpoint	Quarantine invalid event; reconcile committed publication
L1-I05	World Observation Layer	CorrectionReceived	Domain event	Introduces source correction, withdrawal, or supersession without destroying prior history.	At-least-once delivery; consumer deduplication and checkpoint	Quarantine invalid event; reconcile committed publication
L2-I01	Intelligence Layer	TransformEvidence	Command	Runs a declared transformation plan against immutable evidence.	Idempotent acceptance and one authoritative effect	Reject conflict; retry only with same idempotency boundary
L2-I02	Intelligence Layer	IntelligenceObjectProposed	Domain event	Publishes a claim, event, entity, relationship, or assessment candidate with lineage.	At-least-once delivery; consumer deduplication and checkpoint	Quarantine invalid event; reconcile committed publication
L2-I03	Intelligence Layer	ContradictionDetected	Domain event	Links incompatible assertions without collapsing them into a single truth state.	At-least-once delivery; consumer deduplication and checkpoint	Quarantine invalid event; reconcile committed publication
L2-I04	Intelligence Layer	ReviewRequested	Workflow event	Routes ambiguous, high-impact, low-confidence, or policy-sensitive output to accountable humans.	At-least-once delivery; consumer deduplication and checkpoint	Quarantine invalid event; reconcile committed publication
L2-I05	Intelligence Layer	TransformationEvaluated	Domain event	Records quality, safety, cost, latency, and fitness for the producing version.	At-least-once delivery; consumer deduplication and checkpoint	Quarantine invalid event; reconcile committed publication
L3-I01	Enterprise Memory	CommitMemory	Command	Persists a governed object version with policy, provenance, classification, and temporal metadata.	Idempotent acceptance and one authoritative effect	Reject conflict; retry only with same idempotency boundary
L3-I02	Enterprise Memory	RetrieveContext	Query	Returns policy-filtered memory and explanation links for an explicit purpose.	No state change; consistency and staleness explicit	Return partial/state only with declared product state
L3-I03	Enterprise Memory	MemoryCorrected	Domain event	Versions a correction and identifies affected downstream objects.	At-least-once delivery; consumer deduplication and checkpoint	Quarantine invalid event; reconcile committed publication

ID	LAYER	INTERFACE	STYLE	SEMANTIC CONTRACT	RELIABILITY	FAILURE
L3-I04	Enterprise Memory	RetentionActionDude	Domain event	Initiates archive, review, legal hold, deletion, or customer export workflow.	At-least-once delivery; consumer deduplication and checkpoint	Quarantine invalid event; reconcile committed publication
L3-I05	Enterprise Memory	DeletionVerified	Audit event	Proves scope, authorization, execution, and residual references for a deletion action.	At-least-once delivery; consumer deduplication and checkpoint	Quarantine invalid event; reconcile committed publication
L4-I01	Knowledge Graph	ResolveIdentity	Command / query	Returns or proposes canonical identity with evidence, confidence, and stewardship state.	Idempotent acceptance and one authoritative effect	Reject conflict; retry only with same idempotency boundary
L4-I02	Knowledge Graph	CommitGraphRevision	Command	Applies an atomic, validated node, edge, ontology, and provenance change set.	Idempotent acceptance and one authoritative effect	Reject conflict; retry only with same idempotency boundary
L4-I03	Knowledge Graph	GraphChanged	Domain event	Publishes affected identities, relationships, temporal scopes, and downstream subscriptions.	At-least-once delivery; consumer deduplication and checkpoint	Quarantine invalid event; reconcile committed publication
L4-I04	Knowledge Graph	TraverseContext	Query	Returns policy-filtered paths with provenance, time, truth state, and confidence.	No state change; consistency and staleness explicit	Return partial/state only with declared product state
L4-I05	Knowledge Graph	OntologyChangeProposed	Workflow event	Initiates compatibility, migration, domain, and governance review.	At-least-once delivery; consumer deduplication and checkpoint	Quarantine invalid event; reconcile committed publication
L5-I01	Digital Twins	CreateTwinDefinition	Command	Declares boundary, identity, evidence, behaviors, owners, validation, and limitations.	Idempotent acceptance and one authoritative effect	Reject conflict; retry only with same idempotency boundary
L5-I02	Digital Twins	ReconcileTwin	Command	Produces a new observed-state snapshot from evidence and graph revisions.	Idempotent acceptance and one authoritative effect	Reject conflict; retry only with same idempotency boundary
L5-I03	Digital Twins	BranchTwin	Command	Creates an isolated synthetic branch for scenario or simulation use.	Idempotent acceptance and one authoritative effect	Reject conflict; retry only with same idempotency boundary
L5-I04	Digital Twins	TwinStateChanged	Domain event	Publishes version, changed variables, confidence, freshness, and dependency impacts.	At-least-once delivery; consumer deduplication and checkpoint	Quarantine invalid event; reconcile committed publication
L5-I05	Digital Twins	ValidateTwin	Command / event	Evaluates state and model fitness inside the declared operating envelope.	At-least-once delivery; consumer deduplication and checkpoint	Quarantine invalid event; reconcile committed publication
L6-I01	Prediction Engine	RequestForecast	Command	Creates or refreshes a forecast under a declared objective, target, horizon, context, and policy.	Idempotent acceptance and one authoritative effect	Reject conflict; retry only with same idempotency boundary
L6-I02	Prediction Engine	ForecastProduced	Domain event	Publishes distribution, confidence, calibration, drivers, lineage, and expiry.	At-least-once delivery; consumer deduplication and checkpoint	Quarantine invalid event; reconcile committed publication
L6-I03	Prediction Engine	ForecastFitnessChanged	Domain event	Signals drift, calibration failure, data shift, or operating-envelope breach.	At-least-once delivery; consumer deduplication and checkpoint	Quarantine invalid event; reconcile committed publication

ID	LAYER	INTERFACE	STYLE	SEMANTIC CONTRACT	RELIABILITY	FAILURE
L6-I04	Prediction Engine	BacktestForecast	Command	Evaluates historical versions without contaminating prior decision context.	Idempotent acceptance and one authoritative effect	Reject conflict; retry only with same idempotency boundary
L6-I05	Prediction Engine	ForecastWithdrawn	Domain event	Marks a version unfit and propagates impact to scenarios, warnings, simulations, and decisions.	At-least-once delivery; consumer deduplication and checkpoint	Quarantine invalid event; reconcile committed publication
L7-I01	Scenario Intelligence	CreateScenario	Command	Declares baseline, branch logic, assumptions, drivers, owners, indicators, and review policy.	Idempotent acceptance and one authoritative effect	Reject conflict; retry only with same idempotency boundary
L7-I02	Scenario Intelligence	BranchScenario	Command	Creates a versioned upside, downside, disruption, or user-defined alternative.	Idempotent acceptance and one authoritative effect	Reject conflict; retry only with same idempotency boundary
L7-I03	Scenario Intelligence	ScenarioIndicatorChanged	Domain event	Updates branch relevance when a monitored signpost moves.	At-least-once delivery; consumer deduplication and checkpoint	Quarantine invalid event; reconcile committed publication
L7-I04	Scenario Intelligence	ScenarioCoherenceFailed	Domain event	Identifies internal inconsistency, invalid assumption, or incompatible dependency.	At-least-once delivery; consumer deduplication and checkpoint	Quarantine invalid event; reconcile committed publication
L7-I05	Scenario Intelligence	ScenarioReviewed	Workflow event	Records human review, dissent, continuation, retirement, or promotion to simulation.	At-least-once delivery; consumer deduplication and checkpoint	Quarantine invalid event; reconcile committed publication
L8-I01	Simulation Engine	SubmitExperiment	Command	Validates declared state, versions, constraints, budgets, policy, and expected outputs.	Idempotent acceptance and one authoritative effect	Reject conflict; retry only with same idempotency boundary
L8-I02	Simulation Engine	SimulationStarted	Domain event	Records resolved artifacts, execution identity, environment, seed, and run state.	At-least-once delivery; consumer deduplication and checkpoint	Quarantine invalid event; reconcile committed publication
L8-I03	Simulation Engine	SimulationCompleted	Domain event	Publishes outputs, impacts, uncertainty, sensitivity, validation, and resource evidence.	At-least-once delivery; consumer deduplication and checkpoint	Quarantine invalid event; reconcile committed publication
L8-I04	Simulation Engine	ChallengeSimulation	Workflow command	Re-runs or disputes assumptions, models, constraints, and interpretation.	Idempotent acceptance and one authoritative effect	Reject conflict; retry only with same idempotency boundary
L8-I05	Simulation Engine	SimulationInvalidated	Domain event	Marks results unfit and identifies dependent decision packages.	At-least-once delivery; consumer deduplication and checkpoint	Quarantine invalid event; reconcile committed publication
L9-I01	Decision Intelligence	OpenDecision	Command	Declares question, accountable authority, scope, objectives, options, stakeholders, and required evidence.	Idempotent acceptance and one authoritative effect	Reject conflict; retry only with same idempotency boundary
L9-I02	Decision Intelligence	DecisionPackageReady	Domain event	Publishes a complete reviewable package with uncertainty, alternatives, dissent, and provenance.	At-least-once delivery; consumer deduplication and checkpoint	Quarantine invalid event; reconcile committed publication
L9-I03	Decision Intelligence	ApproveDecision	Human command	Records authenticated approval, rationale, obligations, overrides, and commitment state.	Exactly-once institutional effect through idempotency and durable workflow	Reject stale authority, context, version, or approval

ID	LAYER	INTERFACE	STYLE	SEMANTIC CONTRACT	RELIABILITY	FAILURE
L9-I04	Decision Intelligence	DecisionCommitted	Domain event	Activates commitments, execution handoff, monitoring conditions, and replay snapshot.	At-least-once delivery; consumer deduplication and checkpoint	Quarantine invalid event; reconcile committed publication
L9-I05	Decision Intelligence	DecisionReopened	Workflow event	Re-enters the decision lifecycle when evidence, assumptions, policy, or monitoring conditions change.	At-least-once delivery; consumer deduplication and checkpoint	Quarantine invalid event; reconcile committed publication
L10-I01	Executive Operating System	PublishBriefing	Command	Creates a live briefing bound to evidence, changes, objectives, decisions, and audience policy.	Idempotent acceptance and one authoritative effect	Reject conflict; retry only with same idempotency boundary
L10-I02	Executive Operating System	MaterialChangeRaised	Domain event	Routes change to accountable roles based on consequence, confidence, urgency, and attention policy.	At-least-once delivery; consumer deduplication and checkpoint	Quarantine invalid event; reconcile committed publication
L10-I03	Executive Operating System	ReviewConvened	Workflow event	Opens a governed review around a declared objective, decision, scenario, commitment, or outcome.	At-least-once delivery; consumer deduplication and checkpoint	Quarantine invalid event; reconcile committed publication
L10-I04	Executive Operating System	ExecutiveActionRequested	Human command	Requests analysis, scenario, simulation, decision, delegation, suppression, or follow-up.	Exactly-once institutional effect through idempotency and durable workflow	Reject stale authority, context, version, or approval
L10-I05	Executive Operating System	AttentionPolicyChanged	Governance event	Versions materiality, escalation, suppression, and notification rules.	At-least-once delivery; consumer deduplication and checkpoint	Quarantine invalid event; reconcile committed publication

Common interface obligations

- Canonical envelope, version negotiation, normalized error model, idempotency or ordering boundary, deadline, policy, telemetry, and audit causation.
- No direct cross-owner database write, unversioned payload, hidden retry, silent partial result, or deployment-specific semantic path.
- Provider and consumer conformance fixtures cover normal, unauthorized, invalid, conflicting, abstained, stale, degraded, replay, and recovery behavior.

C Canonical Object Schema Profiles

Each canonical intelligence object inherits the universal header. This appendix defines the minimum domain payload and lifecycle profile; domain schemas may add governed fields without removing universal identity, time, truth, provenance, policy, quality, and correction obligations.

COD E	OBJECT	OWNER	MINIMUM DOMAIN PAYLOAD	LIFECYCLE	ENGINEERING MEANING
SRC	Source Contract	Observation	source authority + rights + correction policy	Draft → approved → active → suspended → retired	Authority, rights, purpose, schema, latency, quality, authenticity, correction behavior, and ownership for a source.
OBS	Observation Record	Observation	source identity + observation time + acquisition context	Recorded → validated/quarantined → corrected/superseded	An immutable record that a source produced content or state at a specific observation time.
EVD	Evidence Object	Observation / Intelligence	content digest + custody + transformations + coverage	Captured → verified → admitted/quarantined → retained/deleted	Preserved source material plus chain of custody, transformations, classification, quality, and coverage context.
CLM	Claim	Intelligence	subject + predicate + object/value + evidence + temporal scope	Proposed → reviewed → admitted → disputed/withdrawn/superseded	A source-attributed assertion with truth state, confidence, temporal scope, and supporting or contradicting evidence.
EVT	Event	Intelligence	event type + actors + location + mechanism + time	Proposed → resolved → admitted → corrected/superseded	A temporally scoped occurrence connected to actors, locations, assets, mechanisms, and evidence.
ENT	Entity	Knowledge Graph	canonical identity + aliases + type + stewardship	Proposed → resolved → stewarded → merged/split/retired	A governed identity representing a company, country, person, product, technology, market, asset, objective, or other canonical thing.
REL	Relationship	Knowledge Graph	typed endpoints + direction + temporal scope + provenance	Proposed → validated → active → disputed/superseded	A typed, temporal, provenance-bearing connection between governed entities or intelligence objects.
ASM	Assessment	Intelligence	question + judgment + evidence + method + dissent	Draft → reviewed → published → challenged/superseded	A versioned analytical judgment that separates evidence, method, uncertainty, dissent, and accountable author.
MEM	Memory Entry	Enterprise Memory	memory class + context + retention + correction	Committed → active/archive/hold → corrected/deleted	A durable, permission-aware, versioned organizational record with retention and correction semantics.
GRV	Graph Revision	Knowledge Graph	atomic node/edge/ontology change set + validation	Proposed → validated → committed → superseded	An atomic, validated set of node, edge, ontology, provenance, and temporal changes.
OBJ	Objective	Strategy Graph	intent + owner + horizon + measures + constraints	Proposed → approved → active → achieved/retired	A governed strategic intent with owner, horizon, measures, constraints, dependencies, and review cadence.
RSK	Risk / Opportunity	Strategy Graph	driver + exposure + probability + consequence + response	Candidate → assessed → monitored/treated → closed	A symmetric strategic exposure connected to drivers, affected objectives, scenarios, responses, and owners.

COD E	OBJECT	OWNER	MINIMUM DOMAIN PAYLOAD	LIFECYCLE	ENGINEERING MEANING
TWN	Twin Snapshot	Digital Twins	boundary + state vector + evidence + model + validity	Defined → reconciled → valid/stale → superseded	A versioned state of a declared system boundary, grounded in evidence, graph identity, models, assumptions, and validity.
FCT	Forecast Package	Prediction	target + horizon + distribution + drivers + calibration	Requested → produced → active/expired → withdrawn	A horizon-specific probability distribution with drivers, calibration, sensitivity, model lineage, and failure conditions.
SCN	Scenario Branch	Scenario	baseline + branch assumptions + actors + indicators + impacts	Draft → coherent → reviewed → monitored/retired	A coherent future branch with baseline, assumptions, actors, drivers, indicators, interventions, impacts, and review state.
SIM	Simulation Run	Simulation	initial state + interventions + constraints + seed + outputs	Submitted → running → completed → challenged/invalidated	A reproducible experiment with initial state, model and twin versions, constraints, seeds, interventions, outputs, and sensitivity.
DEC	Decision Package	Decision	question + objectives + options + evidence + human authority	Opened → ready → decided → committed/reopened/closed	A complete choice environment containing objectives, options, evidence, forecasts, scenarios, simulations, uncertainty, dissent, approvals, and monitoring.
CMT	Commitment	Decision / Executive	approved action + owner + resources + obligations + measures	Proposed → approved → active → fulfilled/breached/closed	An approved institutional promise with owner, resources, obligations, execution state, indicators, review dates, and closure criteria.
OUT	Outcome Record	Executive / Learning	observed result + context + indicators + attribution limits	Observed → assessed → reviewed → superseded	Observed consequences of an approved decision, linked to actions, context, later evidence, and evaluation.
LES	Lesson Proposal	Learning	evidence + proposed change + evaluation + approving authority	Proposed → evaluated → approved/rejected → released	A proposed, evidence-backed change to institutional memory, model, policy, workflow, ontology, or practice pending governance.
WRN	Warning Product	Proactive Intelligence	change + mechanism + materiality + window + owner	Candidate → issued → acknowledged → resolved/dismissed	A deduplicated, prioritized alert with evidence, confidence, consequence, response window, owner, and closure criteria.
AGX	Agent Execution	Agent System	agent + plan + grants + tools + budgets + trace + result	Admitted → running → suspended/completed/failed	A bounded execution trace with agent identity, owner, purpose, plan, tools, data scope, budgets, outputs, evaluation, and stop state.
POL	Policy Decision	Control Plane	principal + action + object + purpose + decision + obligations	Evaluated → enforced → expired/superseded	A recorded authorization, obligation, denial, exception, or purpose decision bound to identity, object, action, and time.
AUD	Audit Evidence	Control Plane	actor + authority + action + target + result + proof	Created → sealed → retained/exported → expired	Tamper-evident proof of material access, transformation, model or agent action, approval, decision, release, correction, or export.

Universal schema inheritance

HEADER FAMILY	MANDATORY SCHEMA CONTENT
Identity	Tenant, domain, object ID, object type, schema version, object version, owner, and stable references.
Time	Event, observation, valid, system, issue, expiry, and clock-quality fields where meaningful.
Truth	Truth state, confidence, uncertainty, synthetic marker, lifecycle, withdrawal, and supersession.
Provenance	Sources, evidence, transformations, models, agents, humans, policy decisions, causation, and corrections.
Control	Classification, purpose, rights, residency, retention, access policy, export, deletion, and holds.
Quality	Completeness, freshness, coverage, validity, calibration or fitness, decision-use constraint, and owner.
Operations	Correlation, trace, idempotency, checksum, audit, producer, deployment, and admission evidence.

D Canonical Envelope Field Dictionary

Envelope fields carry the identity, intent, semantics, time, trust, control, operations, and integrity required to preserve meaning across APIs, events, workflows, agents, tools, models, deployments, and external transfer.

FIELD	FAMILY	DEFINITION	PRESENCE	APPLIES TO
message_id	Identity	Globally unique immutable message identifier.	Required	All
tenant_id	Identity	Authenticated customer tenant boundary.	Required	All
domain_id	Identity	Customer Intelligence Domain within the tenant.	Required	All
principal_id	Identity	Authenticated human, workload, agent, device, or source identity.	Required	All
delegation_id	Identity	Reference to constrained delegated authority where applicable.	Conditional	Commands / queries / tools
purpose_id	Intent	Approved institutional purpose for processing.	Required	All protected operations
action	Intent	Normalized requested action or committed transition.	Required	All
side_effect_class	Intent	None, reversible, compensatable, approval-required, irreversible, or prohibited.	Required	Commands / tools
object_type	Semantics	Canonical object or contract type.	Required	All
object_id	Semantics	Stable canonical identity when the operation addresses an object.	Conditional	Object operations
object_version	Semantics	Exact version read, proposed, or committed.	Conditional	Versioned objects
schema_version	Semantics	Payload schema identifier and version.	Required	All
ontology_version	Semantics	Semantic ontology version required to interpret content.	Conditional	Semantic products
truth_state	Semantics	Observed, claimed, inferred, assessed, decided, synthetic, withdrawn, or superseded.	Conditional	Intelligence objects
event_time	Time	When the represented event occurred in the world.	Conditional	Event-bearing objects
observation_time	Time	When a source was observed.	Conditional	Observations / evidence
valid_time	Time	World-time interval for which the assertion applies.	Conditional	Temporal objects
system_time	Time	When The Eye admitted or changed the version.	Required	Committed products
issued_at	Time	Envelope creation time with clock quality.	Required	All
deadline	Time	Latest useful or permitted completion time.	Conditional	Commands / queries / tasks
expires_at	Time	Time after which the envelope, grant, or result is invalid.	Conditional	Temporary authority / results
source_refs	Trust	References to source and Evidence Objects.	Conditional	Derived intelligence
lineage_ref	Trust	Provenance graph root for the product or transition.	Conditional	Derived / committed products
confidence	Trust	Calibrated confidence representation with method.	Conditional	Inferred / assessed products
quality_state	Trust	Decision-active quality, freshness, coverage, and completeness state.	Required	Products
classification	Control	Sensitivity and handling classification.	Required	All protected content
policy_version	Control	Policy set used for admission or effect.	Required	All protected operations
obligations	Control	Machine-enforceable duties returned by policy.	Conditional	Allow-with-obligations
residency_profile	Control	Placement and processing sovereignty constraints.	Conditional	Governed data
retention_profile	Control	Retention, hold, archive, correction, and deletion policy.	Conditional	Persistent content
correlation_id	Operations	End-to-end business or workflow correlation identifier.	Required	All

FIELD	FAMILY	DEFINITION	PRESENCE	APPLIES TO
causation_id	Operations	Immediate predecessor command, event, task, or action.	Conditional	Derived interactions
trace_id	Operations	Distributed trace root.	Required	Runtime interactions
idempotency_key	Operations	Caller-selected duplicate-effect boundary.	Conditional	Commands / side effects
sequence	Operations	Monotonic sequence within the declared partition.	Conditional	Ordered streams
partition_key	Operations	Ordering and routing domain.	Conditional	Events / streams
payload_digest	Integrity	Canonical digest of payload and bound metadata.	Required	All
signature_ref	Integrity	Signature or verification material for origin and integrity.	Conditional	Cross-trust / packages

E Standard Error Code Catalog

Errors are typed product states. They never disclose unauthorized existence, collapse abstention into failure, imply success for an unknown external effect, or invite unsafe retry.

CODE	MACHINE NAME	CONDITION	RETRY	CALLER ACTION
EYE-REQ-001	invalid_request	Request cannot be interpreted under the declared contract.	No	Correct fields or schema; retain correlation.
EYE-REQ-002	unsupported_version	Contract, schema, ontology, model, or package version is unsupported.	No	Negotiate a supported version or migrate.
EYE-IDN-001	authentication_required	No acceptable principal proof is present.	No	Authenticate at required assurance.
EYE-IDN-002	authentication_failed	Credential, audience, issuer, device, or assurance validation failed.	No	Reauthenticate; investigate repeated failure.
EYE-AUT-001	access_denied	Current policy denies the normalized action.	No	Do not retry unchanged; request authorized review.
EYE-AUT-002	policy_indeterminate	Authorization cannot be safely resolved.	Bounded	Retry only under published continuity policy.
EYE-AUT-003	obligation_unmet	A required control obligation cannot be enforced.	No	Satisfy the obligation or change the request.
EYE-TEN-001	domain_context_invalid	Tenant or domain context is absent, conflicting, or untrusted.	No	Correct authenticated routing context.
EYE-STA-001	object_not_found	No authorized object version matches the request.	No	Verify identity and temporal scope.
EYE-STA-002	version_conflict	Expected and current authoritative versions differ.	No	Read current state and resubmit intentionally.
EYE-STA-003	truth_state_prohibited	The operation is not valid for the object's truth state.	No	Use the permitted adjudication or correction path.
EYE-TMP-001	temporal_scope_invalid	Time interval, clock quality, or as-of boundary is invalid.	No	Correct explicit temporal parameters.
EYE-PRV-001	provenance_incomplete	Required source or transformation lineage is missing.	No	Repair lineage or keep product quarantined.
EYE-QUA-001	quality_insufficient	Product quality is below declared decision-use threshold.	Bounded	Use declared degraded path or await refresh.
EYE-QUA-002	result_abstained	The method cannot support a responsible result.	No	Provide more evidence or route to human analysis.
EYE-WFL-001	workflow_conflict	Transition is invalid for current durable state.	No	Refresh instance and use a permitted command.
EYE-WFL-002	human_gate_required	A named human authorization is required before continuation.	No	Complete the bound human gate.
EYE-WFL-003	deadline_exceeded	Work cannot complete within its useful or authorized deadline.	No	Create a new request with reviewed context.
EYE-AGT-001	capability_denied	Agent lacks a valid grant for the tool, data, or effect.	No	Supervisor may issue a bounded grant after review.
EYE-MDL-001	no_conforming_model	No approved model satisfies task and policy constraints.	Bounded	Await approved capacity or change permitted scope.
EYE-DEP-001	dependency_unavailable	Required dependency is unavailable within the operation deadline.	Yes	Retry with server backoff and same idempotency key.
EYE-CAP-001	capacity_exhausted	Admission control cannot safely accept current work.	Yes	Retry after declared interval or reduce scope.
EYE-DEG-001	capability_degraded	Requested capability is constrained by an active degraded mode.	Bounded	Observe guarantees, prohibitions, and retry guidance.
EYE-INT-001	integrity_failure	Digest, signature, ordering, or custody verification failed.	No	Quarantine and investigate; never auto-repair evidence.
EYE-AUD-001	audit_unavailable	Required durable evidence cannot be committed.	Bounded	Follow consequence-specific continuity; do not claim completion.
EYE-EXT-001	external_effect_unknown	Dispatch occurred but external completion cannot be proven.	No	Reconcile by receipt before retry or compensation.

CODE	MACHINE NAME	CONDITION	RETRY	CALLER ACTION
EYE-RCV-001	reconciliation_required	Recovery or failover left state requiring deterministic reconciliation.	No	Complete reconciliation before normal writes.
EYE-GOV-001	conformance_blocked	Artifact or operation lacks current governing evidence.	No	Close evidence or obtain an authorized exception.

Error envelope obligations

- Error code, correlation, causation, trace, request contract version, policy-safe human message, retry classification, deadline, degraded state, and support reference.
- No stack trace, secret, model prompt, protected identifier, unauthorized object existence, cross-tenant signal, or internal topology in caller-visible content.
- Unknown external effects return reconciliation-required state; clients must not retry until the effect boundary is resolved.

F Quality and SLO Catalog

These baseline objectives establish the minimum quality-control vocabulary. Component owners refine populations and targets by consequence and deployment profile without weakening constitutional safety or product transparency.

ID	INDICATOR	MEASURED CONDITION	BASELINE OBJECTIVE	WINDOW	BREACH BEHAVIOR
SLO-001	Canonical write durability	Acknowledged writes survive declared failure domain	99.999%	30 days	No false durable acknowledgement
SLO-002	Canonical read availability	Authorized reads served within product mode	99.95%	30 days	Expose stale or degraded state
SLO-003	Event publication continuity	Committed transitions publish or remain reconcilably pending	99.999%	30 days	No lost committed event
SLO-004	Workflow recovery	Interrupted instances resume from durable state	99.9% < 5 min	30 days	Escalate unresolved effects
SLO-005	Policy decision latency	Authorization decision within service envelope	P99 < 75 ms	24 hours	Restrictive continuity or deny
SLO-006	Policy decision availability	Current policy decisions resolvable	99.99%	30 days	Fail closed by consequence
SLO-007	Audit acceptance	Required records durably accepted before acknowledgement	99.999%	30 days	Block or evidence-pending
SLO-008	Source freshness	Observation completed within source-specific contract	Per source	Rolling	Visible stale state
SLO-009	Provenance completeness	Decision-active products resolve complete required lineage	100%	Per release	Quarantine incomplete product
SLO-010	Graph projection lag	Committed revisions visible to authorized queries	P99 < 60 s	24 hours	Return revision watermark
SLO-011	Memory retrieval freshness	Admitted objects searchable by declared freshness	P99 < 120 s	24 hours	Canonical lookup or stale marker
SLO-012	Twin reconciliation	Observed twin state reconciles admitted evidence	Per twin class	Rolling	Estimated/stale state visible
SLO-013	Forecast timeliness	Forecast packages produced before horizon deadline	Per horizon	Rolling	Prior package marked age/invalidity
SLO-014	Forecast calibration	Probability products meet approved calibration bounds	Per profile	Quarterly	Constrain or withdraw model
SLO-015	Simulation reproducibility	Exact manifests reproduce within tolerance	99.5%	Per release	Mark non-reproducible
SLO-016	Warning timeliness	Material warnings reach accountable owner inside response window	Per severity	Rolling	Escalate routing failure
SLO-017	Human gate integrity	No consequential transition without valid human authority	100%	Continuous	Block transition
SLO-018	Decision replay completeness	Required historical boundary reconstructs	99.9%	Quarterly	Expose explicit missing evidence
SLO-019	Tenant isolation	No unauthorized cross-domain disclosure or effect	100%	Continuous	Stop affected plane and incident
SLO-020	Backup restore verification	Scheduled backups restore and reconcile successfully	100% sampled	Monthly	Escalate recovery risk
SLO-021	Telemetry correlation	Consequential flows preserve trace and causation	99.99%	24 hours	Mark evidence incomplete
SLO-022	Deployment parity	Canonical conformance suite equivalent across profiles	100%	Per release	Block affected deployment

SLO RULE

An error budget controls change velocity; it does not authorize loss of human decision sovereignty, provenance, tenant isolation, policy enforcement, audit, temporal truth, or deployment semantic parity.

G Test and Conformance Matrix

The test architecture supplies independent evidence from local logic through operational acceptance. High-consequence fixtures remain immutable, portable, rights-controlled, and reproducible across supported deployment modes.

ID	SUITE	PRIMARY SCOPE	EXECUTION	AUTHORITY
TS-001	Unit and property	Local logic, invariants, state transitions, boundary values	Every change	Owning component
TS-002	Schema conformance	Canonical objects, envelopes, errors, compatibility	Every change	Contract authority
TS-003	Consumer contract	Provider/consumer combinations and degraded behavior	Every candidate	Interface owners
TS-004	Temporal truth	Event, observation, valid, system time and as-of queries	Every candidate	Data authority
TS-005	Provenance	Backward explanation, forward impact, correction propagation	Every candidate	Trust authority
TS-006	Policy	Authorization, obligations, purpose, classification, retention	Every candidate	Policy authority
TS-007	Tenant isolation	API, store, queue, cache, vector, graph, model, telemetry	Continuous	Security
TS-008	Workflow durability	Crash, retry, idempotency, compensation, migration	Every candidate	Workflow platform
TS-009	Human authority	Gate, role, quorum, separation, stale context, replay	Every candidate	Decision governance
TS-010	Agent containment	Capability, tool, network, data, budget, supervisor, escape	Every agent	Agent platform / Security
TS-011	Model fitness	Quality, calibration, slices, drift, abstention, fallback	Every model/task	AI assurance
TS-012	Data quality	Completeness, freshness, accuracy, duplicates, corrections	Continuous	Data owners
TS-013	Performance	Latency, throughput, freshness, fairness, saturation, cost	Every release	Performance engineering
TS-014	Resilience	Failure domains, partitions, backpressure, recovery, blast radius	Every release	SRE
TS-015	Disaster recovery	Restore, reconcile, RPO, RTO, keys, sovereignty	Quarterly	Continuity authority
TS-016	Security adversarial	Identity, policy, supply chain, exfiltration, insider	Every release	Security assurance
TS-017	Privacy lifecycle	Minimization, subject action, retention, deletion, exit	Every release	Privacy authority
TS-018	Deployment parity	Same canonical fixtures across SaaS, private, on-prem	Every release	Architecture assurance
TS-019	Upgrade and rollback	Migrations, compatibility, partial failure, reconciliation	Every release	Release engineering
TS-020	Operational acceptance	Runbooks, alerts, incident, support, recovery, closure	Before acceptance	Operations authority

Evidence record

EVIDENCE DIMENSION	REQUIRED CONTENT
Identity	Requirement, test, fixture, artifact, configuration, schema, policy, model, agent, workflow, environment, and deployment versions.
Execution	Runner, identity, time, seed, inputs, dependencies, commands, result, logs, traces, resource, and exception context.
Validity	Population, coverage, tolerances, exclusions, known limitations, contamination state, expiry, and reviewer.
Integrity	Digests, signatures, attestations, custody, immutable storage, and independent reproduction instructions.

H Configuration Namespace

Configuration names encode authority and scope. Every item is typed, versioned, signed, reviewed, promoted, observable, historically resolvable, and removable through the same controlled delivery system as code.

NAMESPACE	CONTENT	OWNING AUTHORITY	PERMITTED SCOPES
eye.identity.*	Identity assurance, issuers, sessions, workload identity	Security identity authority	Deployment / tenant
eye.policy.*	Authorization policy bundles, precedence, obligations, cache bounds	Policy authority	Global / deployment / tenant
eye.domain.*	Customer Intelligence Domain placement and isolation	Platform authority	Tenant / domain
eye.crypto.*	Cryptographic profiles, key references, rotation and custody	Cryptographic authority	Deployment / tenant / domain
eye.network.*	Trust zones, service channels, ingress and egress grants	Network security	Deployment / cell / workload
eye.observation.*	Source contracts, schedules, quotas, correction behavior	Observation authority	Domain / source
eye.objects.*	Object schemas, admission rules, lifecycle and projections	Data contract authority	Global / domain
eye.graph.*	Ontology, revision, inference, traversal and projection limits	Knowledge authority	Global / domain
eye.twins.*	Twin classes, mappings, reconciliation and branch policy	Twin authority	Domain / twin class
eye.models.*	Model profiles, routing, fallback, context and retention	AI platform authority	Deployment / domain / task
eye.agents.*	Agent admission, capabilities, budgets, tools and supervision	Agent platform authority	Domain / agent / task
eye.workflows.*	Definitions, timers, retries, gates, compensation and migration	Workflow authority	Global / domain / definition
eye.quality.*	Indicators, SLOs, thresholds, freshness and product constraints	Quality authority	Component / product
eye.features.*	Controlled feature lifecycle and eligible scope	Product and release authority	Deployment / tenant / role
eye.capacity.*	Admission, quotas, priorities, resource and cost budgets	SRE authority	Cell / tenant / workload
eye.telemetry.*	Signal schemas, sampling, redaction, residency and retention	Observability authority	Deployment / tenant / component
eye.retention.*	Retention, hold, archive, correction, deletion and residual policy	Data governance	Tenant / object class
eye.release.*	Candidate, rollout, stop, rollback and acceptance controls	Release authority	Environment / deployment

Prohibited configuration patterns

- Unowned free-form keys, secrets, customer content, unbounded global overrides, runtime shell mutation, deployment-specific semantic defaults, or settings without schema and expiry.
- Feature flags that bypass authorization, human gates, audit, provenance, tenant isolation, evaluation, migration, or operational acceptance.
- Manual production-only values that cannot be reproduced, reviewed, promoted, rolled back, or resolved during Decision Replay.

I Requirement and Constitutional Traceability

Traceability proves that implementation obligations resolve upward to architecture and constitutional authority and downward to release evidence. Requirement ranges below remain stable within edition 1.0.

CH .	ENGINEERING SUBJECT	REQUIREMENT RANGE	COUN T	CONSTITUTIONAL INHERITANCE	EVIDENCE FAMILIES
1	Engineering Authority and Scope	ES-01-001–ES-01-005	5	C-001, C-003, C-008, C-051, C-052	Contract + test + release + operational evidence
2	Specification Hierarchy and Conformance	ES-02-001–ES-02-005	5	C-004, C-035, C-049, C-051, C-052	Contract + test + release + operational evidence
3	Normative Requirement Model	ES-03-001–ES-03-005	5	C-005, C-011, C-035, C-049, C-052	Contract + test + release + operational evidence
4	Product Boundary and Engineering Decomposition	ES-04-001–ES-04-005	5	C-001, C-003, C-007, C-008, C-038	Contract + test + release + operational evidence
5	Engineering Principles	ES-05-001–ES-05-005	5	C-002, C-004, C-005, C-010, C-025, C-031, C-043, C-044	Contract + test + release + operational evidence
6	Cross-Volume Traceability	ES-06-001–ES-06-005	5	C-005, C-012, C-025, C-035, C-036, C-051, C-052	Contract + test + release + operational evidence
7	Runtime Topology	ES-07-001–ES-07-005	5	C-035, C-037, C-039, C-041, C-042, C-043	Contract + test + release + operational evidence
8	Customer Intelligence Domain	ES-08-001–ES-08-005	5	C-016, C-037, C-039, C-040, C-041	Contract + test + release + operational evidence
9	Intelligence Cells	ES-09-001–ES-09-005	5	C-038, C-041, C-042, C-043, C-049	Contract + test + release + operational evidence
10	Shared Control Services	ES-10-001–ES-10-005	5	C-035, C-037, C-040, C-041, C-042, C-043	Contract + test + release + operational evidence
11	Component Ownership and Service Boundaries	ES-11-001–ES-11-005	5	C-008, C-035, C-036, C-043, C-049	Contract + test + release + operational evidence
12	State Ownership and Persistence	ES-12-001–ES-12-005	5	C-010, C-011, C-015, C-016, C-025, C-037, C-043	Contract + test + release + operational evidence
13	Cross-Cutting Control Enforcement	ES-13-001–ES-13-005	5	C-035, C-036, C-039, C-040, C-041, C-043, C-049	Contract + test + release + operational evidence
14	Deployment Semantic Parity	ES-14-001–ES-14-005	5	C-007, C-038, C-041, C-042, C-043, C-049	Contract + test + release + operational evidence
15	Canonical Service Contract	ES-15-001–ES-15-005	5	C-035, C-038, C-042, C-043, C-049	Contract + test + release + operational evidence
16	Synchronous API Profile	ES-16-001–ES-16-005	5	C-035, C-038, C-039, C-043, C-049	Contract + test + release + operational evidence
17	Command Profile	ES-17-001–ES-17-005	5	C-011, C-012, C-025, C-036, C-043	Contract + test + release + operational evidence
18	Query and Consistency Profile	ES-18-001–ES-18-005	5	C-005, C-011, C-016, C-025, C-035, C-039, C-049	Contract + test + release + operational evidence
19	Event Profile	ES-19-001–ES-19-005	5	C-011, C-012, C-025, C-036, C-038, C-043	Contract + test + release + operational evidence
20	Canonical Contract Envelope	ES-20-001–ES-20-005	5	C-005, C-010, C-011, C-012, C-035, C-038, C-039, C-042	Contract + test + release + operational evidence
21	Error, Abstention, and Partial-Result Model	ES-21-001–ES-21-005	5	C-005, C-010, C-021, C-024, C-035, C-043, C-049	Contract + test + release + operational evidence
22	Idempotency, Ordering, Retry, and Compensation	ES-22-001–ES-22-005	5	C-011, C-025, C-036, C-043, C-049	Contract + test + release + operational evidence

CH .	ENGINEERING SUBJECT	REQUIREMENT RANGE	COUN T	CONSTITUTIONAL INHERITANCE	EVIDENCE FAMILIES
23	Versioning, Compatibility, and Contract Registry	ES-23-001–ES-23-005	5	C-011, C-035, C-038, C-042, C-045, C-049	Contract + test + release + operational evidence
24	Canonical Intelligence Object Header	ES-24-001–ES-24-005	5	C-010, C-011, C-012, C-016, C-035, C-042	Contract + test + release + operational evidence
25	Identity, Naming, and Tenancy	ES-25-001–ES-25-005	5	C-009, C-010, C-016, C-035, C-037, C-039	Contract + test + release + operational evidence
26	Temporal Truth Engineering	ES-26-001–ES-26-005	5	C-005, C-011, C-019, C-025, C-049	Contract + test + release + operational evidence
27	Truth State, Confidence, and Uncertainty	ES-27-001–ES-27-005	5	C-005, C-010, C-018, C-021, C-022, C-023, C-049	Contract + test + release + operational evidence
28	Provenance and Lineage Engineering	ES-28-001–ES-28-005	5	C-005, C-012, C-015, C-025, C-035, C-036	Contract + test + release + operational evidence
29	Policy, Classification, Retention, and Rights	ES-29-001–ES-29-005	5	C-016, C-035, C-037, C-039, C-040, C-041	Contract + test + release + operational evidence
30	Correction, Supersession, and Projection Rebuild	ES-30-001–ES-30-005	5	C-011, C-012, C-016, C-025, C-033, C-043	Contract + test + release + operational evidence
31	World Observation Layer	ES-31-001–ES-31-010	10	C-012, C-013, C-014, C-035, C-039, C-043	Contract + test + release + operational evidence
32	Intelligence Layer	ES-32-001–ES-32-010	10	C-005, C-010, C-011, C-012, C-030, C-045	Contract + test + release + operational evidence
33	Enterprise Memory	ES-33-001–ES-33-010	10	C-011, C-015, C-016, C-025, C-037, C-040, C-043	Contract + test + release + operational evidence
34	Knowledge Graph	ES-34-001–ES-34-010	10	C-007, C-009, C-010, C-011, C-012, C-017, C-035	Contract + test + release + operational evidence
35	Digital Twins	ES-35-001–ES-35-010	10	C-010, C-011, C-018, C-019, C-023, C-035, C-043	Contract + test + release + operational evidence
36	Prediction Engine	ES-36-001–ES-36-010	10	C-005, C-020, C-021, C-030, C-045, C-049	Contract + test + release + operational evidence
37	Scenario Intelligence	ES-37-001–ES-37-010	10	C-005, C-022, C-023, C-032, C-033, C-046	Contract + test + release + operational evidence
38	Simulation Engine	ES-38-001–ES-38-010	10	C-005, C-018, C-019, C-023, C-035, C-043, C-045	Contract + test + release + operational evidence
39	Decision Intelligence	ES-39-001–ES-39-010	10	C-004, C-005, C-024, C-025, C-026, C-036, C-050	Contract + test + release + operational evidence
40	Executive Operating System	ES-40-001–ES-40-010	10	C-003, C-004, C-027, C-032, C-033, C-047, C-048	Contract + test + release + operational evidence
41	Durable Workflow Runtime	ES-41-001–ES-41-005	5	C-004, C-011, C-025, C-029, C-043, C-049	Contract + test + release + operational evidence
42	Human Gates and Approvals	ES-42-001–ES-42-005	5	C-004, C-005, C-024, C-025, C-026, C-029, C-035	Contract + test + release + operational evidence
43	Multi-Agent Execution Runtime	ES-43-001–ES-43-005	5	C-004, C-005, C-012, C-027, C-028, C-029, C-035	Contract + test + release + operational evidence
44	Agent Tool and Capability Control	ES-44-001–ES-44-005	5	C-004, C-012, C-028, C-029, C-035, C-037, C-040	Contract + test + release + operational evidence
45	Model Gateway and Inference Runtime	ES-45-001–ES-45-005	5	C-005, C-012, C-027, C-030, C-035, C-041, C-045	Contract + test + release + operational evidence
46	Evaluation and Fitness	ES-46-001–ES-46-005	5	C-005, C-021, C-028, C-030, C-031, C-035, C-045, C-049	Contract + test + release + operational evidence
47	Proactive Warning Runtime	ES-47-001–ES-47-005	5	C-002, C-005, C-020, C-022, C-026, C-032, C-033, C-034	Contract + test + release + operational evidence

CH .	ENGINEERING SUBJECT	REQUIREMENT RANGE	COUN T	CONSTITUTIONAL INHERITANCE	EVIDENCE FAMILIES
48	Decision Replay and Governed Learning	ES-48-001–ES-48-005	5	C-004, C-005, C-011, C-025, C-031, C-044, C-045	Contract + test + release + operational evidence
49	Identity and Authentication	ES-49-001–ES-49-005	5	C-004, C-035, C-036, C-037, C-038, C-039, C-040	Contract + test + release + operational evidence
50	Authorization and Policy Enforcement	ES-50-001–ES-50-005	5	C-004, C-035, C-037, C-038, C-039, C-040, C-042, C-047	Contract + test + release + operational evidence
51	Tenant and Domain Isolation	ES-51-001–ES-51-005	5	C-016, C-035, C-037, C-038, C-039, C-040, C-041, C-047	Contract + test + release + operational evidence
52	Cryptography, Keys, and Secrets	ES-52-001–ES-52-005	5	C-035, C-036, C-037, C-038, C-039, C-041, C-048	Contract + test + release + operational evidence
53	Network Trust and Controlled Egress	ES-53-001–ES-53-005	5	C-035, C-036, C-037, C-038, C-040, C-041, C-048	Contract + test + release + operational evidence
54	Privacy, Minimization, and Sovereignty	ES-54-001–ES-54-005	5	C-016, C-035, C-037, C-039, C-040, C-041, C-048	Contract + test + release + operational evidence
55	Audit and Non-Repudiation	ES-55-001–ES-55-005	5	C-004, C-005, C-012, C-025, C-035, C-036, C-049	Contract + test + release + operational evidence
56	Software Supply Chain and Marketplaces	ES-56-001–ES-56-005	5	C-028, C-035, C-040, C-041, C-042, C-045, C-046, C-048	Contract + test + release + operational evidence
57	Quality Contracts and SLOs	ES-57-001–ES-57-005	5	C-005, C-020, C-021, C-032, C-033, C-043, C-049	Contract + test + release + operational evidence
58	Performance and Capacity Engineering	ES-58-001–ES-58-005	5	C-032, C-033, C-042, C-043, C-049, C-050	Contract + test + release + operational evidence
59	Availability and Resilience	ES-59-001–ES-59-005	5	C-037, C-041, C-042, C-043, C-049, C-050	Contract + test + release + operational evidence
60	Degraded Service Semantics	ES-60-001–ES-60-005	5	C-004, C-005, C-033, C-043, C-049, C-050	Contract + test + release + operational evidence
61	Backup, Recovery, and Disaster Recovery	ES-61-001–ES-61-005	5	C-011, C-025, C-037, C-041, C-042, C-043, C-048, C-050	Contract + test + release + operational evidence
62	Observability and Telemetry	ES-62-001–ES-62-005	5	C-005, C-012, C-035, C-036, C-041, C-043, C-049, C-050	Contract + test + release + operational evidence
63	Incident, Change, and Problem Management	ES-63-001–ES-63-005	5	C-004, C-005, C-025, C-031, C-035, C-043, C-045, C-049	Contract + test + release + operational evidence
64	Operational Data Lifecycle	ES-64-001–ES-64-005	5	C-011, C-012, C-016, C-025, C-035, C-039, C-041, C-048	Contract + test + release + operational evidence
65	Repository and Module Standards	ES-65-001–ES-65-005	5	C-008, C-035, C-036, C-046, C-048, C-051, C-052	Contract + test + release + operational evidence
66	Configuration and Feature Control	ES-66-001–ES-66-005	5	C-004, C-035, C-042, C-043, C-045, C-049, C-051	Contract + test + release + operational evidence
67	Test Architecture	ES-67-001–ES-67-005	5	C-004, C-005, C-011, C-035, C-042, C-043, C-045, C-049, C-052	Contract + test + release + operational evidence
68	CI/CD and Release Gates	ES-68-001–ES-68-005	5	C-035, C-042, C-045, C-046, C-048, C-049, C-051, C-052	Contract + test + release + operational evidence
69	Environment Promotion	ES-69-001–ES-69-005	5	C-035, C-037, C-041, C-042, C-045, C-048, C-049	Contract + test + release + operational evidence

CH	ENGINEERING SUBJECT	REQUIREMENT RANGE	COUNT	CONSTITUTIONAL INHERITANCE	EVIDENCE FAMILIES
70	Deployment Packaging	ES-70-001–ES-70-005	5	C-035, C-041, C-042, C-043, C-046, C-048, C-049, C-050	Contract + test + release + operational evidence
71	Operational Acceptance	ES-71-001–ES-71-005	5	C-004, C-005, C-035, C-043, C-045, C-049, C-051, C-052	Contract + test + release + operational evidence
72	Engineering Governance and Exceptions	ES-72-001–ES-72-005	5	C-004, C-008, C-035, C-042, C-049, C-051, C-052	Contract + test + release + operational evidence

Constitutional implementation matrix

ID	INVARIANT	BINDING STATEMENT	PRIMARY VOLUME 4 ANCHORS
C-001	Category integrity	The Eye shall be conceived, designed, sold, and operated as a Strategic Intelligence Operating System. It shall not be reduced to a chatbot, dashboard, BI tool, copilot, or reporting product.	Ch. 1, Ch. 4, Ch. 5
C-002	Closed strategic loop	The product shall implement one continuous operating loop: Observe, Understand, Predict, Simulate, Decide, and Learn. No stage may be treated as a disconnected feature island.	Ch. 5, Ch. 47
C-003	Product boundary	Interfaces such as chat, dashboards, reports, and APIs may exist, but they are access modes to the operating system rather than the product definition.	Ch. 1, Ch. 4, Ch. 40, Ch. 2
C-004	Human decision sovereignty	A named human authority shall make every final strategic or high-impact decision. The system may advise, prepare, and execute approved workflows, but it shall not silently assume decision rights.	Ch. 2, Ch. 5, Ch. 41, Ch. 42, Ch. 43, Ch. 44, Ch. 48, Ch. 49, Ch. 50, Ch. 55, Ch. 60, Ch. 63, Ch. 66, Ch. 67, Ch. 71, Ch. 72, Ch. 39, Ch. 40, Ch. 1
C-005	Explain every recommendation	Every material forecast, scenario, simulation result, risk, opportunity, and recommendation shall expose evidence, assumptions, uncertainty, lineage, alternatives, and accountable system versions.	Ch. 3, Ch. 5, Ch. 6, Ch. 18, Ch. 20, Ch. 21, Ch. 26, Ch. 27, Ch. 28, Ch. 42, Ch. 43, Ch. 45, Ch. 46, Ch. 47, Ch. 48, Ch. 55, Ch. 57, Ch. 60, Ch. 62, Ch. 63, Ch. 67, Ch. 71, Ch. 32, Ch. 36, Ch. 37, Ch. 38, Ch. 39
C-006	Enterprise operating standard	The common platform shall satisfy enterprise requirements for governance, security, scale, auditability, reliability, and lifecycle management from its first production architecture.	Ch. 1, Ch. 2
C-007	Universal core, governed specialization	A single domain-neutral core shall serve all customer classes. Domain ontologies, models, workflows, agents, and policies may extend the core without forking its constitutional semantics.	Ch. 4, Ch. 14, Ch. 34, Ch. 6
C-008	Ten-layer architecture	The ten named architecture layers are the canonical product structure. Cross-cutting planes may support them but shall not replace, collapse, or rename them without constitutional amendment.	Ch. 1, Ch. 4, Ch. 11, Ch. 65, Ch. 72, Ch. 31
C-009	Knowledge Graph centrality	The Knowledge Graph shall be the connected semantic heart of The Eye and the shared context substrate for memory, twins, reasoning, prediction, scenarios, simulations, and decisions.	Ch. 25, Ch. 34
C-010	Truth-state separation	Observed evidence, extracted claims, inferred relationships, model outputs, human judgments, and synthetic scenario data shall remain explicitly distinguishable throughout the system.	Ch. 5, Ch. 12, Ch. 20, Ch. 21, Ch. 24, Ch. 25, Ch. 27, Ch. 32, Ch. 34, Ch. 35
C-011	Temporal truth	Material facts and relationships shall preserve event time, observation time, validity time, and system-record time so that past world states can be reconstructed without hindsight contamination.	Ch. 3, Ch. 12, Ch. 17, Ch. 18, Ch. 19, Ch. 20, Ch. 22, Ch. 23, Ch. 24, Ch. 26, Ch. 30, Ch. 41, Ch. 48, Ch. 61, Ch. 64, Ch. 67, Ch. 32, Ch. 33, Ch. 34, Ch. 35
C-012	End-to-end provenance	Every material intelligence object shall carry source, chain of custody, transformation history, ownership, rights, quality, and version lineage back to its evidence.	Ch. 6, Ch. 17, Ch. 19, Ch. 20, Ch. 24, Ch. 28, Ch. 30, Ch. 43, Ch. 44, Ch. 45, Ch. 55, Ch. 62, Ch. 64, Ch. 31, Ch. 32, Ch. 34
C-013	Observation breadth	The World Observation Layer shall support the full canonical source universe and allow governed custom sources without architectural rework.	Ch. 31, Ch. 32
C-014	Freshness and quality contracts	Sources, derived data, twins, models, and intelligence products shall declare measurable freshness, completeness, quality, and expiry expectations.	Ch. 31, Ch. 32
C-015	Durable institutional memory	The Eye shall preserve institutional context, decisions, assumptions, evidence, outcomes, and lessons as durable memory, subject to lawful retention, deletion, and customer policy.	Ch. 12, Ch. 28, Ch. 33, Ch. 34
C-016	Memory governance	Memory shall be permission-aware, tenant-isolated, versioned, retrievable, correctable, and protected from unverified model output becoming accepted organizational truth.	Ch. 8, Ch. 12, Ch. 18, Ch. 24, Ch. 25, Ch. 29, Ch. 30, Ch. 51, Ch. 54, Ch. 64, Ch. 33

ID	INVARIANT	BINDING STATEMENT	PRIMARY VOLUME 4 ANCHORS
C-017	Strategy Graph	Objectives, hypotheses, decisions, initiatives, capabilities, resources, indicators, risks, opportunities, and outcomes shall be connected in a governed Strategy Graph anchored to the Knowledge Graph.	Ch. 34
C-018	Grounded Digital Twins	Every Digital Twin shall be grounded in evidence, graph entities, temporal state, declared assumptions, and calibrated models; it shall never present synthetic state as observed fact.	Ch. 27, Ch. 35, Ch. 38
C-019	Twin versionability	Digital Twins shall support time travel, branching, comparison, replay, reconciliation, and versioned evolution across actual and simulated states.	Ch. 26, Ch. 35, Ch. 38
C-020	Canonical prediction horizons	The Prediction Engine shall support 30-day, 90-day, 180-day, 1-year, 3-year, and 5-year horizons with horizon-appropriate methods and uncertainty.	Ch. 47, Ch. 57, Ch. 36
C-021	Probabilistic forecasting	Forecasts shall communicate distributions, confidence, calibration, drivers, sensitivity, and failure conditions rather than false precision or unsupported certainty.	Ch. 21, Ch. 27, Ch. 46, Ch. 57, Ch. 36
C-022	Scenario plurality	Scenario Intelligence shall maintain multiple coherent futures, including baseline, upside, downside, disruption, and user-defined alternatives; no single forecast shall masquerade as destiny.	Ch. 27, Ch. 47, Ch. 37
C-023	Reproducible simulation	Every simulation shall declare initial state, assumptions, interventions, model versions, constraints, random seeds where applicable, outputs, and sensitivity so results can be reproduced and challenged.	Ch. 27, Ch. 35, Ch. 37, Ch. 38
C-024	Decision completeness	Decision Intelligence shall compare viable options against explicit objectives, constraints, risks, opportunities, trade-offs, second-order effects, reversibility, and information value.	Ch. 21, Ch. 42, Ch. 39
C-025	Decision Replay	The Eye shall preserve the state of knowledge available at decision time and support replay against later evidence and outcomes without rewriting the historical record.	Ch. 5, Ch. 6, Ch. 12, Ch. 17, Ch. 18, Ch. 19, Ch. 22, Ch. 26, Ch. 28, Ch. 30, Ch. 41, Ch. 42, Ch. 48, Ch. 55, Ch. 61, Ch. 63, Ch. 64, Ch. 33, Ch. 39
C-026	Objective alignment	Recommendations shall be evaluated against declared objectives, stakeholder obligations, risk appetite, policy, and strategic constraints rather than an opaque global score.	Ch. 42, Ch. 47, Ch. 39
C-027	Executive Operating System	The executive surface shall organize strategic attention, decisions, scenarios, evidence, objectives, commitments, and outcomes. It shall not degrade into a passive dashboard collection.	Ch. 43, Ch. 45, Ch. 40
C-028	Bounded agent authority	Every agent shall have a declared purpose, tools, data scope, permissions, budgets, escalation rules, prohibited actions, evaluation suite, and accountable owner.	Ch. 43, Ch. 44, Ch. 46, Ch. 56
C-029	Supervised orchestration	Planner, Supervisor, and Workflow agents shall enforce policy, separation of duties, checkpoints, timeouts, retries, human gates, and complete execution traces across multi-agent work.	Ch. 41, Ch. 42, Ch. 43, Ch. 44
C-030	Model pluralism	The platform shall support multiple models and reasoning approaches behind governed interfaces so customers retain control over cost, sovereignty, performance, and vendor dependence.	Ch. 45, Ch. 46, Ch. 32, Ch. 36
C-031	No silent self-modification	Continuous learning shall never permit unreviewed production models, agents, prompts, policies, ontologies, or decision logic to modify themselves silently.	Ch. 5, Ch. 46, Ch. 48, Ch. 63
C-032	Proactive intelligence	The system shall surface material changes, weak signals, risks, and opportunities before users ask, while respecting relevance, authority, attention, and notification policy.	Ch. 47, Ch. 57, Ch. 58, Ch. 37, Ch. 40
C-033	Early-warning discipline	Warnings shall be deduplicated, prioritized, evidence-backed, confidence-aware, time-bounded, and linked to affected objectives, assets, decisions, and response playbooks.	Ch. 30, Ch. 47, Ch. 57, Ch. 58, Ch. 60, Ch. 37, Ch. 40
C-034	Decomposable Strategic Health Score	Any Strategic Health Score shall expose component measures, weights, evidence, confidence, trend, and sensitivity. The score shall never replace the underlying analysis.	Ch. 47
C-035	Trust as a system property	Trust, provenance, identity, authorization, policy, safety, and audit shall operate across every layer and intelligence object, not as a downstream compliance add-on.	Ch. 2, Ch. 3, Ch. 6, Ch. 7, Ch. 10, Ch. 11, Ch. 13, Ch. 15, Ch. 16, Ch. 18, Ch. 20, Ch. 21, Ch. 23, Ch. 24, Ch. 25, Ch. 28, Ch. 29, Ch. 42, Ch. 43, Ch. 44, Ch. 45, Ch. 46, Ch. 49, Ch. 50, Ch. 51, Ch. 52, Ch. 53, Ch. 54, Ch. 55, Ch. 56, Ch. 62, Ch. 63, Ch. 64, Ch. 65, Ch. 66, Ch. 67, Ch. 68, Ch. 69, Ch. 70, Ch. 71, Ch. 72, Ch. 31, Ch. 34, Ch. 35, Ch. 38
C-036	Non-repudiable accountability	Material access, transformations, agent actions, approvals, decisions, overrides, releases, and exports shall produce tamper-evident audit records attributable to identities and versions.	Ch. 6, Ch. 11, Ch. 13, Ch. 17, Ch. 19, Ch. 22, Ch. 28, Ch. 49, Ch. 52, Ch. 53, Ch. 55, Ch. 62, Ch. 65, Ch. 39

ID	INVARIANT	BINDING STATEMENT	PRIMARY VOLUME 4 ANCHORS
C-037	Customer data control	Customers retain governance over their data, models, twins, policies, retention, residency, encryption, exports, and deletion within the chosen deployment and contractual boundary.	Ch. 7, Ch. 8, Ch. 10, Ch. 12, Ch. 25, Ch. 29, Ch. 44, Ch. 49, Ch. 50, Ch. 51, Ch. 52, Ch. 53, Ch. 54, Ch. 59, Ch. 61, Ch. 69, Ch. 33
C-038	Open interoperability	The Eye shall integrate through governed APIs, events, connectors, common data formats, and export paths; core customer knowledge shall not be trapped behind proprietary-only access.	Ch. 4, Ch. 9, Ch. 14, Ch. 15, Ch. 16, Ch. 19, Ch. 20, Ch. 23, Ch. 49, Ch. 50, Ch. 51, Ch. 52, Ch. 53
C-039	Zero-trust security	Identity, device, workload, agent, model, source, and data access shall be continuously authenticated, authorized, scoped, logged, and re-evaluated under least privilege.	Ch. 7, Ch. 8, Ch. 13, Ch. 16, Ch. 18, Ch. 20, Ch. 25, Ch. 29, Ch. 49, Ch. 50, Ch. 51, Ch. 52, Ch. 54, Ch. 64, Ch. 31
C-040	Privacy and minimization	Collection and use shall be purpose-bound, proportionate, policy-enforced, and minimized, with support for consent, legal basis, access, correction, retention, and deletion obligations.	Ch. 8, Ch. 10, Ch. 13, Ch. 29, Ch. 44, Ch. 49, Ch. 50, Ch. 51, Ch. 53, Ch. 54, Ch. 56, Ch. 33
C-041	Sovereignty by design	Data residency, key ownership, network isolation, local inference, model choice, administrative control, and disconnected operation shall be supported where customer policy requires them.	Ch. 7, Ch. 8, Ch. 9, Ch. 10, Ch. 13, Ch. 14, Ch. 29, Ch. 45, Ch. 51, Ch. 52, Ch. 53, Ch. 54, Ch. 56, Ch. 59, Ch. 61, Ch. 62, Ch. 64, Ch. 69, Ch. 70
C-042	Deployment semantic parity	SaaS, Private Cloud, and On-Premise deployments shall preserve the same core semantics, governance model, APIs, artifacts, and intelligence lifecycle, with documented capability differences only where unavoidable.	Ch. 7, Ch. 9, Ch. 10, Ch. 14, Ch. 15, Ch. 20, Ch. 23, Ch. 24, Ch. 50, Ch. 56, Ch. 58, Ch. 59, Ch. 61, Ch. 66, Ch. 67, Ch. 68, Ch. 69, Ch. 70, Ch. 72
C-043	Resilient operation	The platform shall fail safely, degrade intelligibly, recover predictably, preserve evidence and audit continuity, and expose when data, models, or dependencies are unavailable or stale.	Ch. 5, Ch. 7, Ch. 9, Ch. 10, Ch. 11, Ch. 12, Ch. 13, Ch. 14, Ch. 15, Ch. 16, Ch. 17, Ch. 19, Ch. 21, Ch. 22, Ch. 30, Ch. 41, Ch. 57, Ch. 58, Ch. 59, Ch. 60, Ch. 61, Ch. 62, Ch. 63, Ch. 66, Ch. 67, Ch. 70, Ch. 71, Ch. 31, Ch. 33, Ch. 35, Ch. 38
C-044	Governed learning loop	Learning from outcomes, feedback, overrides, and new evidence shall pass through evaluation, approval, release, monitoring, rollback, and tenant-isolation controls.	Ch. 5, Ch. 48
C-045	Continuous evaluation	Models, agents, forecasts, extractors, ontologies, simulations, and recommendations shall be evaluated for accuracy, calibration, robustness, drift, bias, safety, cost, and operational fitness.	Ch. 23, Ch. 45, Ch. 46, Ch. 48, Ch. 56, Ch. 63, Ch. 66, Ch. 67, Ch. 68, Ch. 69, Ch. 71, Ch. 32, Ch. 36, Ch. 38
C-046	Governed marketplaces	Scenario and Agent Marketplace packages shall be signed, versioned, inspectable, permission-scoped, licensed, evaluated, sandboxed, monitored, revocable, and traceable to publishers.	Ch. 56, Ch. 65, Ch. 68, Ch. 70, Ch. 37
C-047	Experience integrity	The product experience shall expose context, uncertainty, provenance, decision state, and consequence at the point of use; critical meaning shall not be hidden behind ornamental visualization.	Ch. 50, Ch. 51, Ch. 40
C-048	Role and accessibility clarity	Experiences shall be accessible and role-appropriate for executives, analysts, operators, domain experts, administrators, and governance teams without creating separate truths.	Ch. 52, Ch. 53, Ch. 54, Ch. 56, Ch. 61, Ch. 64, Ch. 65, Ch. 68, Ch. 69, Ch. 70, Ch. 40
C-049	Operational quality contract	Reliability, performance, scale, freshness, accuracy, calibration, explainability, reproducibility, security, accessibility, and cost shall be measured as product behavior, not assumed as implementation detail.	Ch. 2, Ch. 3, Ch. 9, Ch. 11, Ch. 13, Ch. 14, Ch. 15, Ch. 16, Ch. 18, Ch. 21, Ch. 22, Ch. 23, Ch. 26, Ch. 27, Ch. 41, Ch. 46, Ch. 55, Ch. 57, Ch. 58, Ch. 59, Ch. 60, Ch. 62, Ch. 63, Ch. 66, Ch. 67, Ch. 68, Ch. 69, Ch. 70, Ch. 71, Ch. 72, Ch. 36
C-050	Responsible-use boundary	The Eye shall not autonomously execute decisions that materially determine life, liberty, legal status, access to essential services, employment, credit, or equivalent high-impact rights and interests.	Ch. 58, Ch. 59, Ch. 60, Ch. 61, Ch. 62, Ch. 70, Ch. 39
C-051	Constitutional change control	Changes to mission, product category, human authority, principles, canonical layers, core semantics, or constitutional invariants require explicit amendment, impact analysis, ratification, and versioned publication.	Ch. 1, Ch. 2, Ch. 6, Ch. 65, Ch. 66, Ch. 68, Ch. 71, Ch. 72
C-052	Documentation inheritance	Volumes 1 through 10 shall inherit this constitution, use its canonical terms, trace material requirements to invariant IDs, and identify rather than conceal any unresolved conflict.	Ch. 1, Ch. 2, Ch. 3, Ch. 6, Ch. 65, Ch. 67, Ch. 68, Ch. 71, Ch. 72

J Architecture Decision Implementation Obligations

The accepted Volume 3 decisions below are engineering obligations. Implementation detail may evolve behind them, but releases cannot contradict their semantics or treat a local exception as architecture change.

ADR	DECISION	ARCHITECTURE CONSEQUENCE	VOLUME 4 IMPLEMENTATION OBLIGATION	INHERITANCE
ADR-0001	Canonical ten-layer architecture	The ten named layers are the product structure; control planes support but do not replace them.	Implement through an identified state owner, versioned contract, enforcement point, conformance suite, release evidence, and operational control. The release manifest SHALL identify any active bounded exception.	C-008
ADR-0002	Knowledge Graph centrality	Shared semantic identity and relationships are governed through the Knowledge Graph.	Implement through an identified state owner, versioned contract, enforcement point, conformance suite, release evidence, and operational control. The release manifest SHALL identify any active bounded exception.	C-009
ADR-0003	Canonical intelligence object header	All material objects carry identity, version, time, truth, provenance, control, quality, and operational correlation.	Implement through an identified state owner, versioned contract, enforcement point, conformance suite, release evidence, and operational control. The release manifest SHALL identify any active bounded exception.	C-010–C-012
ADR-0004	Bitemporal system record	Valid and system time are preserved alongside event and observation time.	Implement through an identified state owner, versioned contract, enforcement point, conformance suite, release evidence, and operational control. The release manifest SHALL identify any active bounded exception.	C-011
ADR-0005	Immutable version history	Corrections create new versions and propagation events; historical state is not overwritten.	Implement through an identified state owner, versioned contract, enforcement point, conformance suite, release evidence, and operational control. The release manifest SHALL identify any active bounded exception.	C-011, C-025
ADR-0006	Customer Intelligence Domain	Customer evidence, knowledge, models, agents, policies, decisions, and audit are isolated by a stable domain boundary.	Implement through an identified state owner, versioned contract, enforcement point, conformance suite, release evidence, and operational control. The release manifest SHALL identify any active bounded exception.	C-037, C-039
ADR-0007	Human final authority	Decision Intelligence may recommend and prepare; a named human makes final high-impact decisions.	Implement through an identified state owner, versioned contract, enforcement point, conformance suite, release evidence, and operational control. The release manifest SHALL identify any active bounded exception.	C-004
ADR-0008	Durable workflow and human gates	Consequential state transitions use durable workflows with explicit roles, evidence, approval, retry, and audit.	Implement through an identified state owner, versioned contract, enforcement point, conformance suite, release evidence, and operational control. The release manifest SHALL identify any active bounded exception.	C-004, C-029
ADR-0009	Bounded agent execution	Agents operate through capability grants, supervision, budgets, evaluation, and stop conditions.	Implement through an identified state owner, versioned contract, enforcement point, conformance suite, release evidence, and operational control. The release manifest SHALL identify any active bounded exception.	C-028, C-029
ADR-0010	Model gateway and pluralism	Models remain replaceable, governed dependencies resolved by task and deployment policy.	Implement through an identified state owner, versioned contract, enforcement point, conformance suite, release evidence, and operational control. The release manifest SHALL identify any active bounded exception.	C-030
ADR-0011	Digital Twin state separation	Observed, estimated, forecast, scenario, and simulated twin states remain distinct and versioned.	Implement through an identified state owner, versioned contract, enforcement point, conformance suite, release evidence, and operational control. The release manifest SHALL identify any active bounded exception.	C-018, C-019
ADR-0012	Probabilistic forecast package	Forecast products carry distributions, calibration, sensitivity, lineage, and failure conditions.	Implement through an identified state owner, versioned contract, enforcement point, conformance suite, release evidence, and operational control. The release manifest SHALL identify any active bounded exception.	C-020, C-021
ADR-0013	Scenario portfolio	Multiple coherent branches remain visible and monitored; no branch becomes destiny.	Implement through an identified state owner, versioned contract, enforcement point, conformance suite, release evidence, and operational control. The release manifest SHALL identify any active bounded exception.	C-022

ADR	DECISION	ARCHITECTURE CONSEQUENCE	VOLUME 4 IMPLEMENTATION OBLIGATION	INHERITANCE
ADR-0014	Reproducible simulation manifest	Simulation binds exact inputs, versions, constraints, environment, seeds, outputs, and validation.	Implement through an identified state owner, versioned contract, enforcement point, conformance suite, release evidence, and operational control. The release manifest SHALL identify any active bounded exception.	C-023
ADR-0015	Decision package completeness	Objectives, options, evidence, uncertainty, trade-offs, dissent, authority, and monitoring form one package.	Implement through an identified state owner, versioned contract, enforcement point, conformance suite, release evidence, and operational control. The release manifest SHALL identify any active bounded exception.	C-024–C-026
ADR-0016	Deployment semantic parity	SaaS, Private Cloud, and On-Premise implement one product contract.	Implement through an identified state owner, versioned contract, enforcement point, conformance suite, release evidence, and operational control. The release manifest SHALL identify any active bounded exception.	C-042
ADR-0017	Sovereignty as policy dimensions	Data, keys, models, administration, telemetry, supply, support, and continuity are independently governed.	Implement through an identified state owner, versioned contract, enforcement point, conformance suite, release evidence, and operational control. The release manifest SHALL identify any active bounded exception.	C-041
ADR-0018	Safe degradation hierarchy	Failure may reduce capability but cannot fabricate currency, confidence, authorization, or approval.	Implement through an identified state owner, versioned contract, enforcement point, conformance suite, release evidence, and operational control. The release manifest SHALL identify any active bounded exception.	C-043, C-049
ADR-0019	Governed marketplace packages	Packages are signed, declared, sandboxed, evaluated, capability-scoped, monitored, and revocable.	Implement through an identified state owner, versioned contract, enforcement point, conformance suite, release evidence, and operational control. The release manifest SHALL identify any active bounded exception.	C-046
ADR-0020	Governed learning release	Lessons become behavior only through evaluation, approval, deployment, monitoring, and rollback.	Implement through an identified state owner, versioned contract, enforcement point, conformance suite, release evidence, and operational control. The release manifest SHALL identify any active bounded exception.	C-031, C-044, C-045

K Engineering Glossary

These definitions are canonical for engineering, assurance, delivery, and operations. Domain specifications may add narrower terms but must not redefine the concepts below.

TERM	ENGINEERING DEFINITION
Acceptance evidence	Signed, versioned proof that a defined requirement or readiness condition was evaluated for an exact artifact and scope.
Admission control	A boundary decision that accepts, queues, sheds, degrades, reschedules, or rejects work before unsafe execution.
Artifact identity	Content digest plus declared type, version, provenance, and signing context that uniquely binds a release input or output.
Authority boundary	The point at which one accountable owner may commit state or action and other parties must interact by contract.
Capability grant	Revocable task-specific authorization for an agent or workload to use a tool, data scope, model, network path, resource, or side effect.
Canonical state	Authoritative versioned state governed by an owning component and preserved for temporal truth and replay.
Compensation	Explicit action that addresses the institutional effects of a committed step when atomic rollback is impossible.
Conformance fixture	Portable canonical input, environment assumptions, expected behavior, tolerances, and evidence used across implementations.
Consequence class	Declared level of potential institutional, human, financial, legal, security, or strategic effect that controls assurance.
Control plane	Cross-cutting service and authority for identity, policy, trust, workflow, audit, evaluation, learning, deployment, or sovereignty.
Correction propagation	Durable identification, invalidation, review, and reconciliation of products affected by corrected or withdrawn information.
Customer Intelligence Domain	Primary logical isolation boundary for a customer's evidence, knowledge, models, agents, policies, decisions, and audit.
Decision-active	Permitted to influence an assessment, warning, recommendation, approval, commitment, or executive action under current policy and quality.
Degraded mode	Named runtime state with explicit guarantees, prohibitions, entry conditions, exit criteria, and consumer signals.
Deployment semantic parity	Equivalent objects, contracts, workflows, authority, controls, audit, quality, and portability across supported deployment modes.
Durable workflow	Persisted execution state that survives interruption and preserves transitions, timers, retries, compensation, human gates, and evidence.
Effect receipt	Verifiable acknowledgement from an external or authoritative boundary that a specific idempotent side effect completed.
Error budget	Allowed unreliability for a quality objective; it never authorizes breach of a constitutional or security control.
Evidence closure	State in which all required verification records are current, authentic, scoped correctly, and linked to the exact artifact.
Failure domain	Boundary within which one defect or outage is expected to remain isolated.
Fitness decision	Governed approval, constraint, shadow status, suspension, withdrawal, or indeterminate result for an exact artifact and use.
Human gate	Durable transition requiring an authenticated named human role to act under verified authority and an exact information boundary.
Idempotency boundary	Scope within which repeated delivery of the same intent cannot create an additional institutional effect.
Immutable promotion	Advancement of the same artifact digest across environments without rebuild or manual replacement.
Intelligence Cell	Scalable and isolated deployment and failure unit for a bounded set of customer or domain workloads.

TERM	ENGINEERING DEFINITION
Material change	Change significant enough by consequence, urgency, confidence, reversibility, exposure, or information value to require accountable attention.
Model gateway	Governed boundary that resolves and invokes approved models according to task, data, quality, sovereignty, cost, and failure policy.
Non-repudiation	Evidence sufficient to attribute a consequential act and verify its integrity, authority, time, and result.
Operational acceptance	Governed authorization to operate an exact capability and scope after cross-domain readiness evidence closes.
Policy obligation	Mandatory handling action returned with an authorization decision, such as masking, logging, localization, approval, or expiry.
Projection	Rebuildable access representation derived from declared canonical state and algorithms.
Quality contract	Measurable agreement defining indicator, population, objective, window, consequence, degraded behavior, and evidence.
Reconciliation	Deterministic comparison and repair that restores consistency across authoritative state, events, workflows, audit, and projections.
Recovery point objective	Maximum acceptable authoritative state loss measured against a mutually consistent recovery boundary.
Recovery time objective	Maximum time to restore and verify a declared product capability, including reconciliation.
Release candidate	Immutable complete artifact set evaluated for promotion as one governed unit.
Release gate	Policy decision over required evidence, authority, exceptions, scope, and artifact identity before promotion.
Residual state	Known copy or effect remaining after deletion, correction, revocation, rollback, migration, or customer exit.
Side-effect class	Classification of an action as none, reversible, compensatable, approval-required, irreversible, or prohibited.
Source of truth	The explicitly declared owner of authoritative state for one object and lifecycle; it is never inferred from convenience or query speed.
State owner	Exactly one accountable component authorized to commit the canonical lifecycle of a state class.
System time	When The Eye admitted or changed an object version, used to reconstruct what the platform knew at a historical instant.
Temporal truth	Preservation of event, observation, valid, and system time without overwriting historical knowledge state.
Truth state	Explicit distinction among observed, claimed, inferred, assessed, decided, synthetic, withdrawn, and superseded information.
Verification class	Inspection, analysis, automated test, fault injection, security test, operational exercise, or accountable review.
Workload identity	Cryptographically verifiable identity of a running service, job, agent, model, device, or integration.

Baseline closure

Volume 4 closes when every deployable behavior can identify its controlling constitutional invariant, architecture layer and component, canonical object and interface, state owner, control and trust boundary, quality contract, failure and recovery semantics, deployment-parity fixture, release evidence, operational authority, and retirement path.

END STATE

The Eye is engineered here as one persistent Strategic Intelligence Operating System: continuously observing, constructing governed understanding, remembering permanently, connecting knowledge, representing the enterprise and world through living twins, forecasting and simulating possible futures, preparing explainable recommendations for named human decision-makers, governing commitments, and learning only through accountable release.